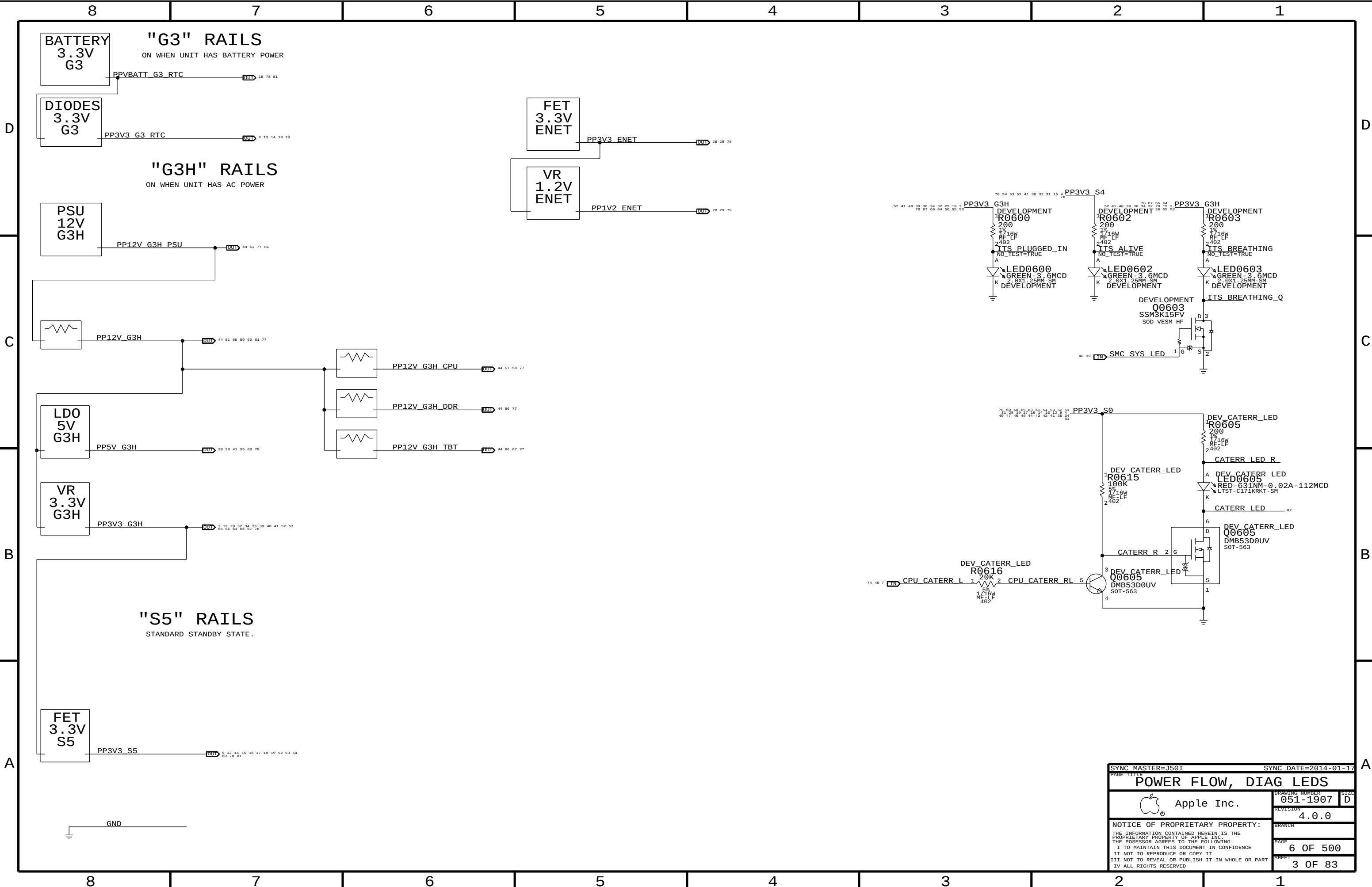
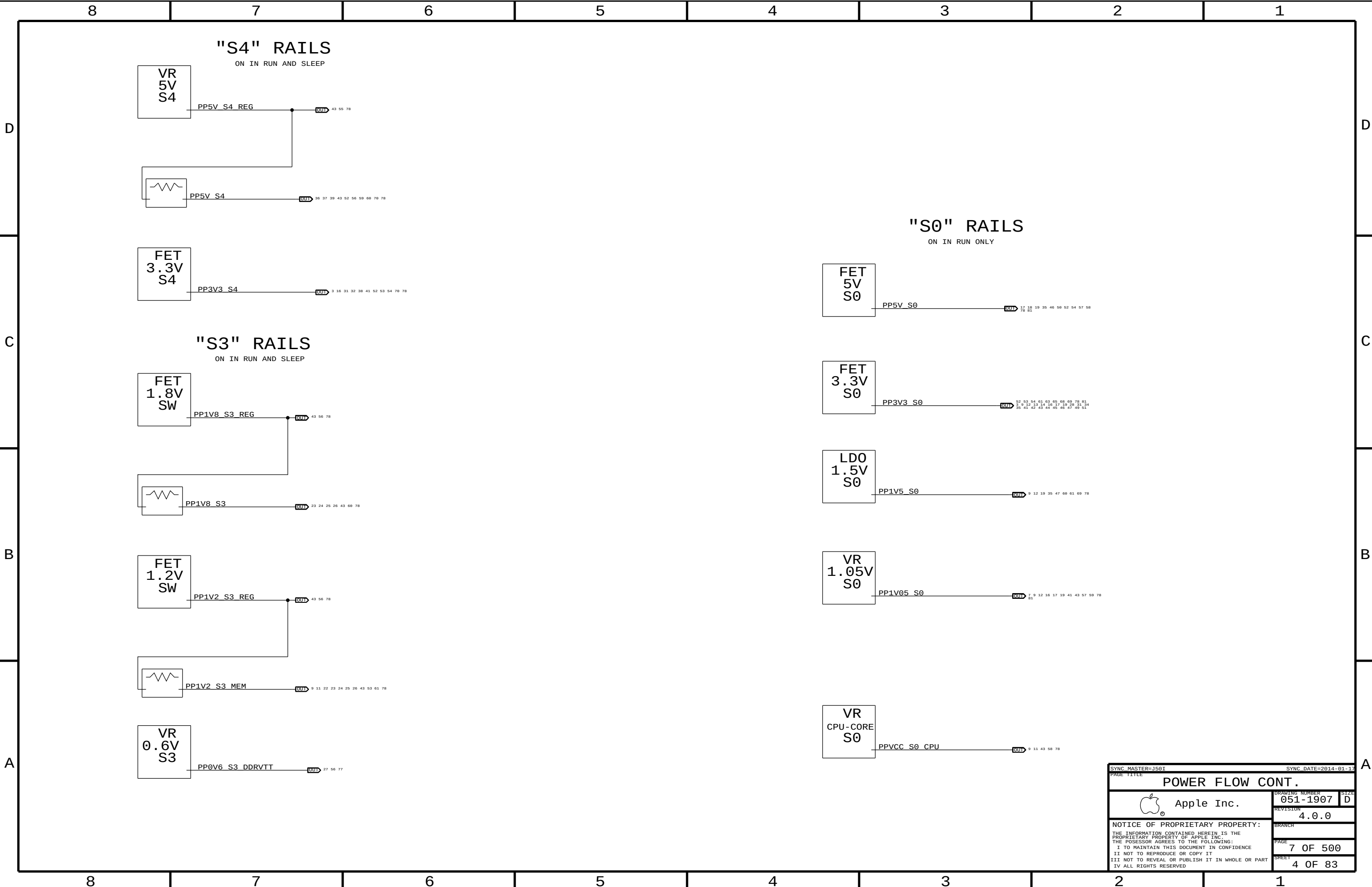
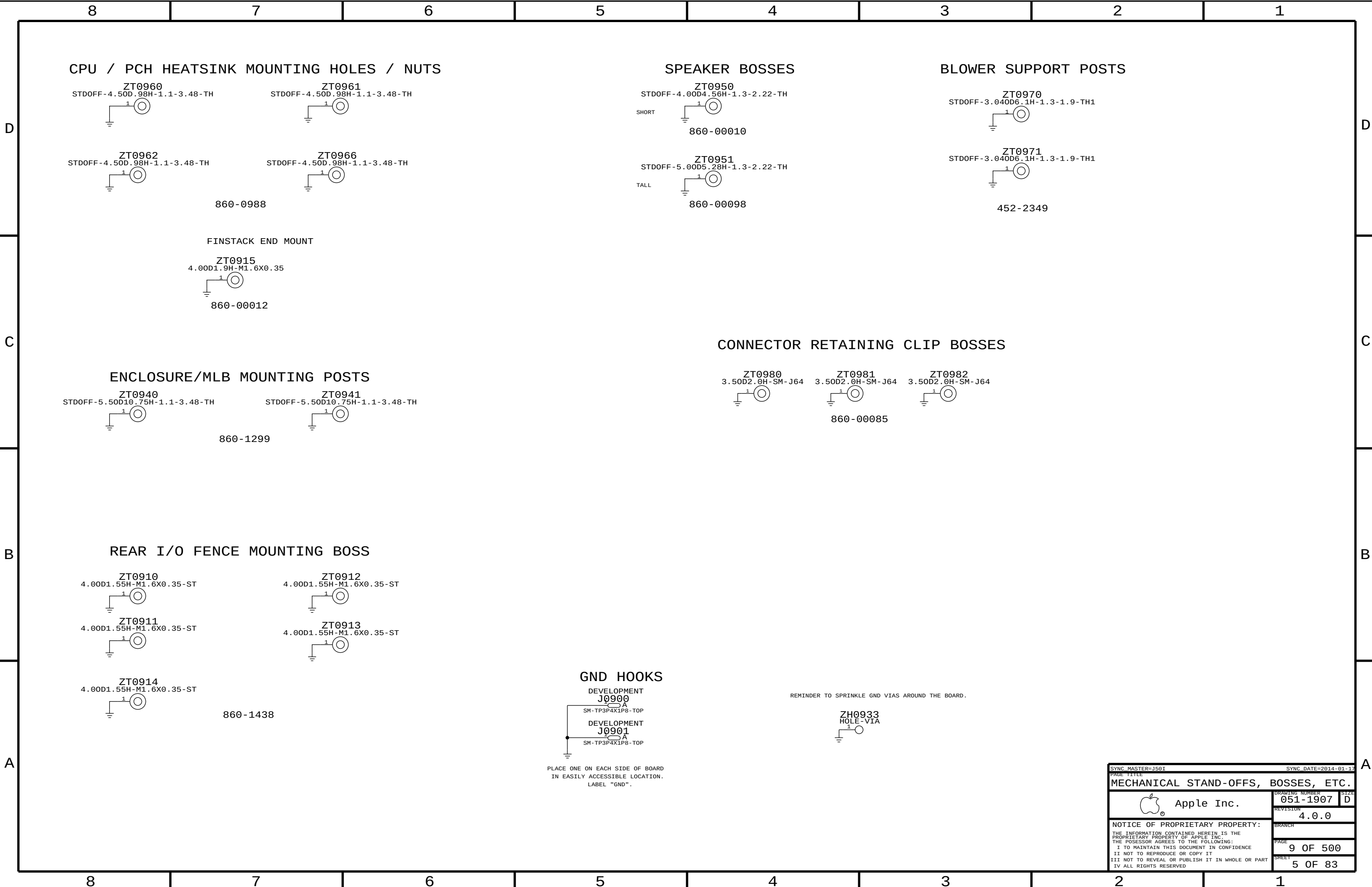


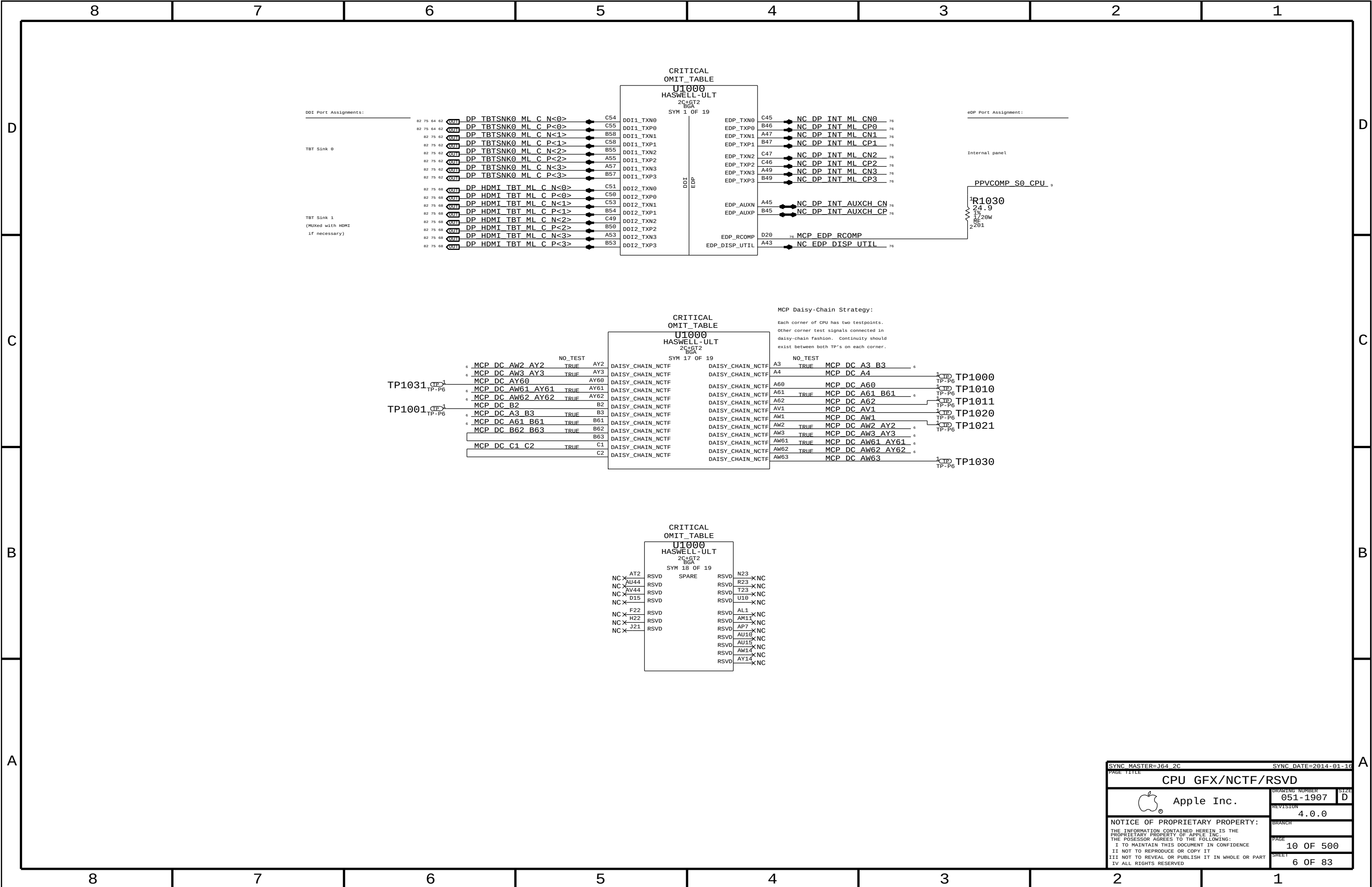


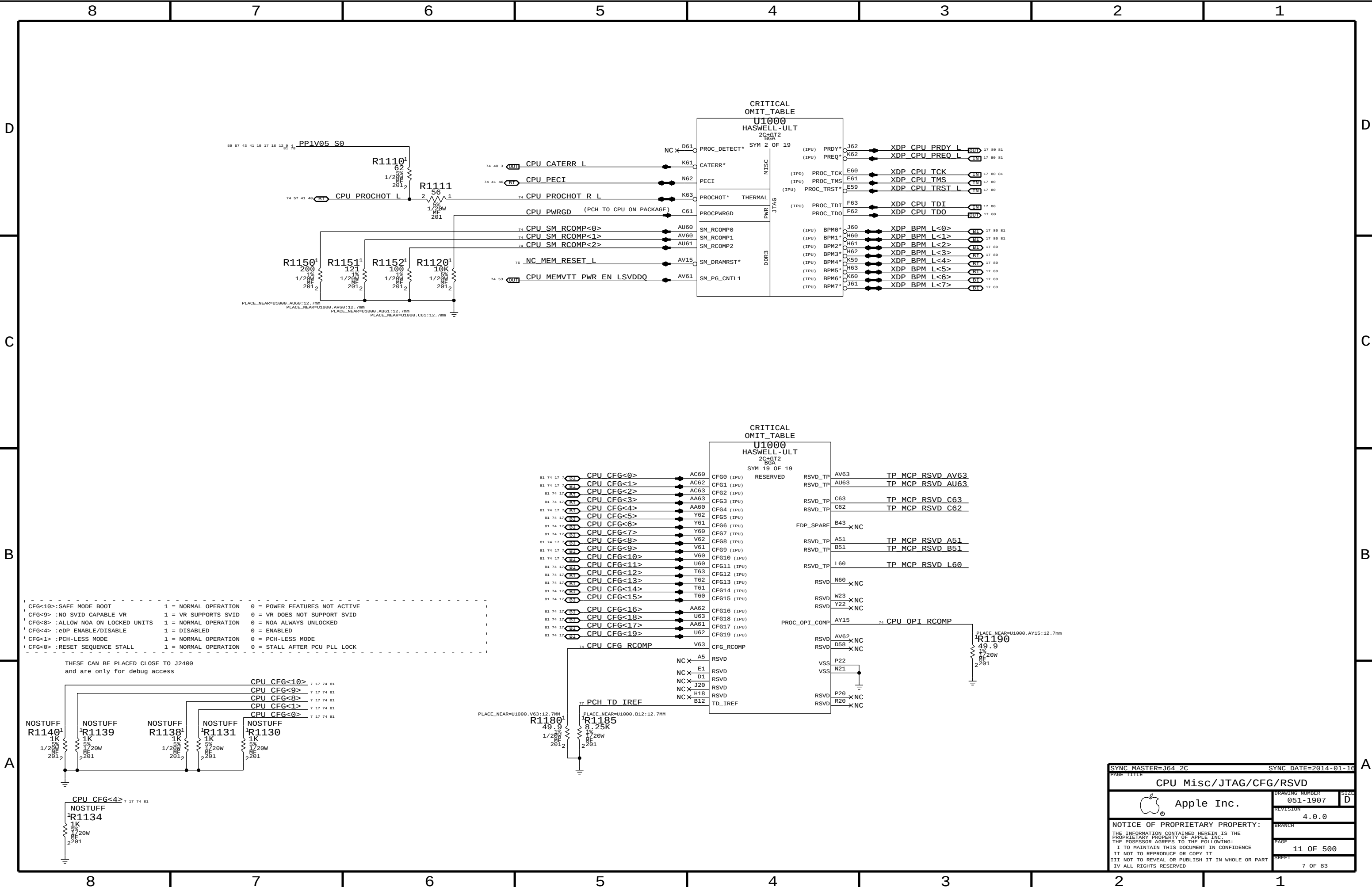


SYNC MASTER=J501		SYNC DATE=2014-01-17	
PAGE TITLE		POWER FLOW CONT.	
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		PAGE	7 OF 500
		SHEET	4 OF 83

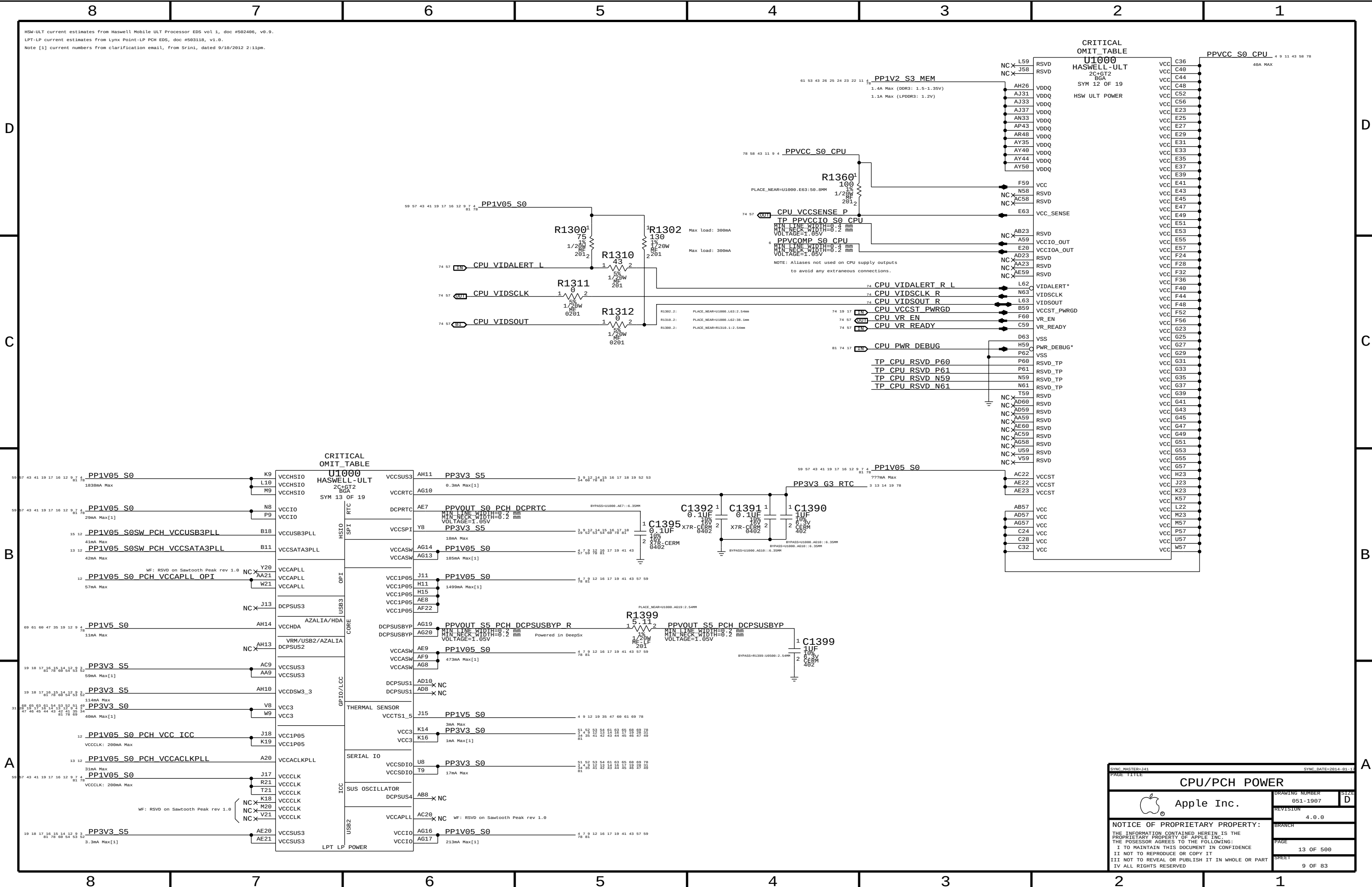


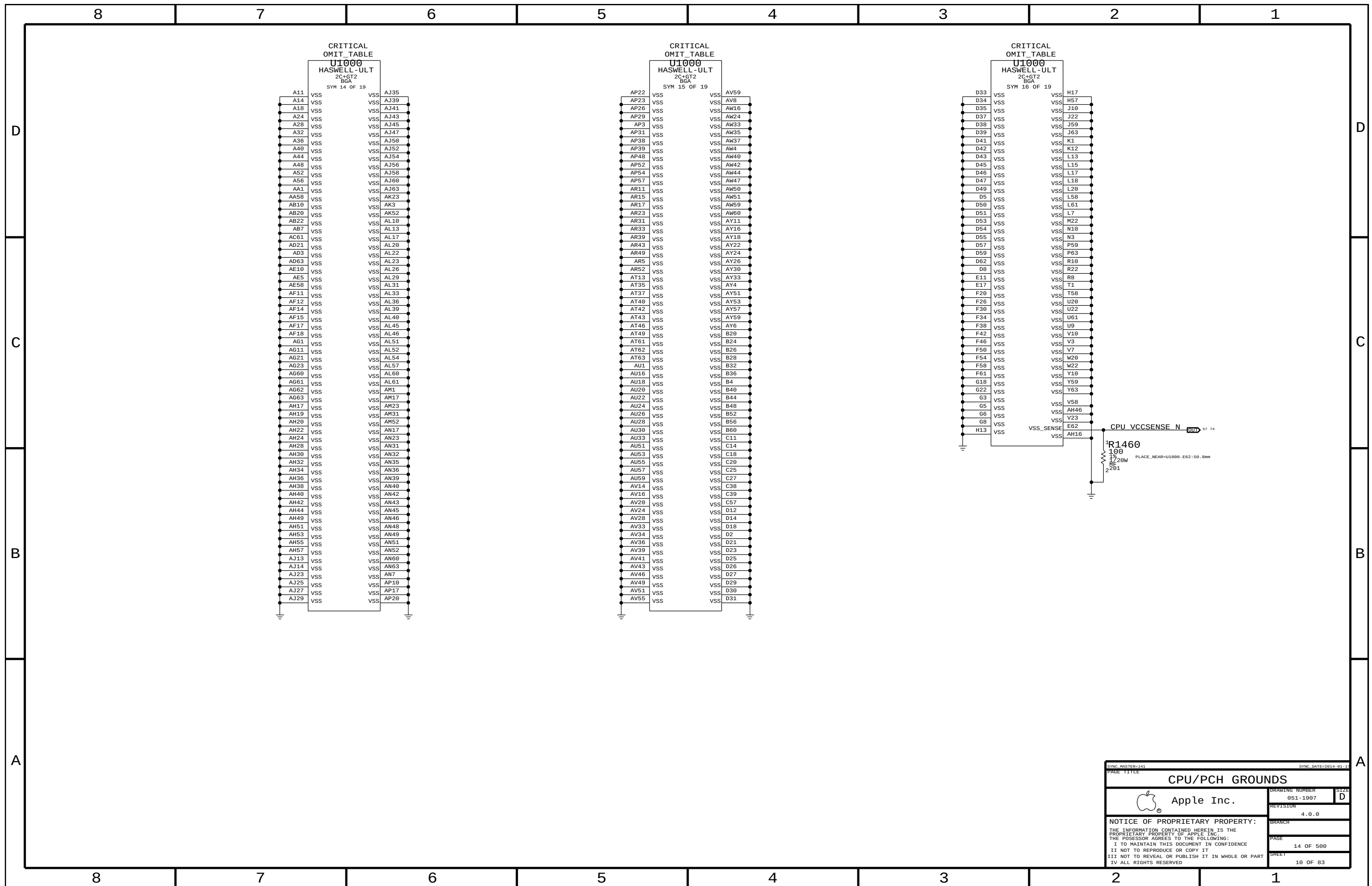
SYNC MASTER=J56I		SYNC DATE=2014-01-17	
PAGE TITLE		MECHANICAL STAND-OFFS, BOSSES, ETC.	
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		PAGE	9 OF 500
		SHEET	5 OF 83





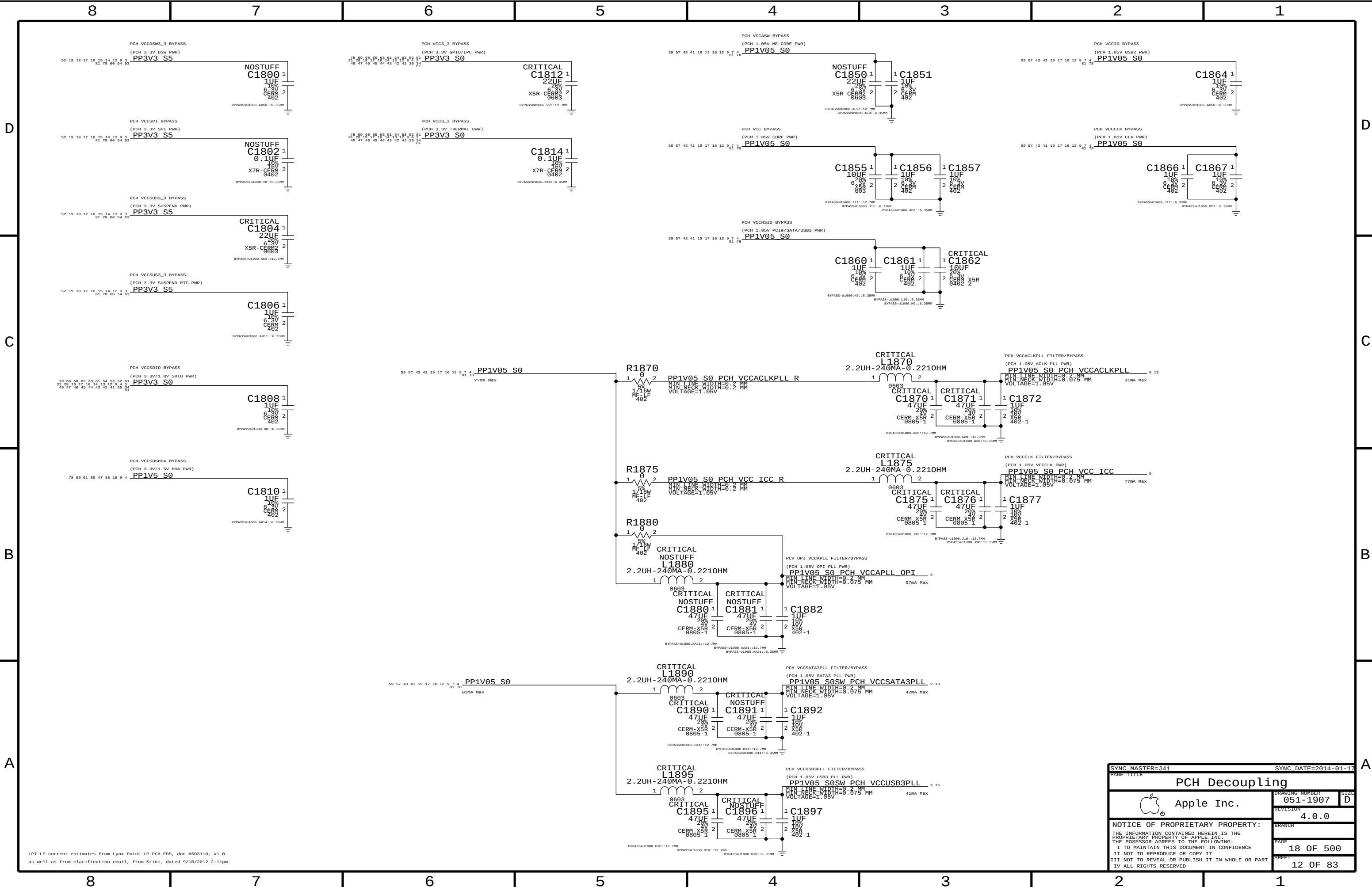






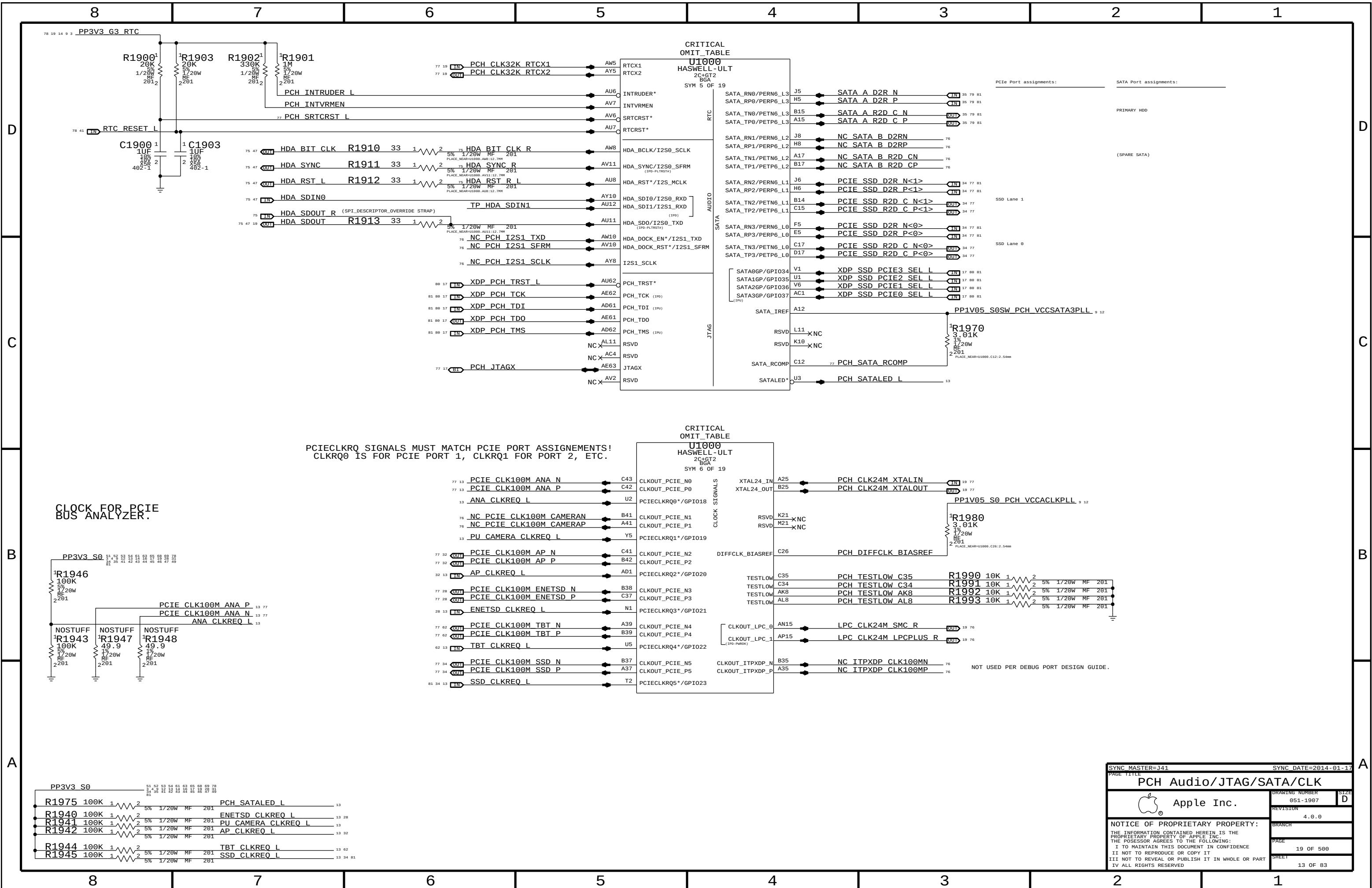


SYNC MASTER=J64 2C		SYNC DATE=2014-01-26	
PAGE TITLE			
CPU Decoupling			
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		PAGE	16 OF 500
		SHEET	11 OF 83



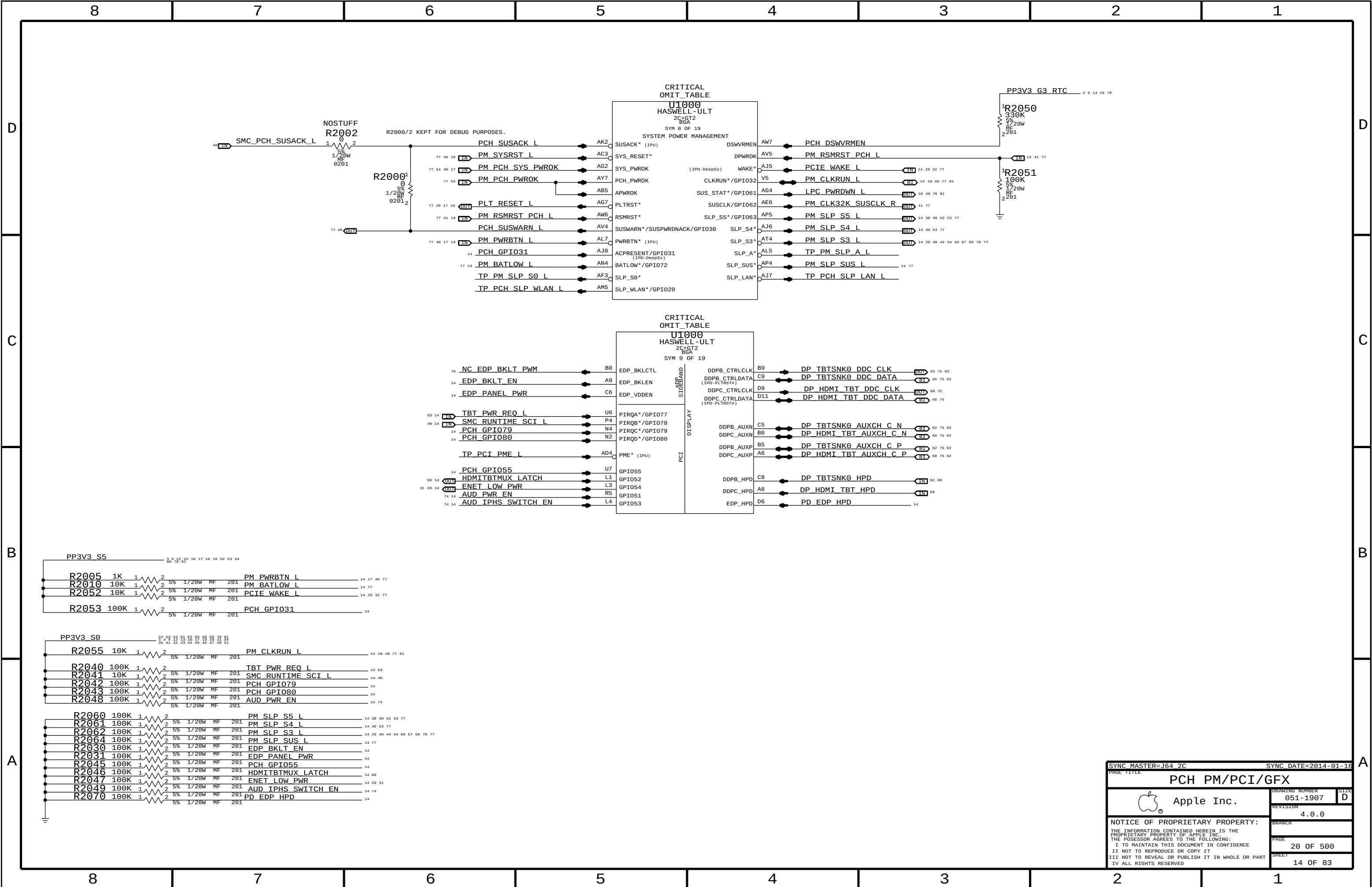
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Srin, dated 9/10/2012 2:11pm.

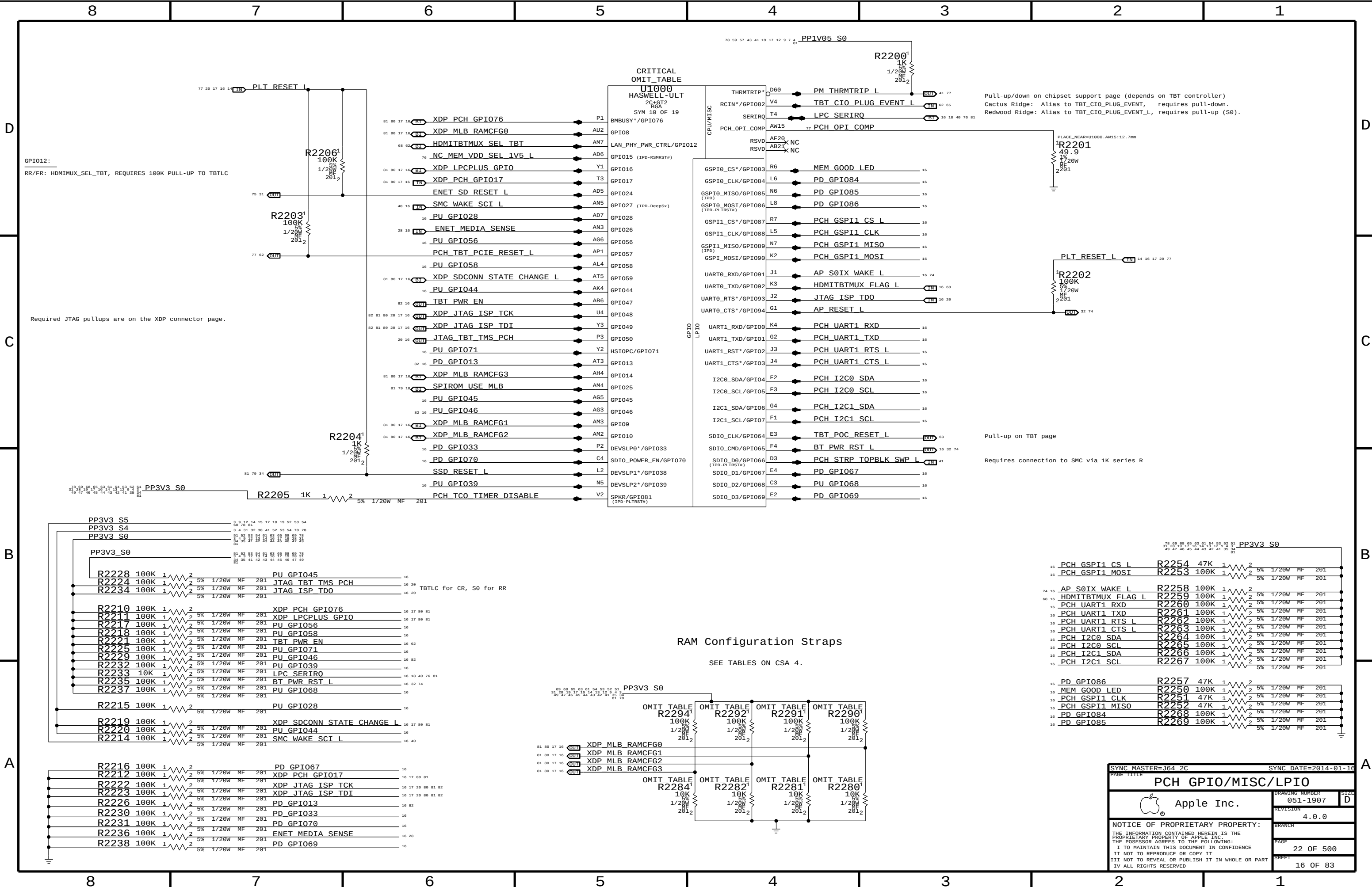
SYNC MASTER=J41		SYNC DATE=2014-01-17	
PAGE TITLE		PCH Decoupling	
 Apple Inc.		DRAWING NUMBER	051-1907
		SIZE	D
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		PAGE	18 OF 500
		SHEET	12 OF 83



CLOCK FOR PCIE
BUS ANALYZER.

SYNC MASTER=J41		SYNC DATE=2014-01-17	
PAGE TITLE			
PCH Audio/JTAG/SATA/CLK		DRAWING NUMBER	SIZE
Apple Inc.		051-1907	D
REVISION		4.0.0	
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Pull-up/down on chipset support page (depends on TBT controller)
Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.
Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).

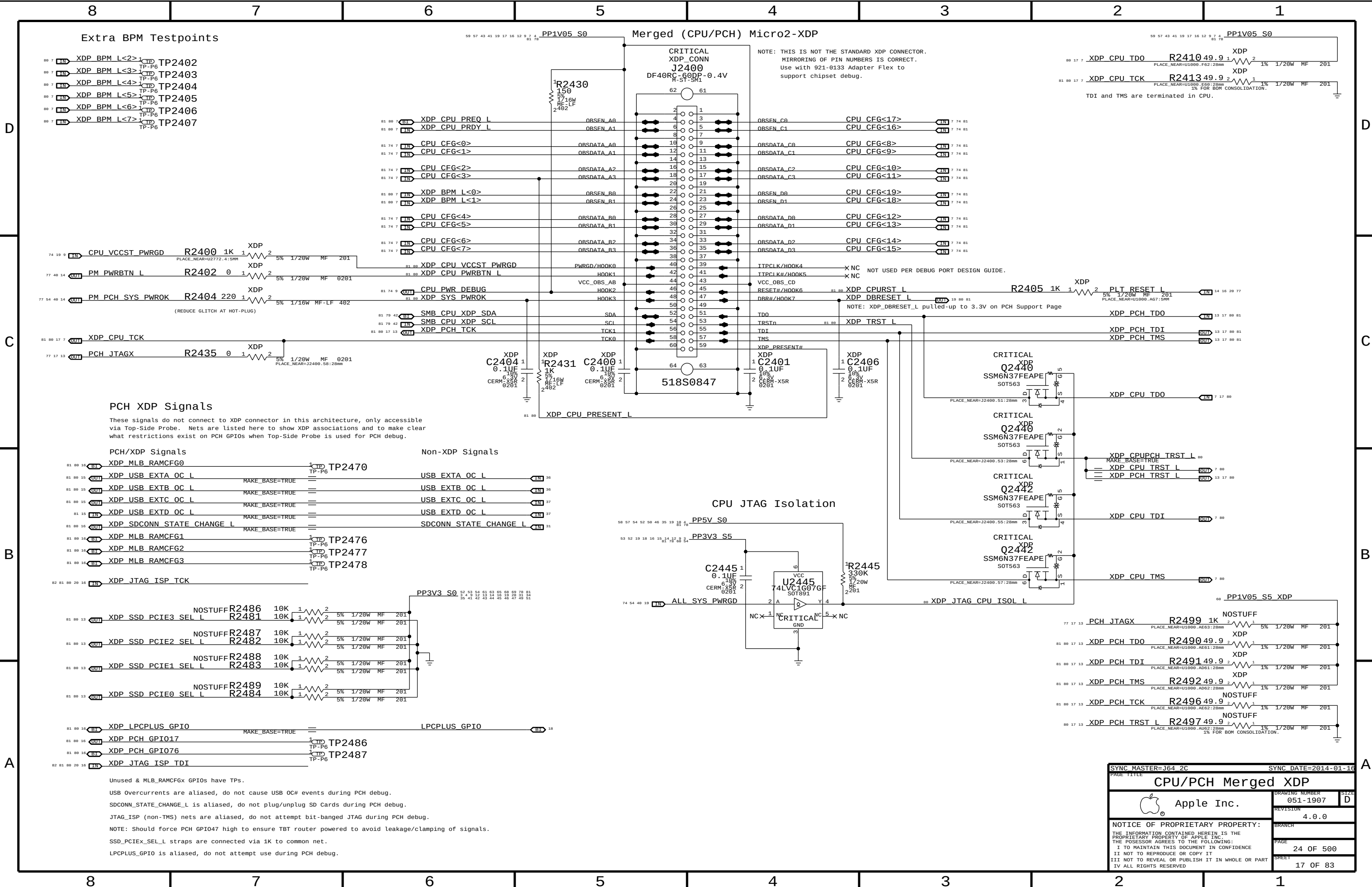
Pull-up on TBT page

Requires connection to SMC via 1K series R

RAM Configuration Straps

SEE TABLES ON CSA 4.

SYNC MASTER=J64 2C		SYNC DATE=2014-01-16	
PAGE TITLE		PCH GPIO/MISC/LPIO	
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		PAGE	22 OF 500
		SHEET	16 OF 83



C

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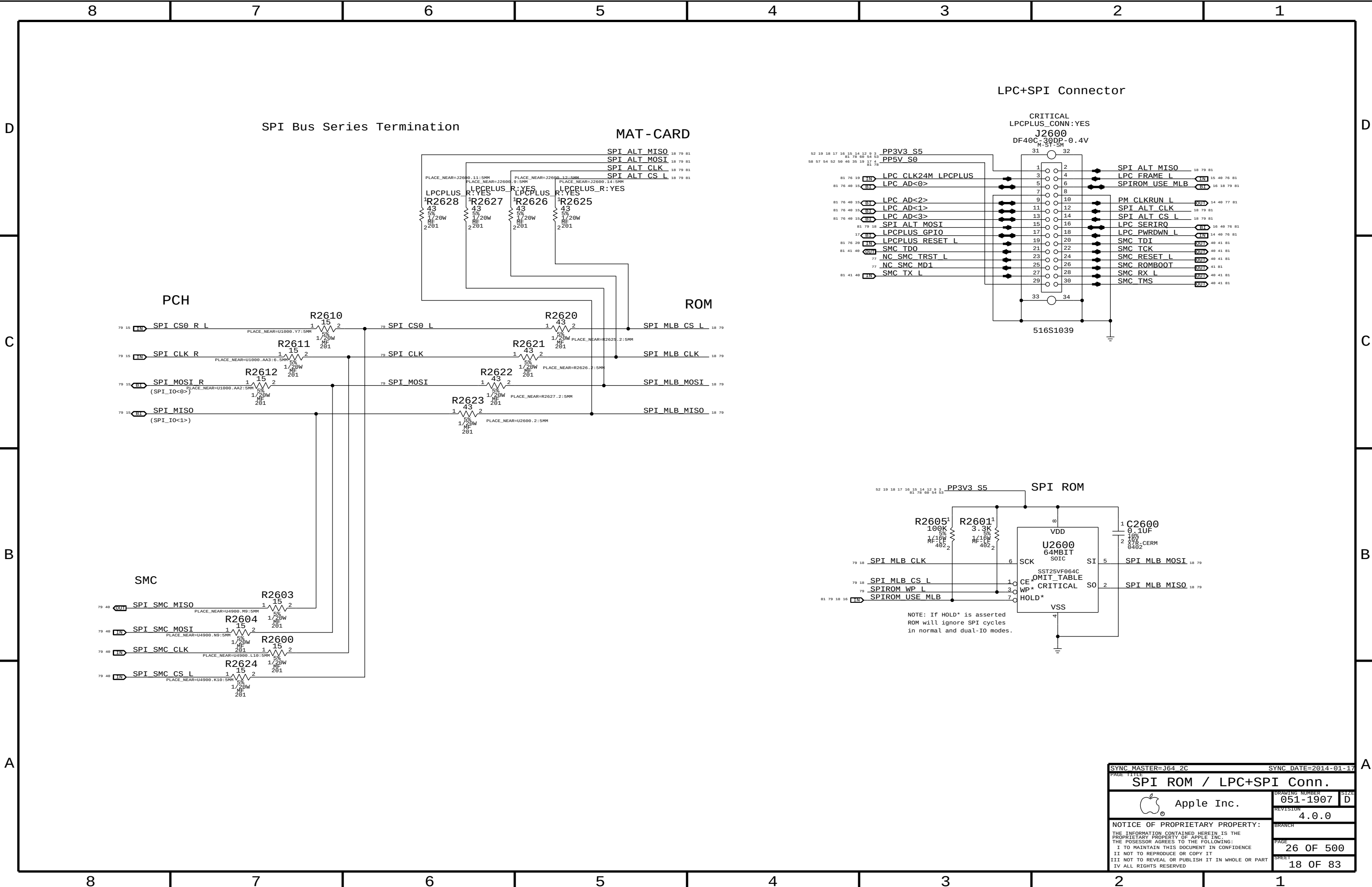
D

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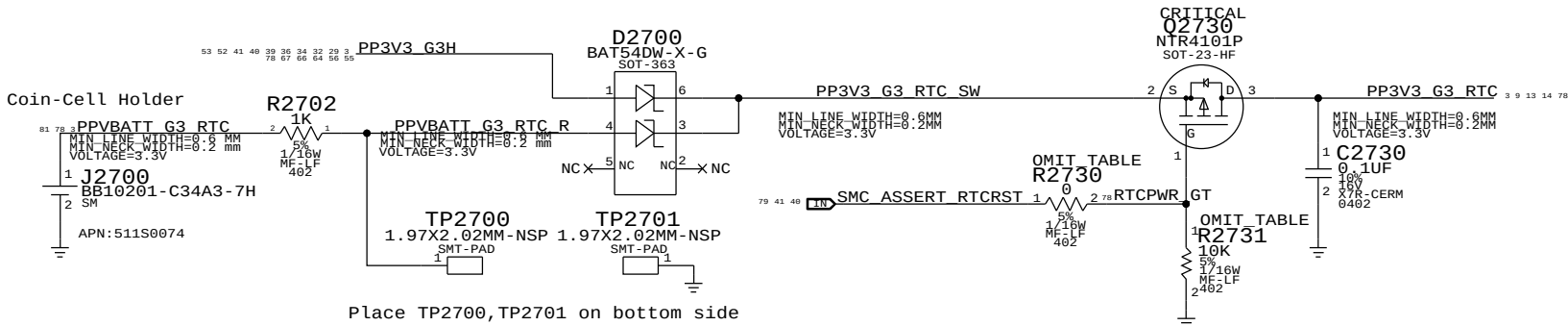
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SYNC MASTER=J64_2C		SYNC DATE=2014-01-16	
PAGE TITLE			
CPU/PCH Merged XDP			
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		SHEET	17 OF 83



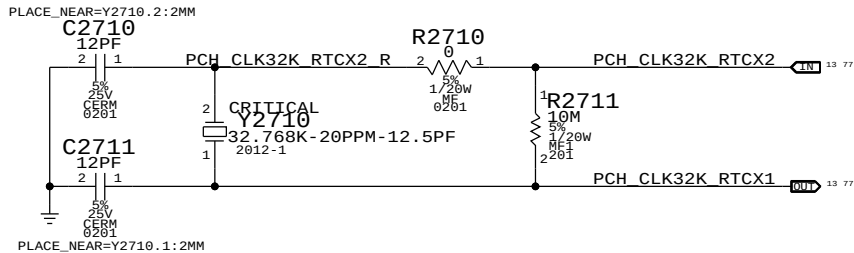
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
742-0062	1	BAT, LI/CF, BR2032, 3V, 190MAH	BAT	CRITICAL	
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
742-0080	742-0062	ALT_CMN		ALTERNATE	

RTC POWER SOURCES

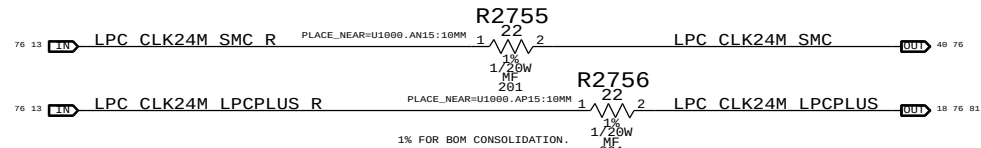


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0090	1	RES, 10K 0HM, 402	R2730	RTCRST:Y
132S0333	1	CAP, 0.1 UF, 402	R2731	RTCRST:Y
116S0004	1	RES, 0 0HM, 402	R2730	RTCRST:N
116S0090	1	RES, 10K 0HM, 402	R2731	RTCRST:N

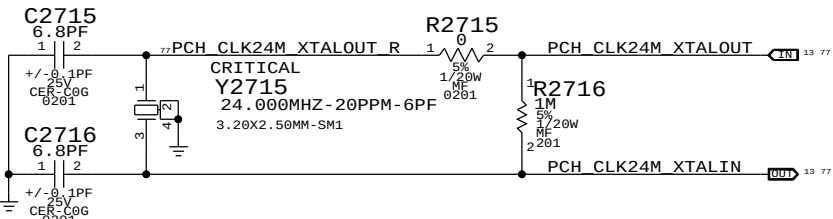
PCH RTC Crystal



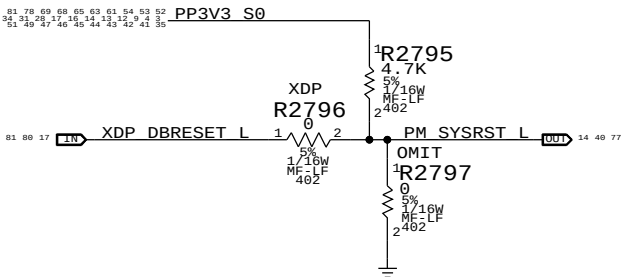
Clock series termination



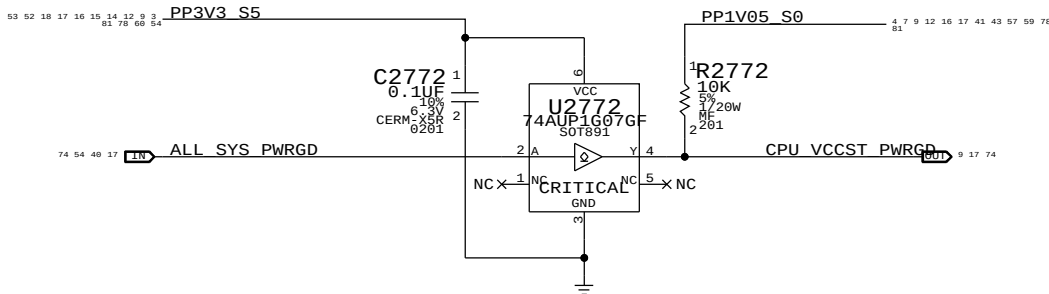
PCH 24MHz Crystal



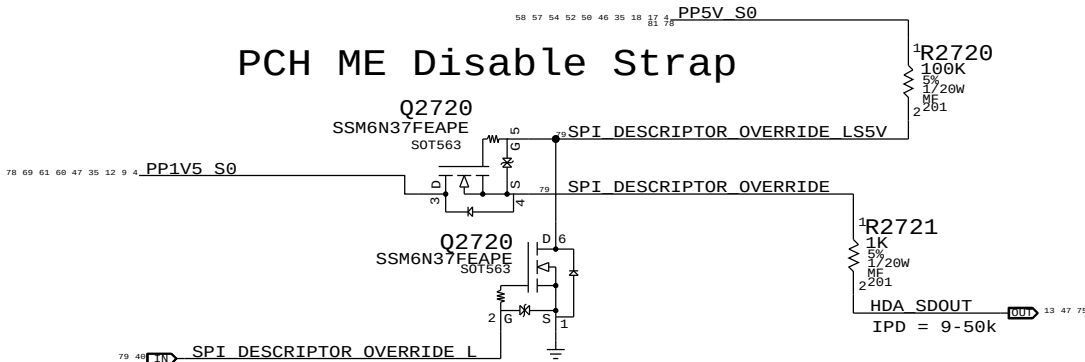
PCH Reset Button



ALL_SYS_PWRGD/CPU_VCCST_PWRGD Level-Shifter

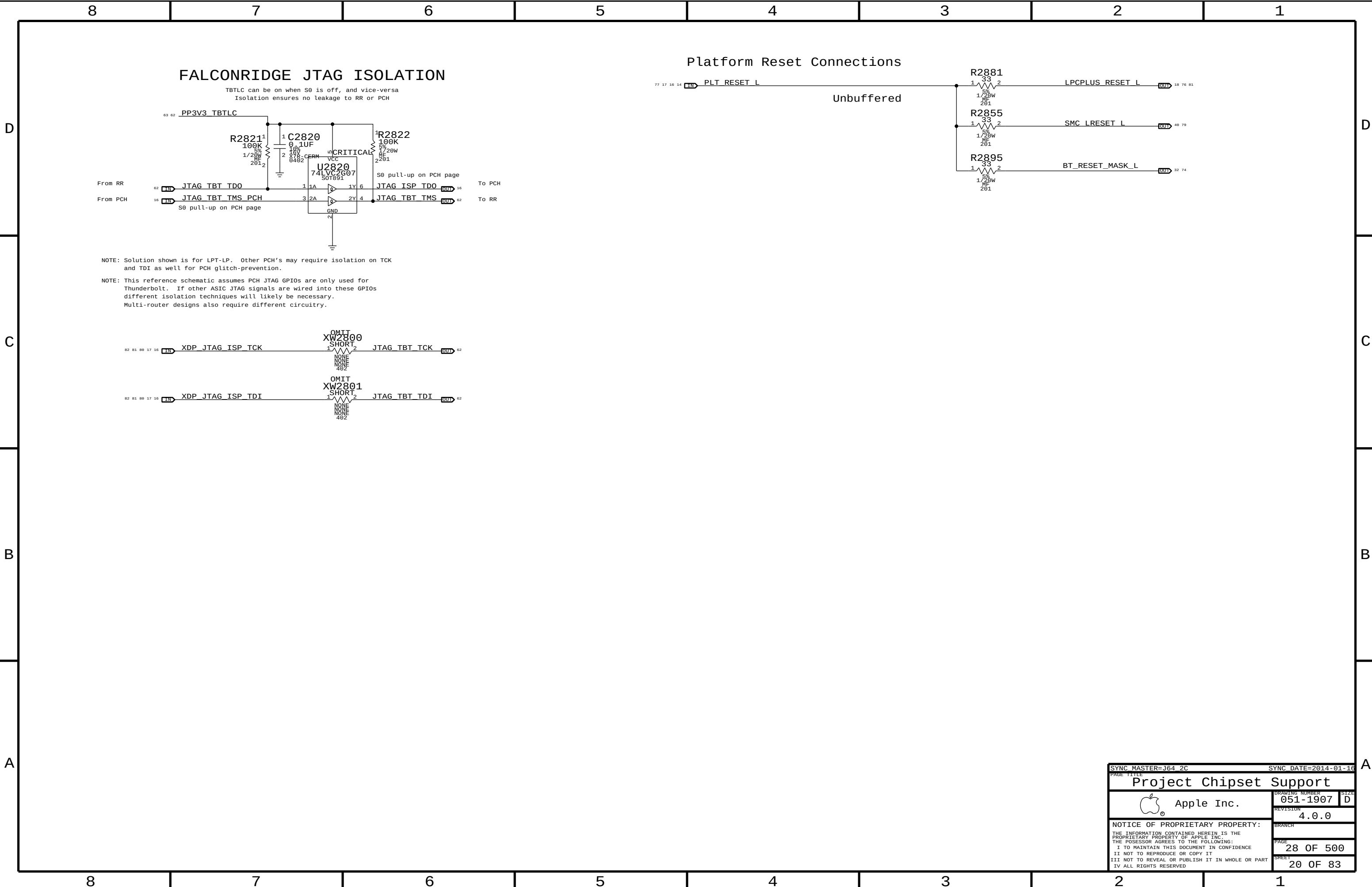


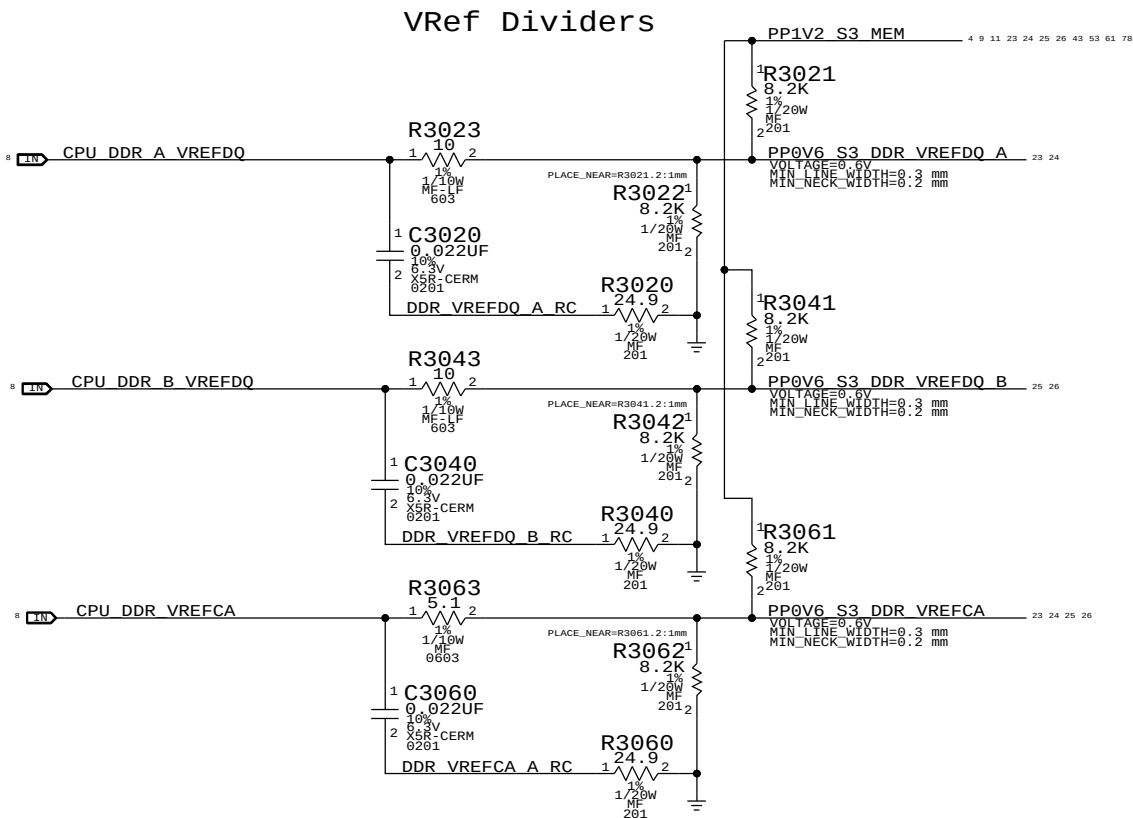
PCH ME Disable Strap



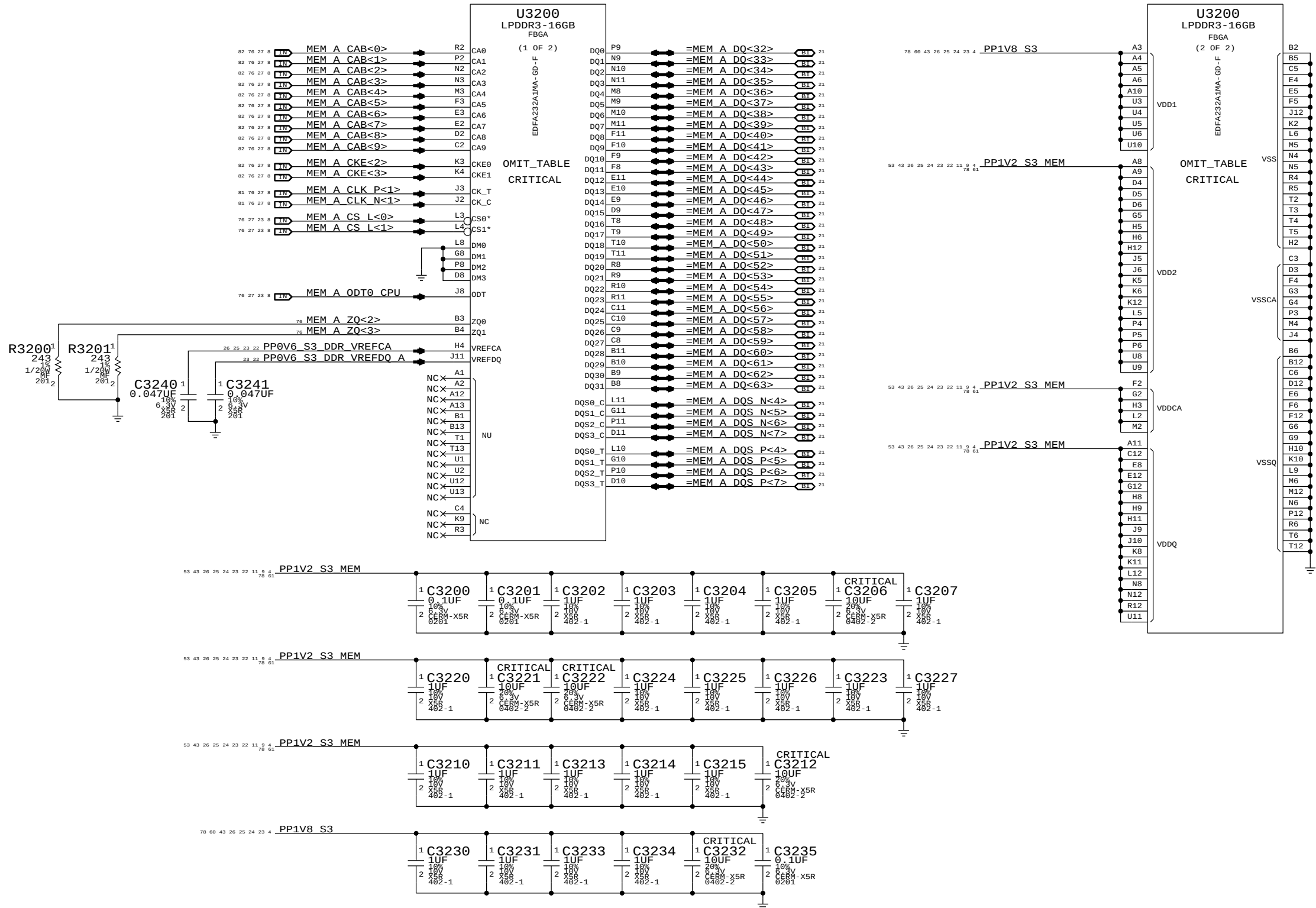
PCH uses HDA_SDOUT as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.

SYNC MASTER=J64_2C	SYNC DATE=2014-01-21
Chipset Support	
Apple Inc.	DRAWING NUMBER 051-1907 SIZE D
REVISION 4.0.0	BRANCH
PAGE 27 OF 500	SHEET
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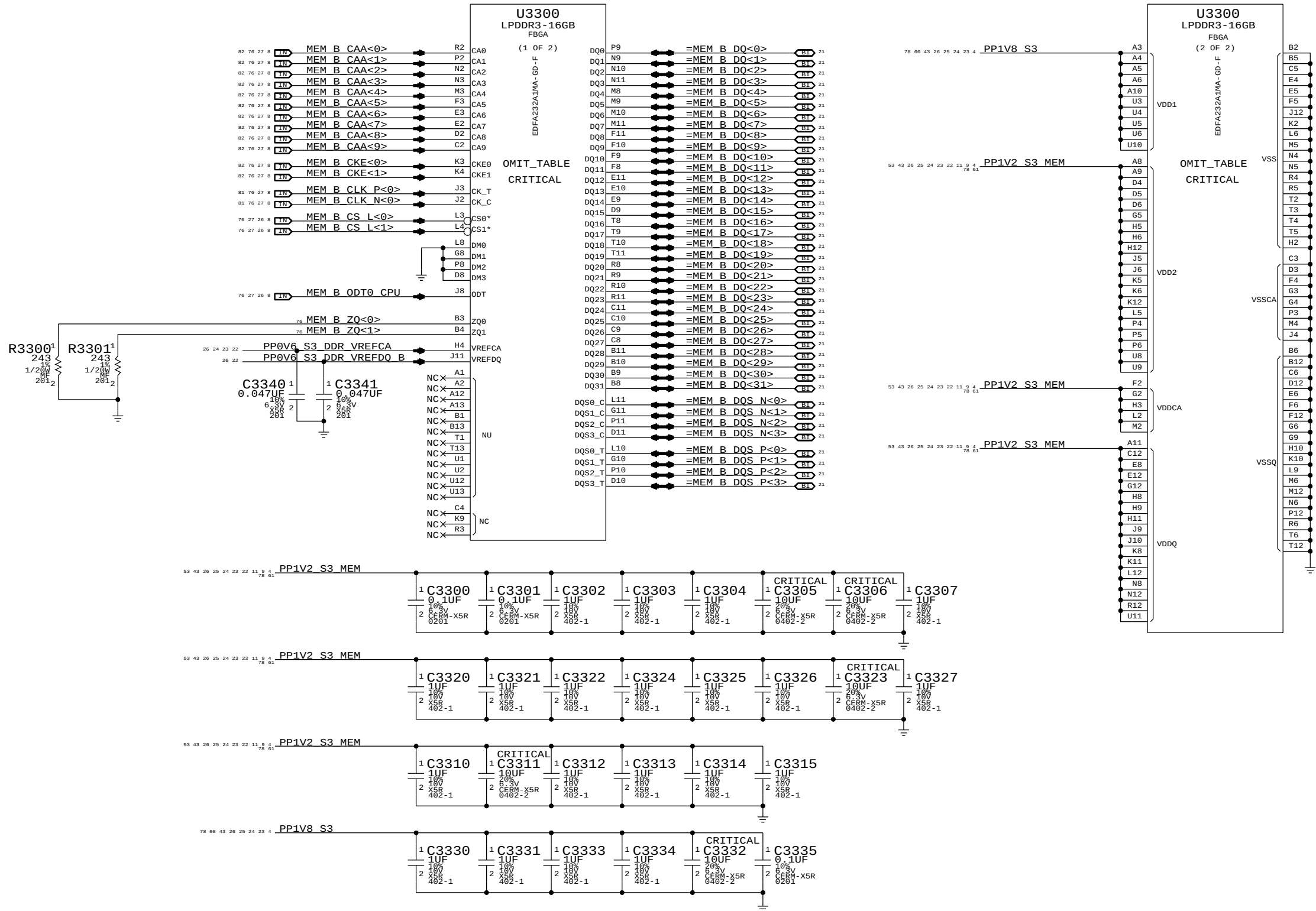




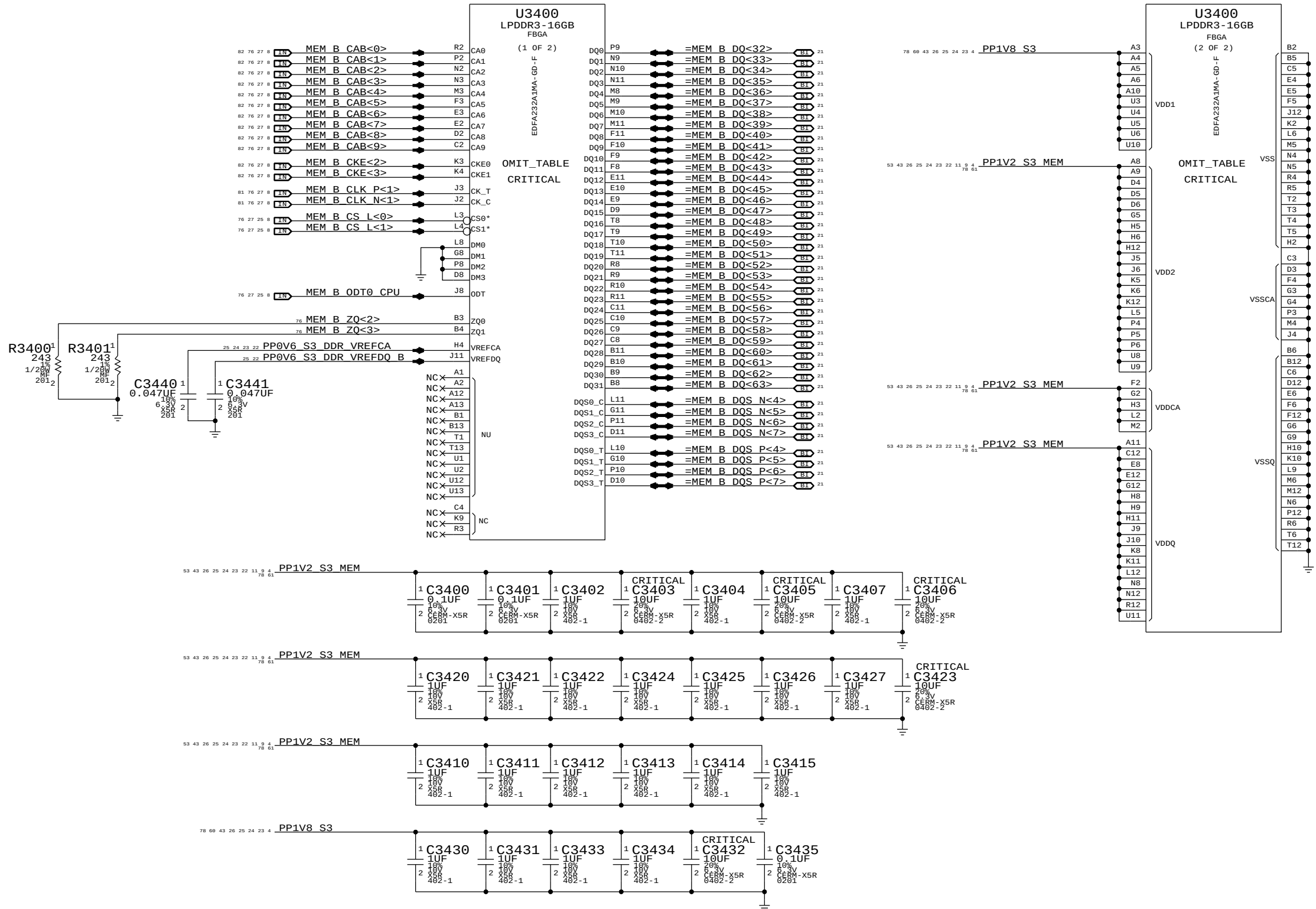
LPDDR3 CHANNEL A (32-63)

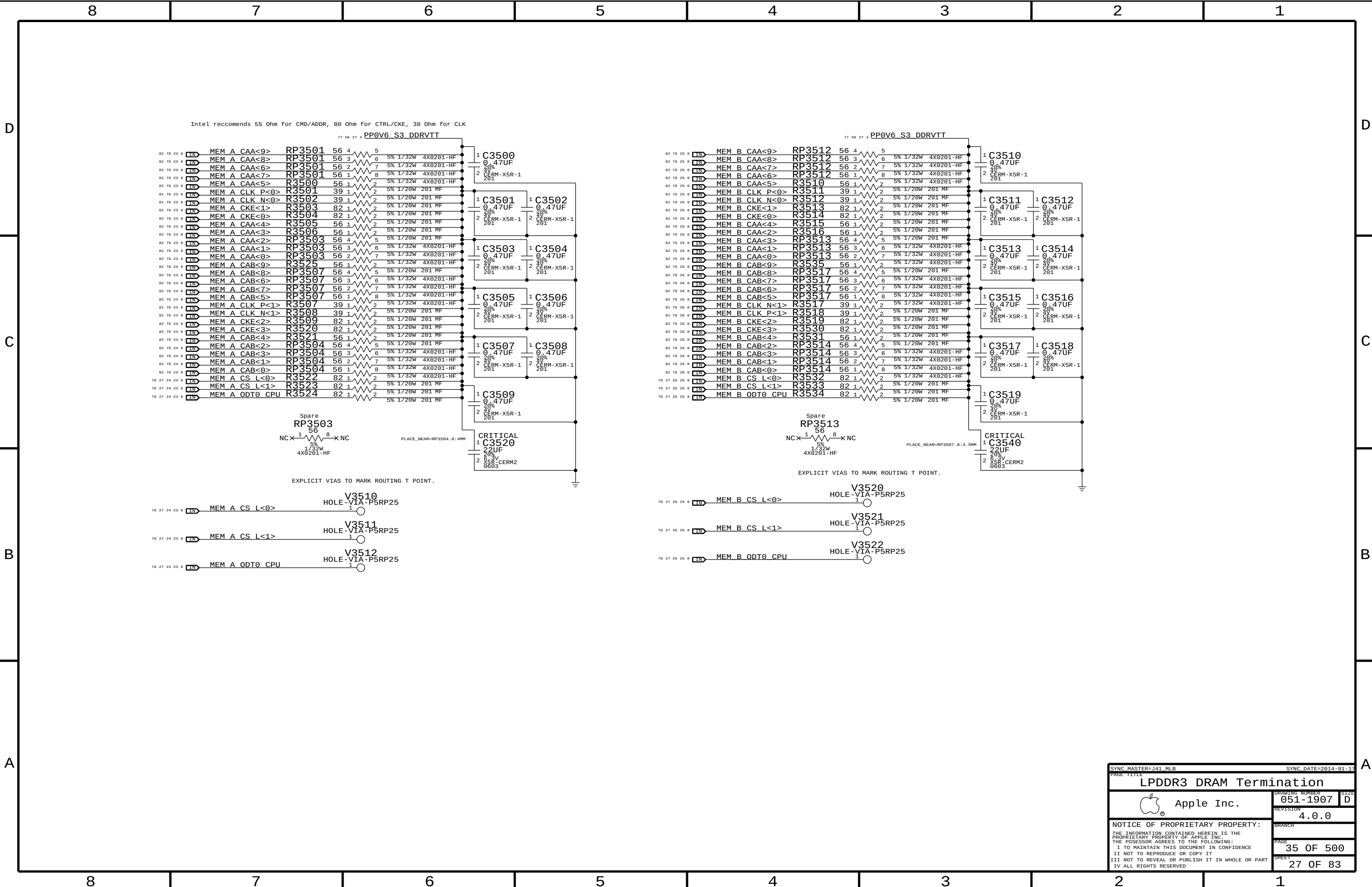


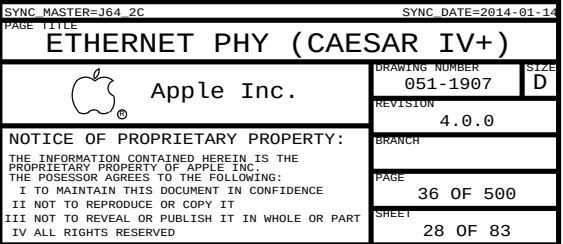
LPDDR3 CHANNEL B (0-31)



LPDDR3 CHANNEL B (32-63)







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A

D

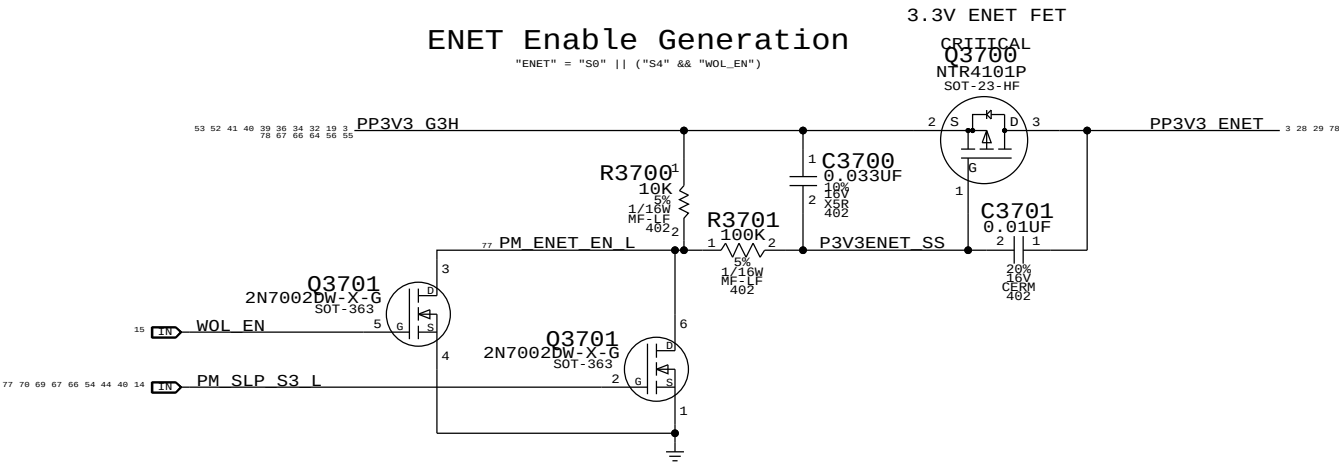
C

B

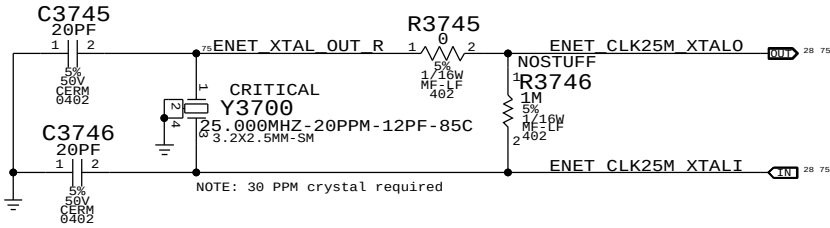
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ENET Enable Generation

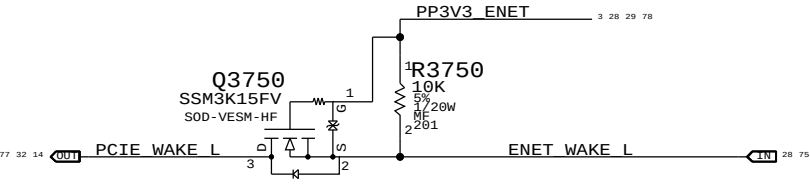
"ENET" = "S0" || ("S4" && "WOL_EN")



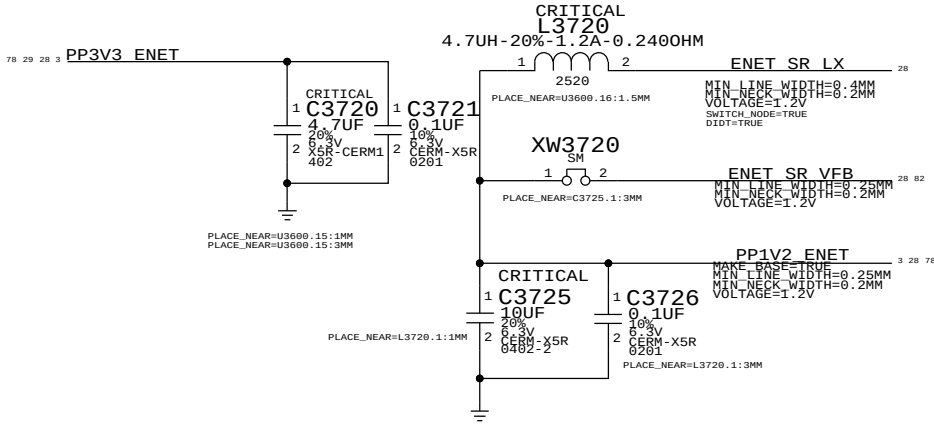
CRYSTAL Ethernet 25MHz Clock



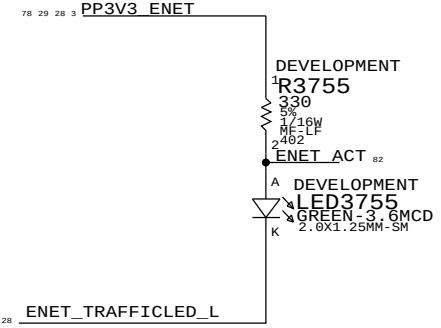
Ethernet WAKE# Isolation



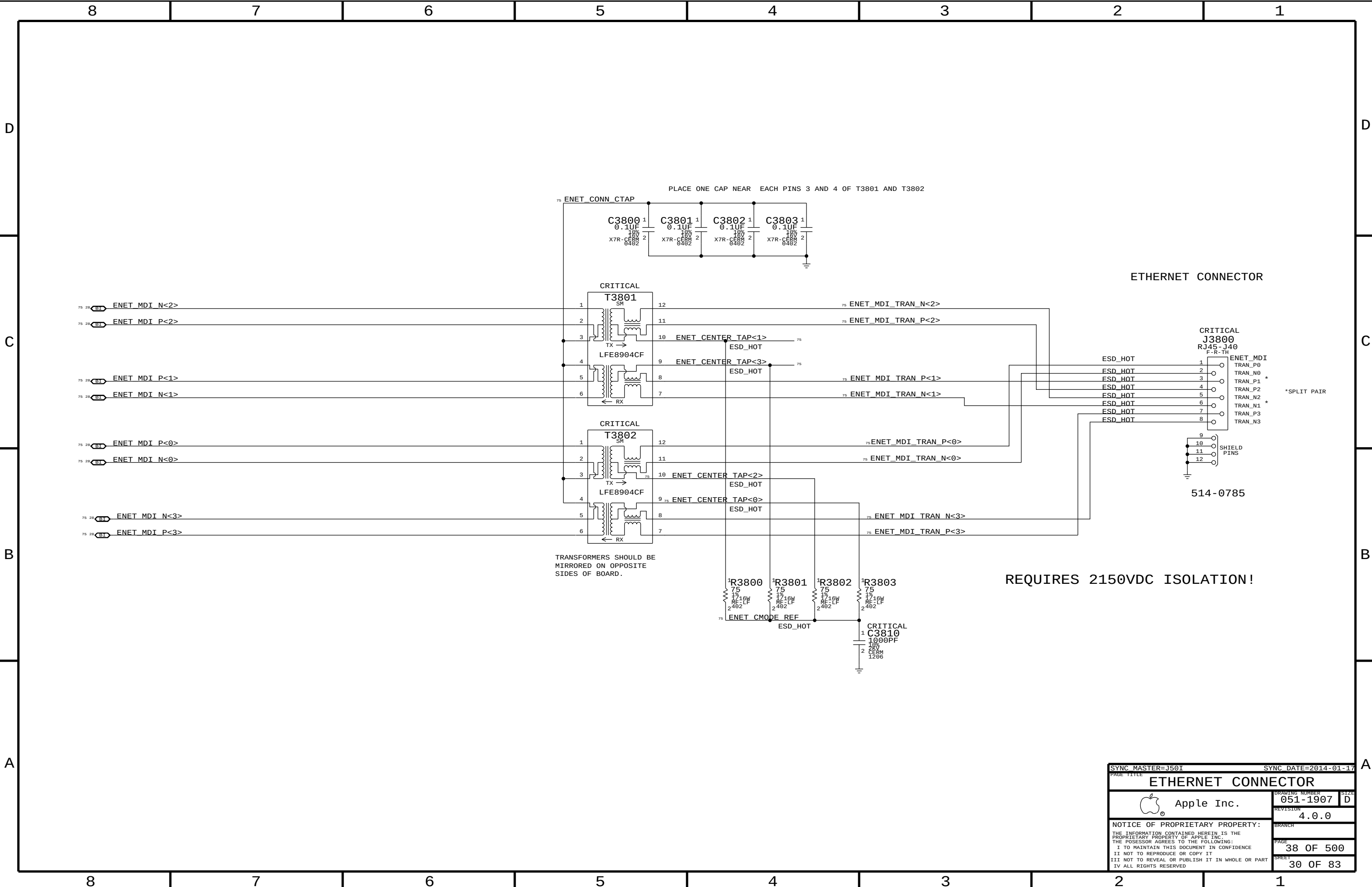
CAESAR IV 1.2V INT.VR CMPTS



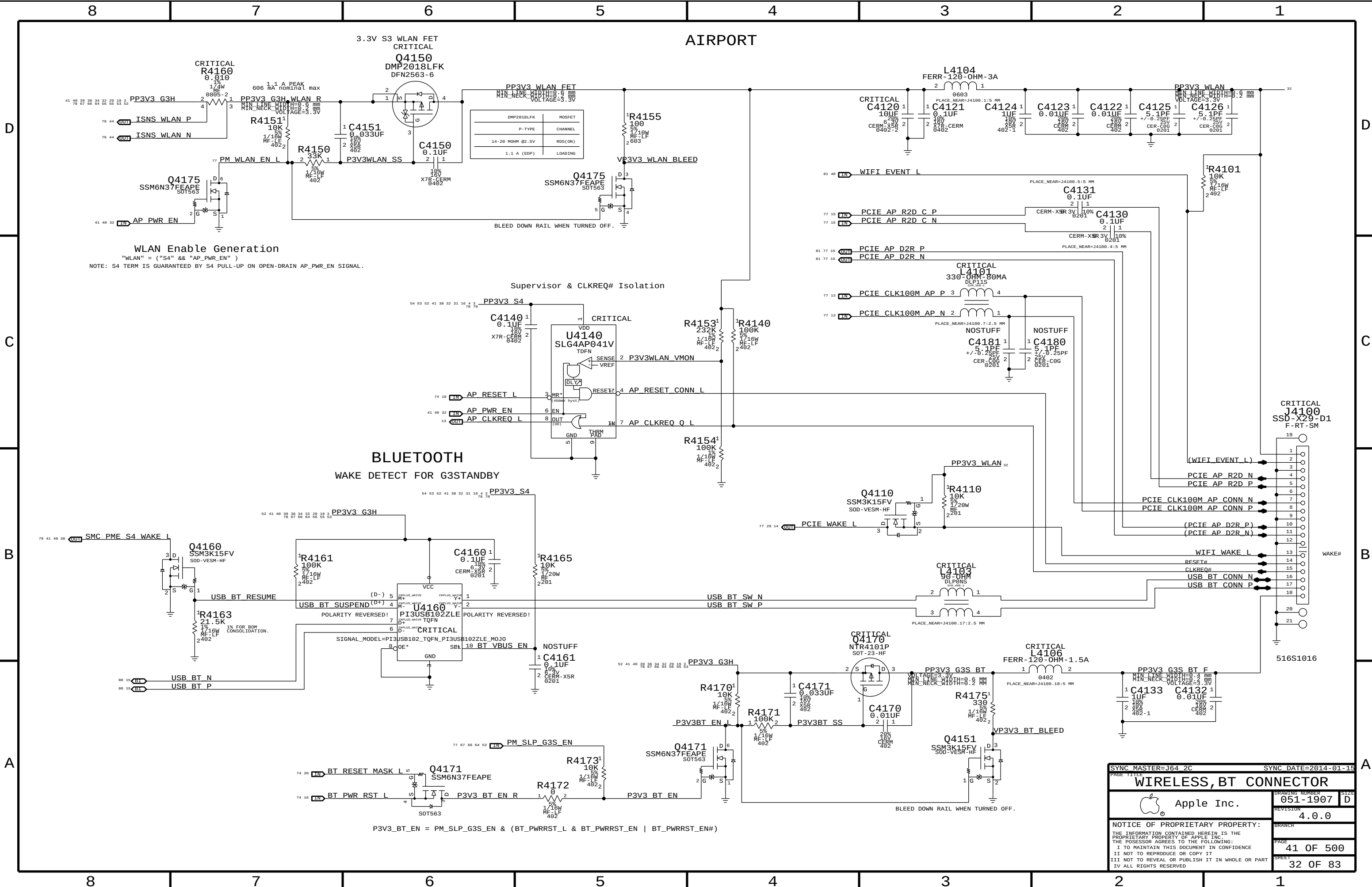
CAESAR IV ACTIVITY LED



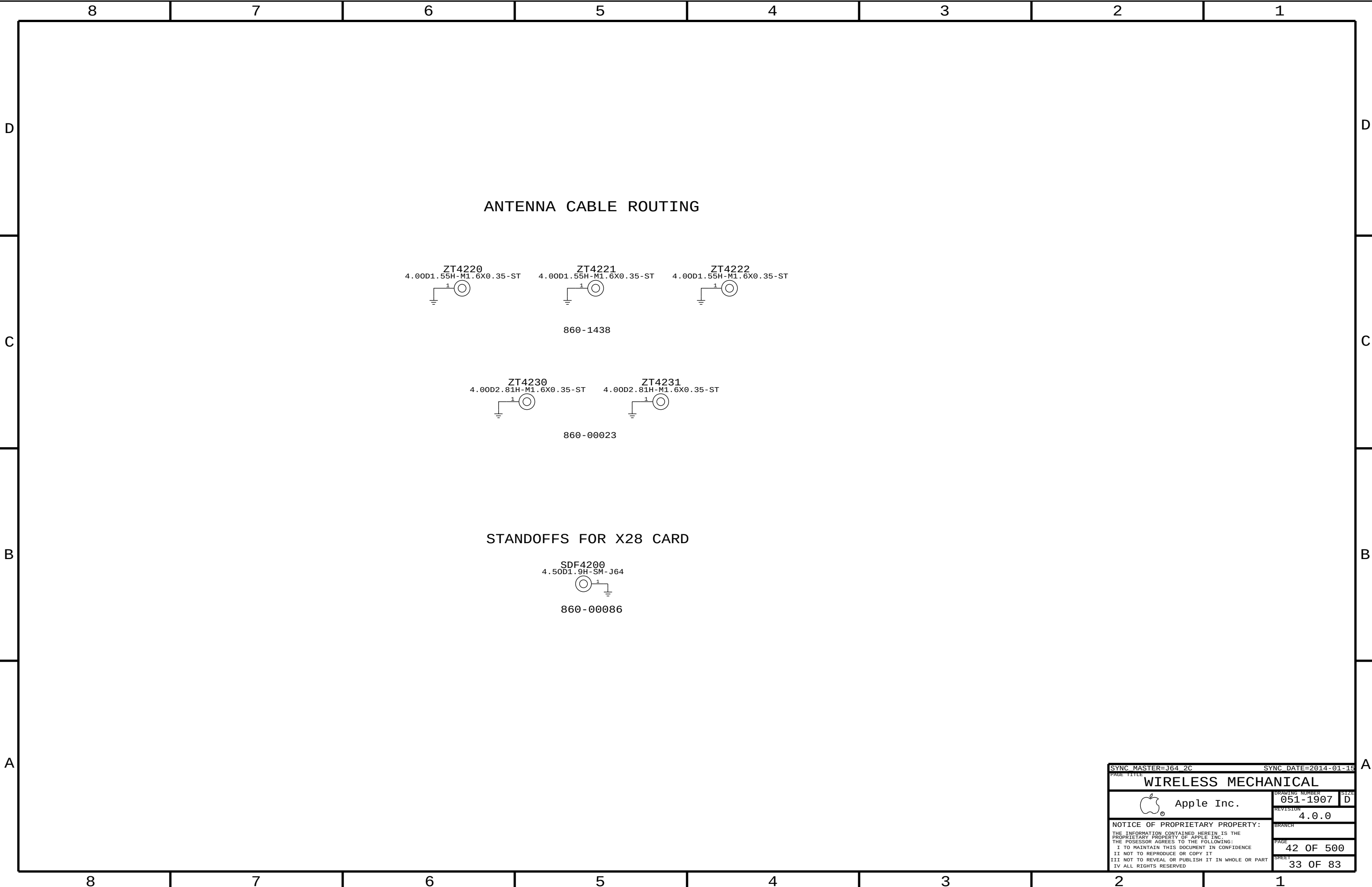
SYNC MASTER=J64 2C		SYNC DATE=2014-01-14	
PAGE TITLE		ETHERNET SUPPORT	
Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
		BRANCH	
		PAGE	37 OF 500
		SHEET	29 OF 83

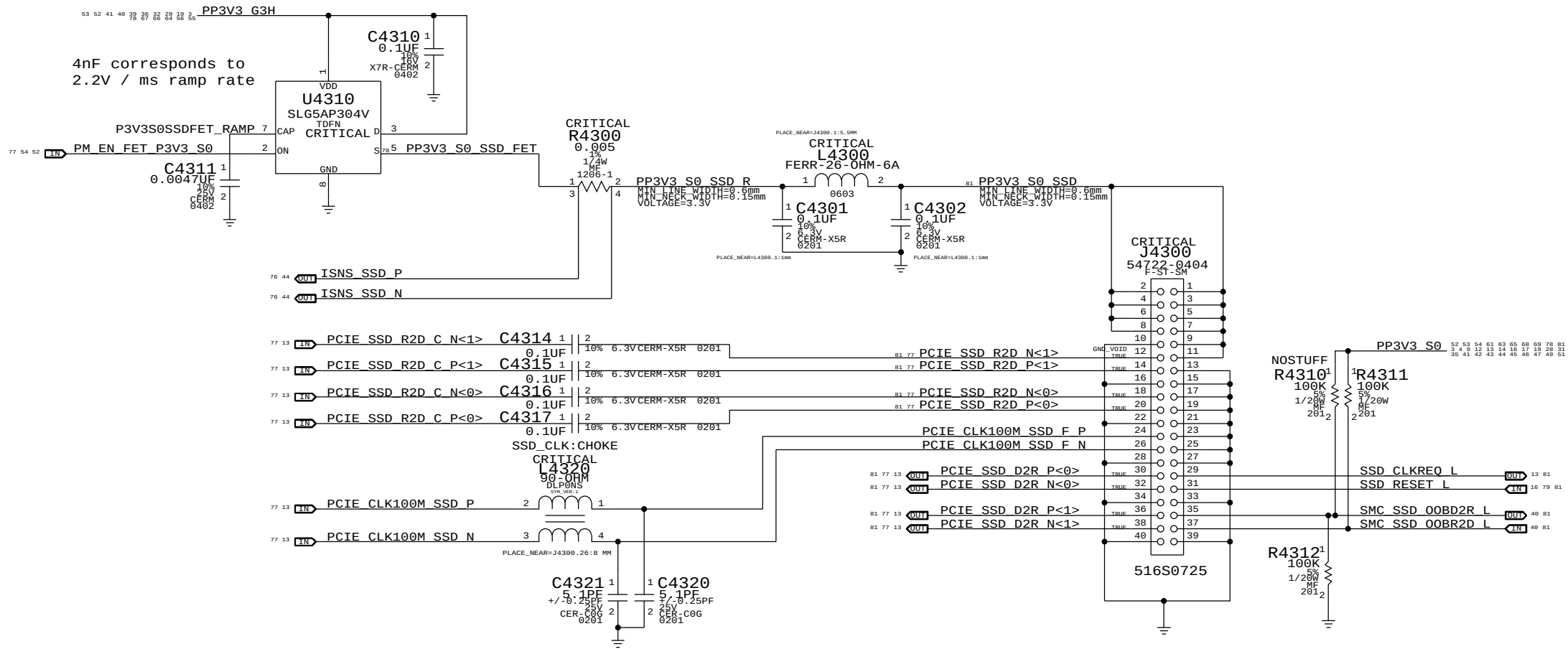


SYNC MASTER=J501		SYNC DATE=2014-01-17	
PAGE TITLE		ETHERNET CONNECTOR	
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		PAGE	38 OF 500
		SHEET	30 OF 83

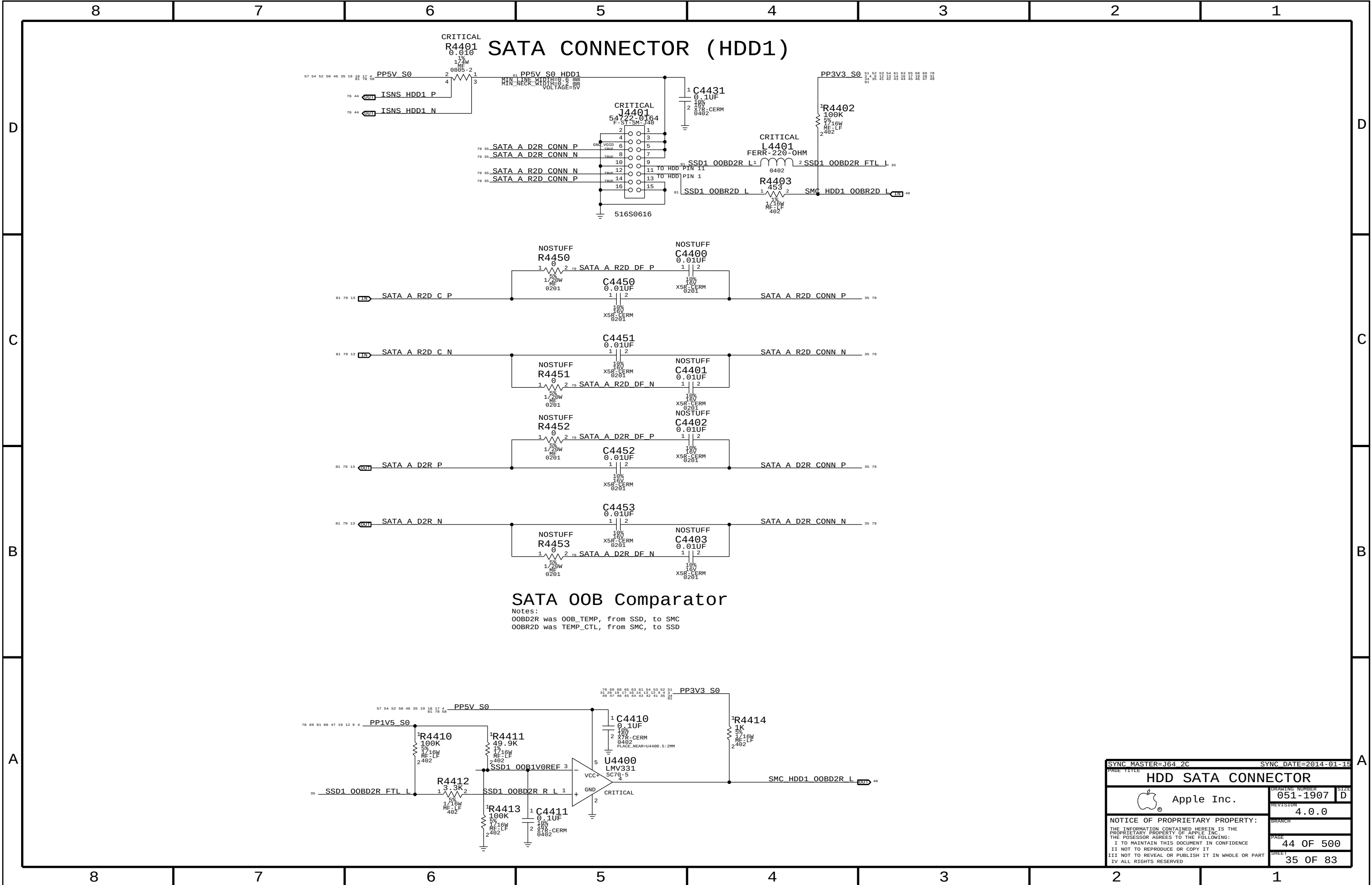


SYNC MASTER=J64 2C		SYNC DATE=2014-01-15	
PAGE TITLE			
WIRELESS, BT CONNECTOR			
 Apple Inc.		DRAWING NUMBER	051-1907
		SIZE	D
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		BRANCH	
		PAGE	41 OF 500
		SHEET	32 OF 83

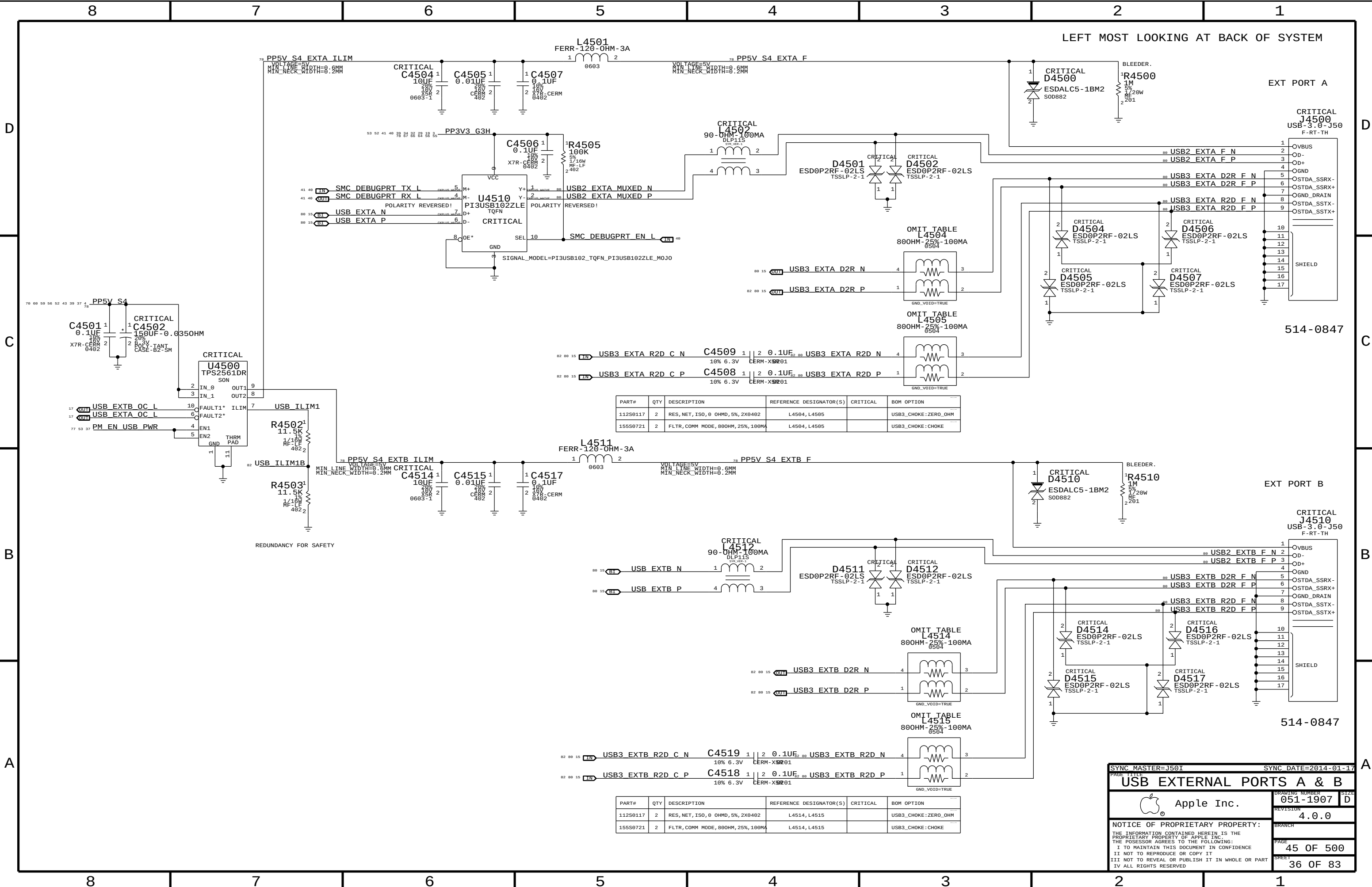




BYPASS CHOKE WITH 2X ZERO-OHM RESISTERS.					
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0201	2	RES, 0.0, 5, 1A, 0201, BLACK	L4320A, L4320B		SSD_CLK:NO_CHOKE



SYNC MASTER=J64 2C		SYNC DATE=2014-01-15	
PAGE TITLE		HDD SATA CONNECTOR	
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
112S0117	2	RES, NET, ISO, 0 OHM, 5%, 2X0402	L4504, L4505		USB3_CHOKE:ZERO_OHM
155S0721	2	FLTR, COMM MODE, 800HM, 25%, 100MA	L4504, L4505		USB3_CHOKE:CHOKE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
112S0117	2	RES, NET, ISO, 0 OHM, 5%, 2X0402	L4514, L4515		USB3_CHOKE:ZERO_OHM
155S0721	2	FLTR, COMM MODE, 800HM, 25%, 100MA	L4514, L4515		USB3_CHOKE:CHOKE

SYNC MASTER=J501

SYNC DATE=2014-01-17

USB EXTERNAL PORTS A & B

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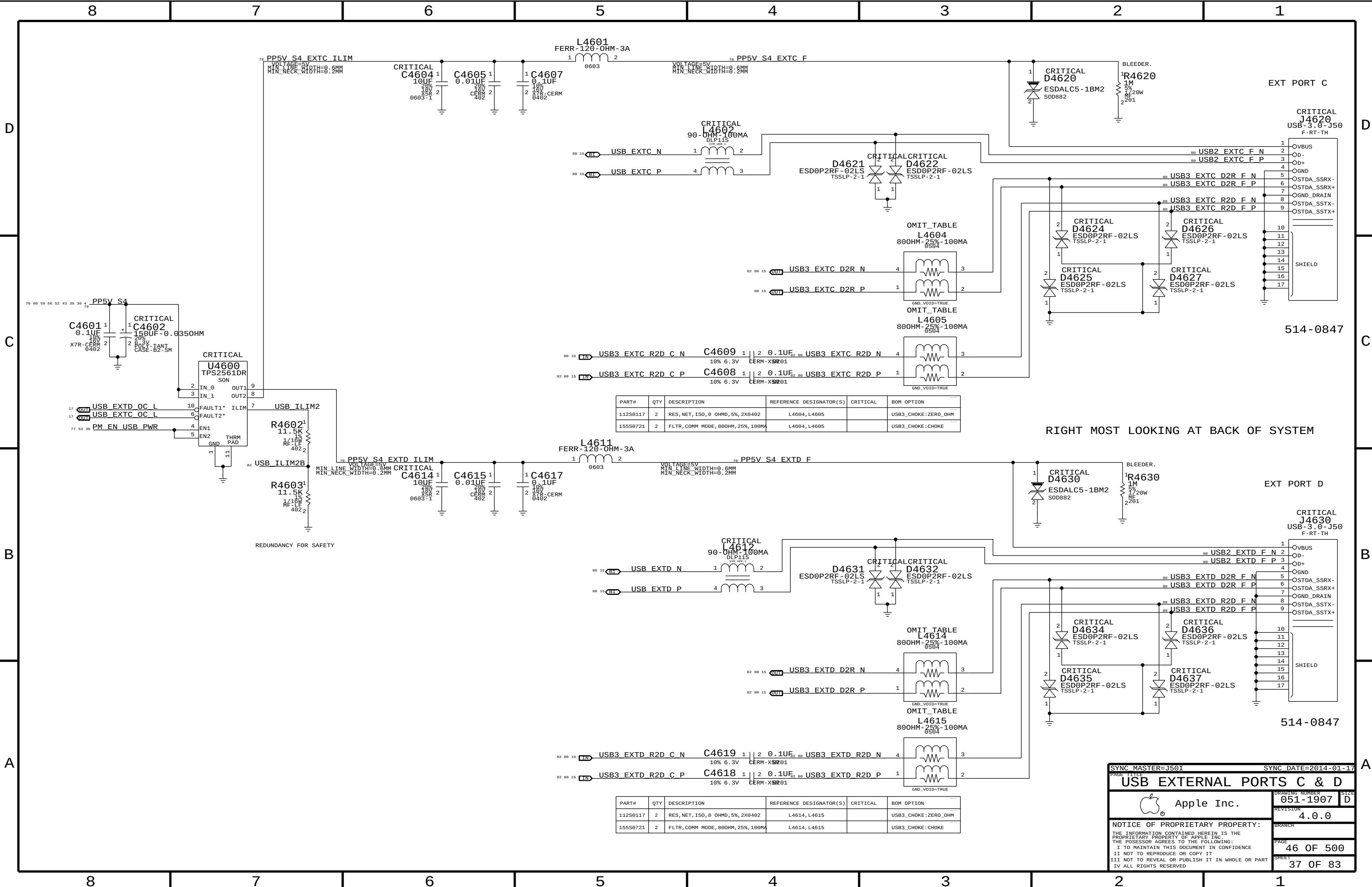
BRANCH

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SHEET

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
112S0117	2	RES, NET, ISO, 0 OHM, 5%, 2X0402	L4604, L4605		USB3_CHOKE:ZERO_OHM
155S0721	2	FLTR, COMM MODE, 800HM, 25%, 100MA	L4604, L4605		USB3_CHOKE:CHOKE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
112S0117	2	RES, NET, ISO, 0 OHM, 5%, 2X0402	L4614, L4615		USB3_CHOKE:ZERO_OHM
155S0721	2	FLTR, COMM MODE, 800HM, 25%, 100MA	L4614, L4615		USB3_CHOKE:CHOKE

SYNC MASTER=J501

SYNC DATE=2014-01-17

USB EXTERNAL PORTS C & D

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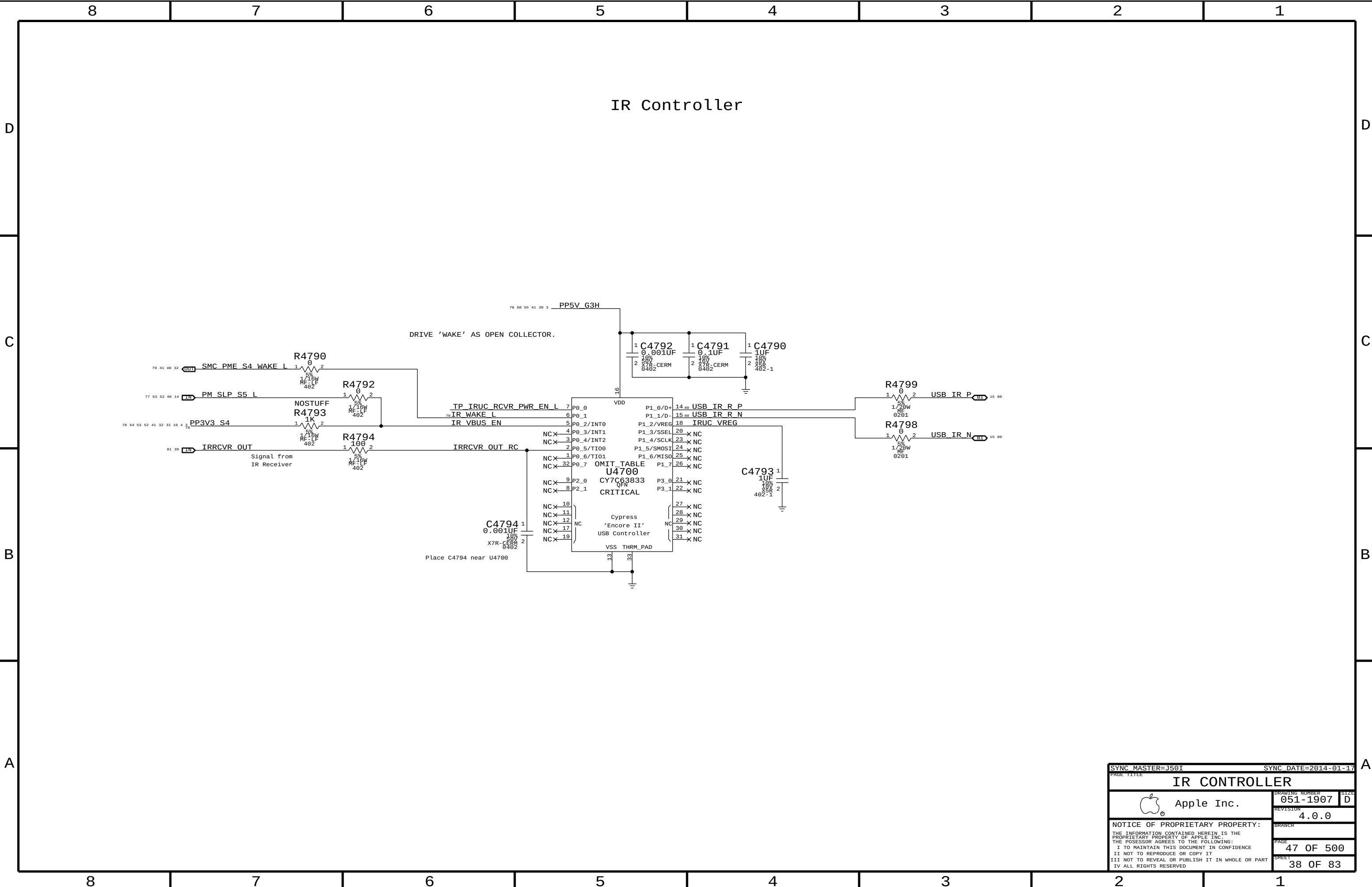
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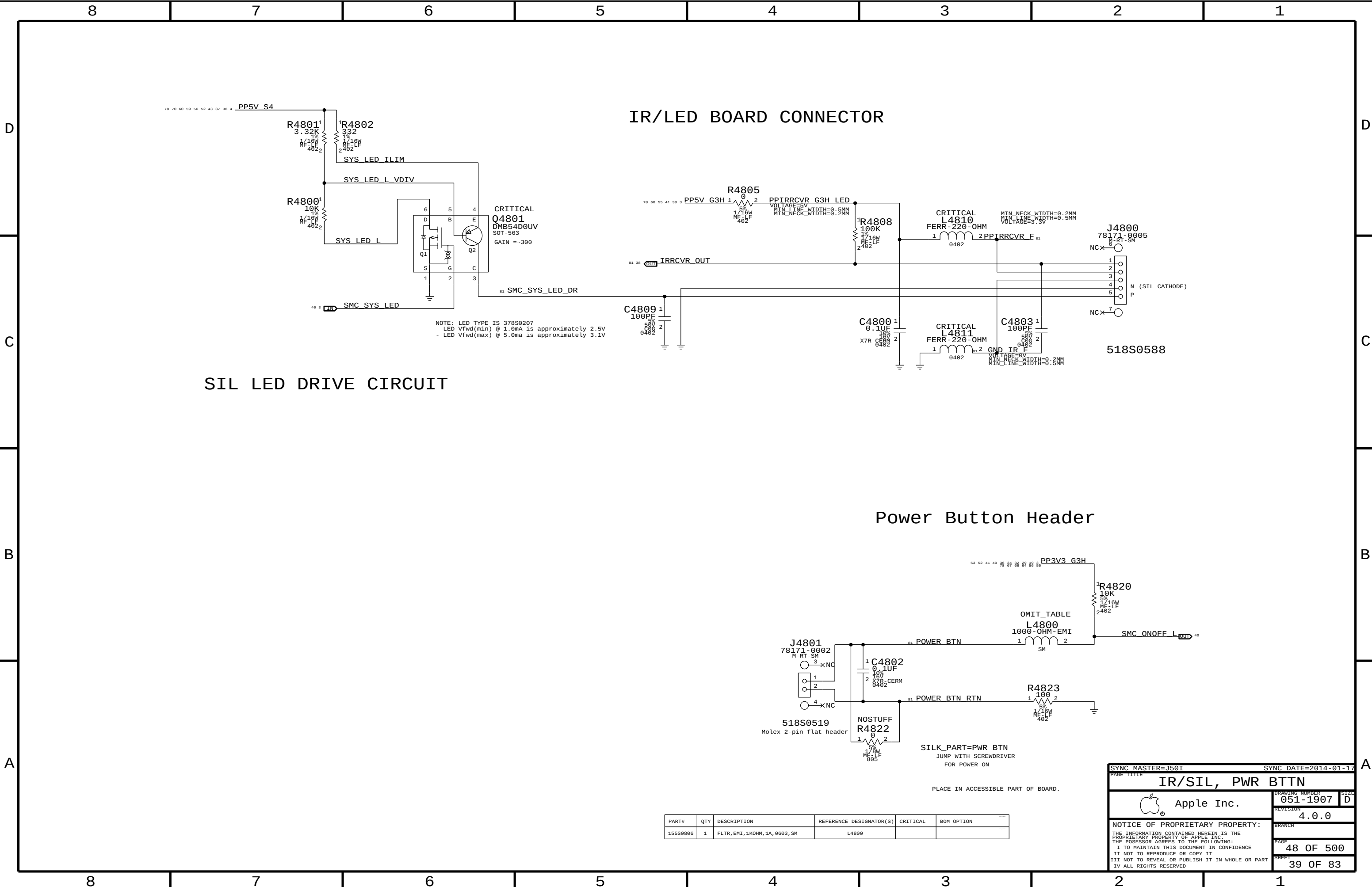
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SIL LED DRIVE CIRCUIT

IR/LED BOARD CONNECTOR

Power Button Header

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S0806	1	FLTR, EMI, 1KOHM, 1A, 0603, SM	L4800		

SYNC MASTER=J501

SYNC DATE=2014-01-17

PAGE TITLE

IR/SIL, PWR BTTN

Apple Inc.

051-1907

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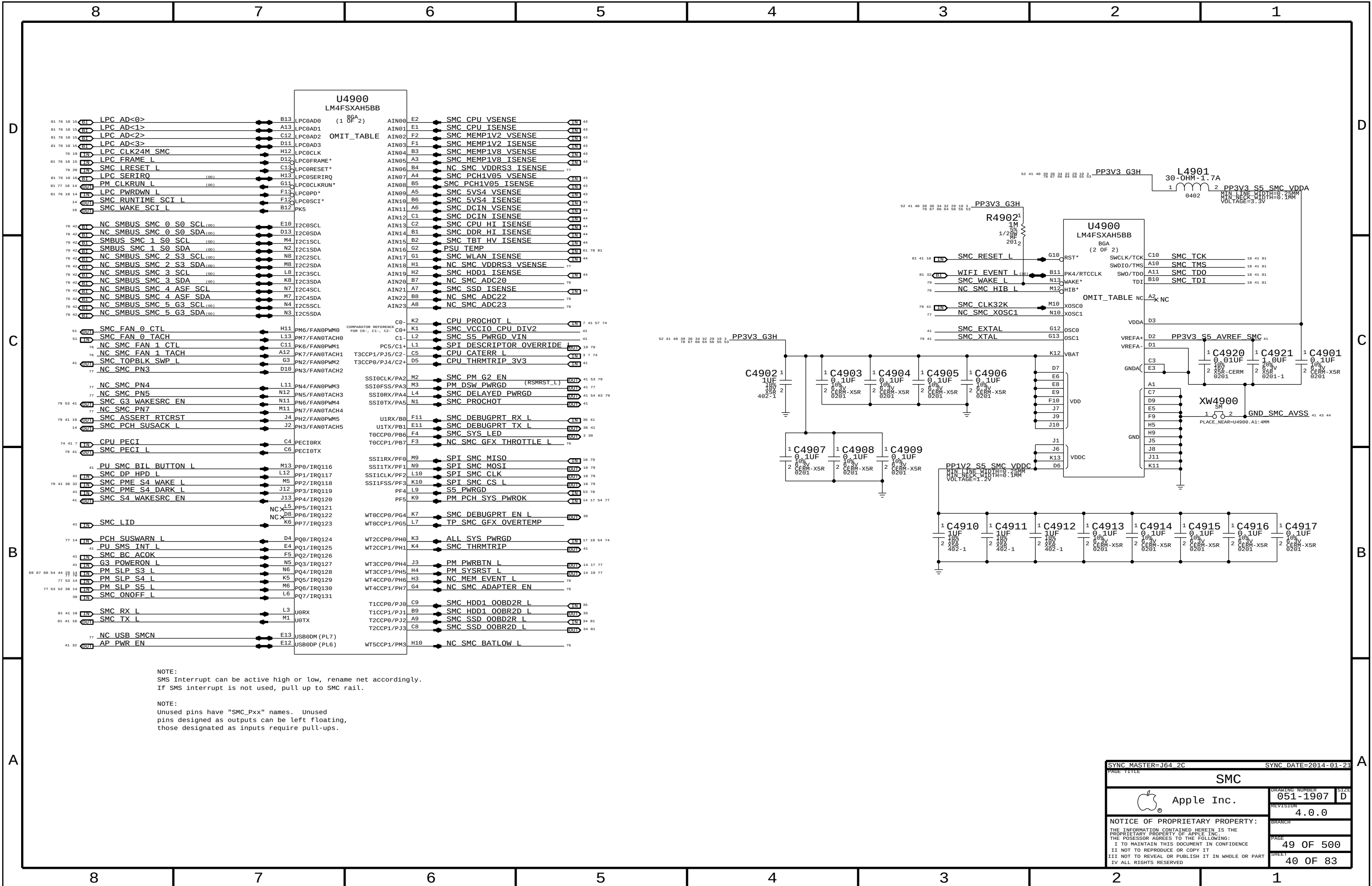
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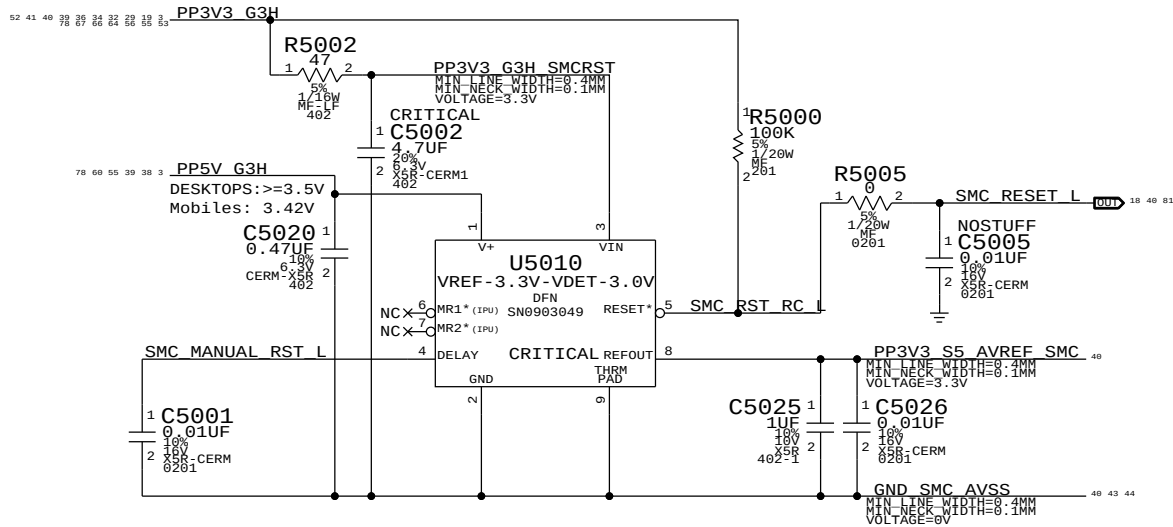


NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

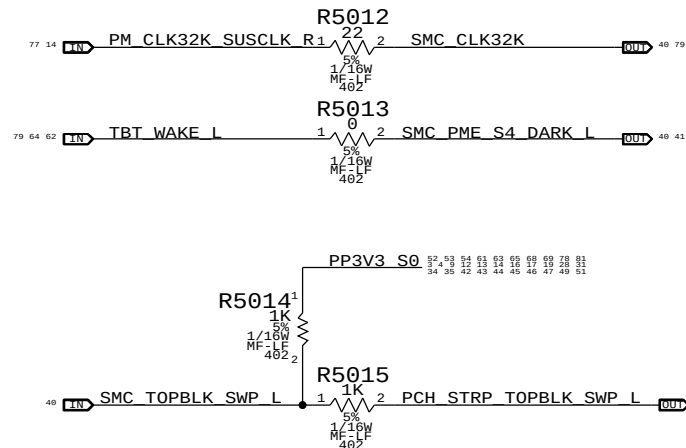
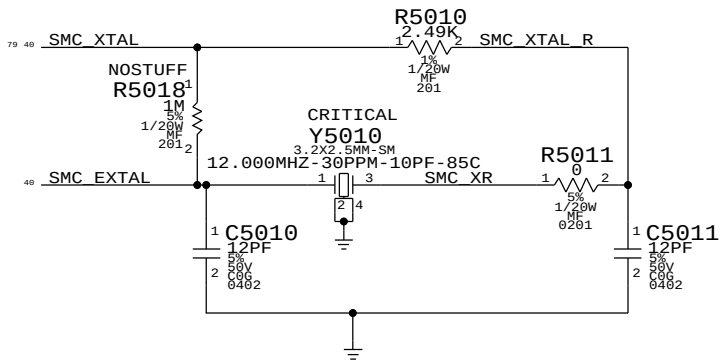
SYNC MASTER=J64 2C		SYNC DATE=2014-01-21	
PAGE TITLE			
SMC			
 Apple Inc.		DRAWING NUMBER	SIZE
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SMC Reset "Button", Supervisor & AVREF Supply

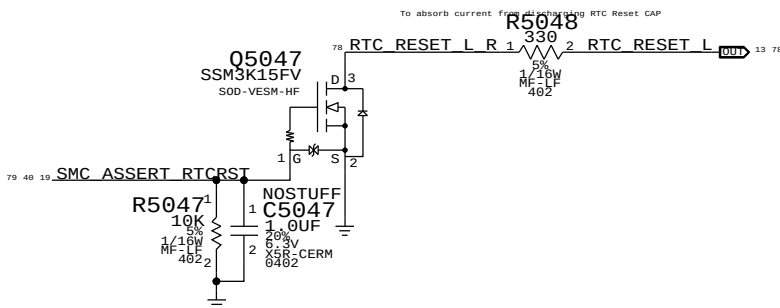


MR1* and MR2* must both be low to cause manual reset.
Used on mobiles to support SMC reset via keyboard.
NOTE: Internal pull-ups are to VIN, not V+.

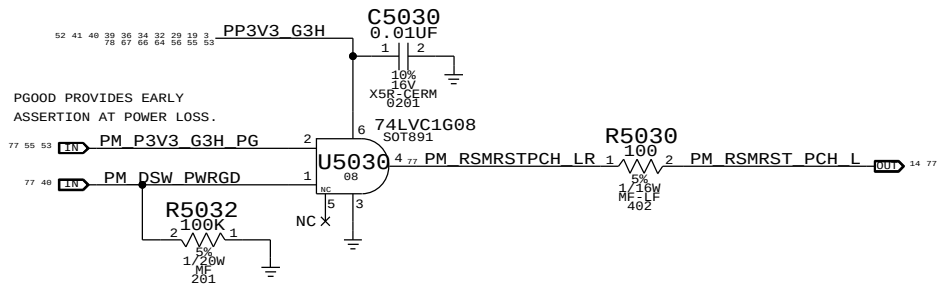
SMC Crystal Circuit



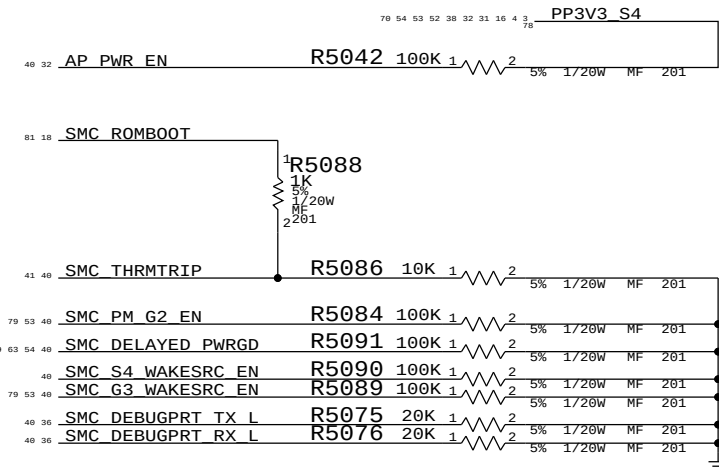
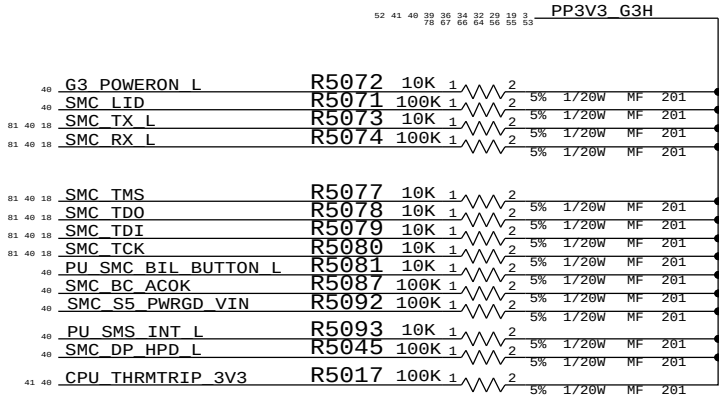
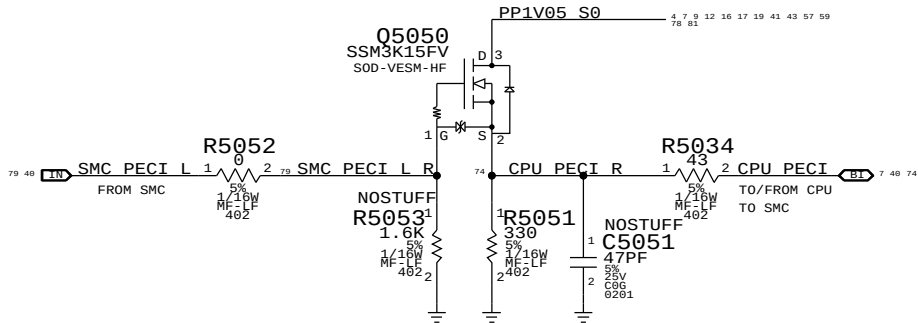
SMC Controlled RTC Reset



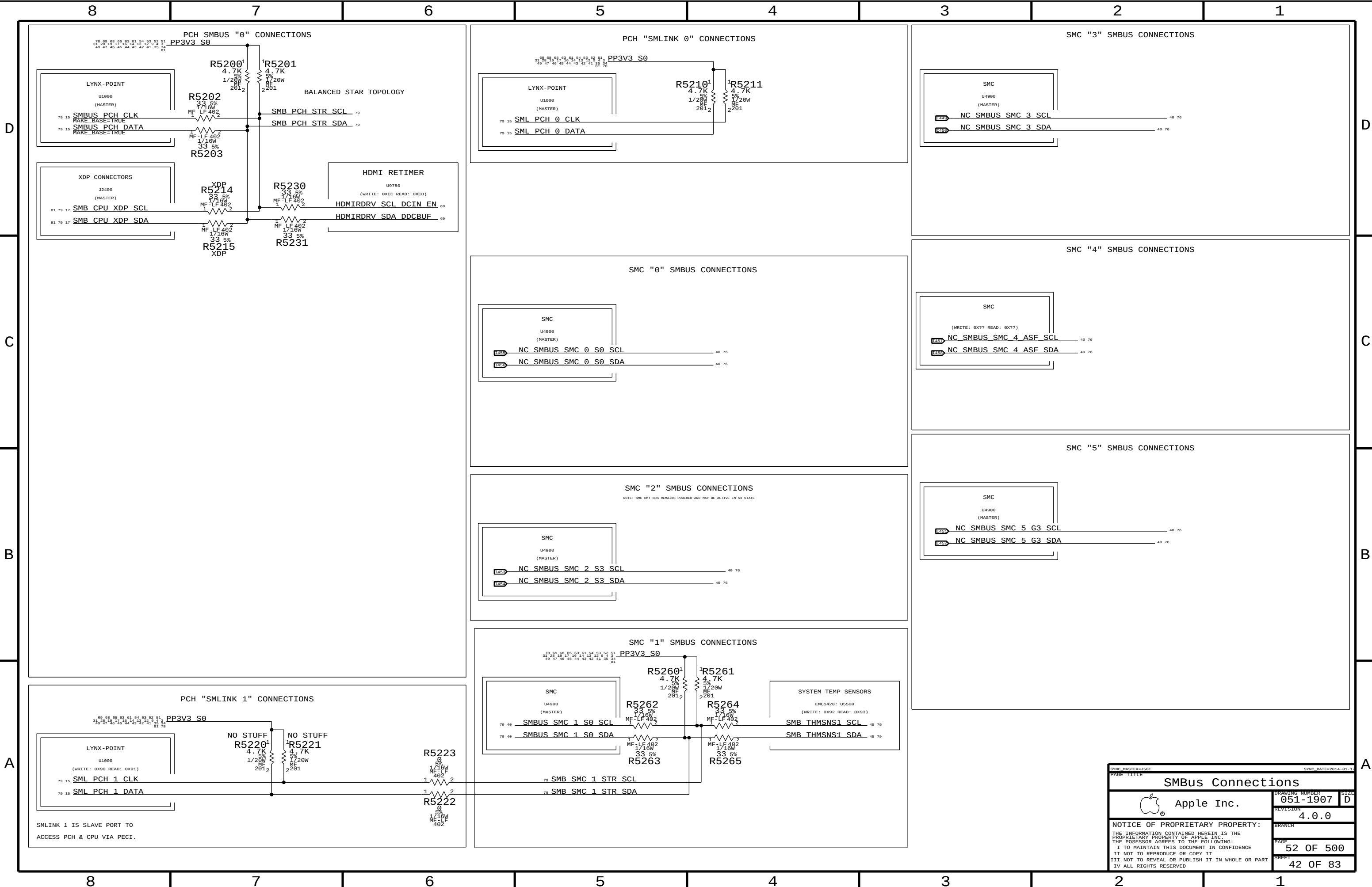
RESUME RESET LOGIC

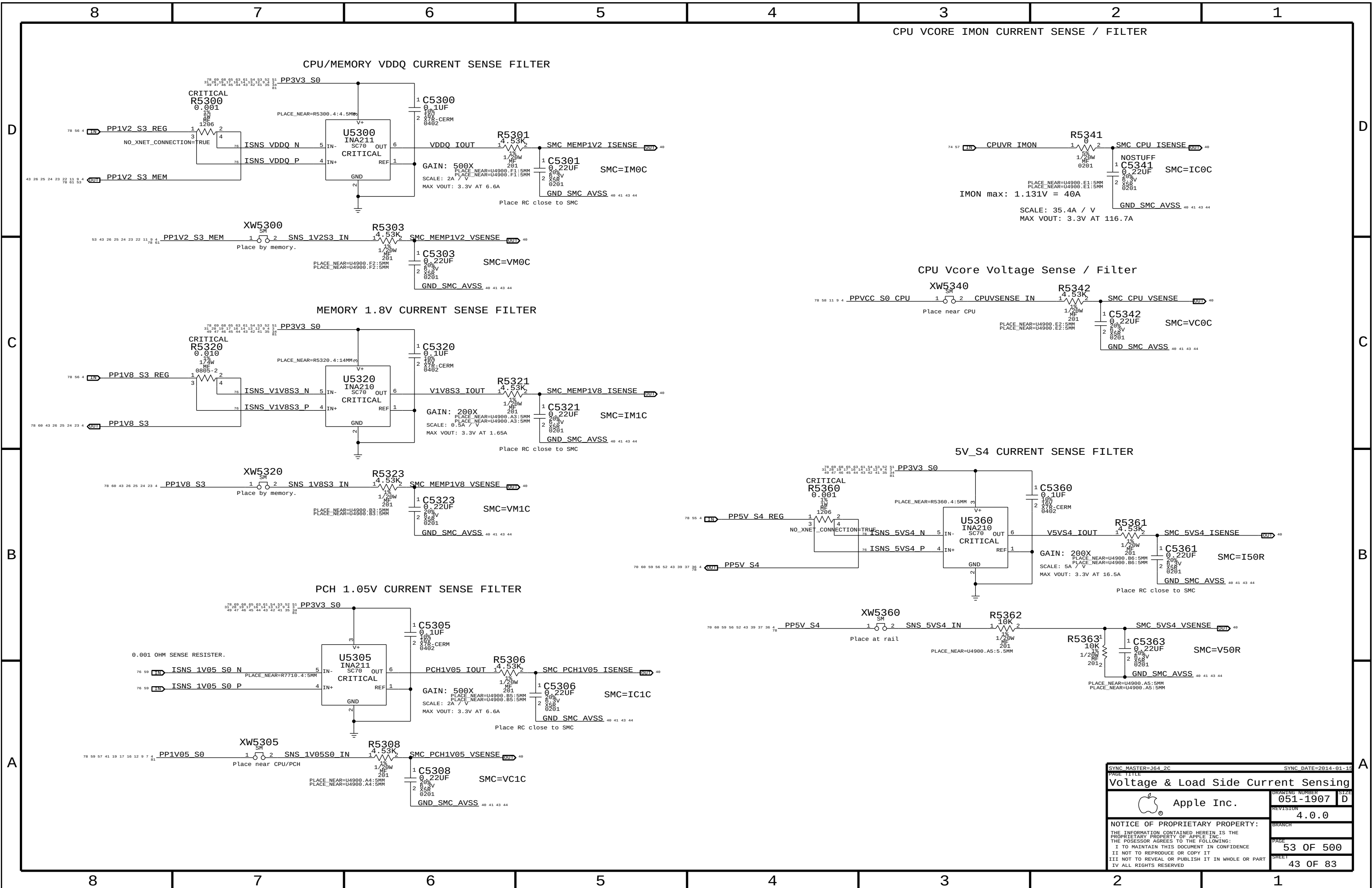


SMC12 Peci Support

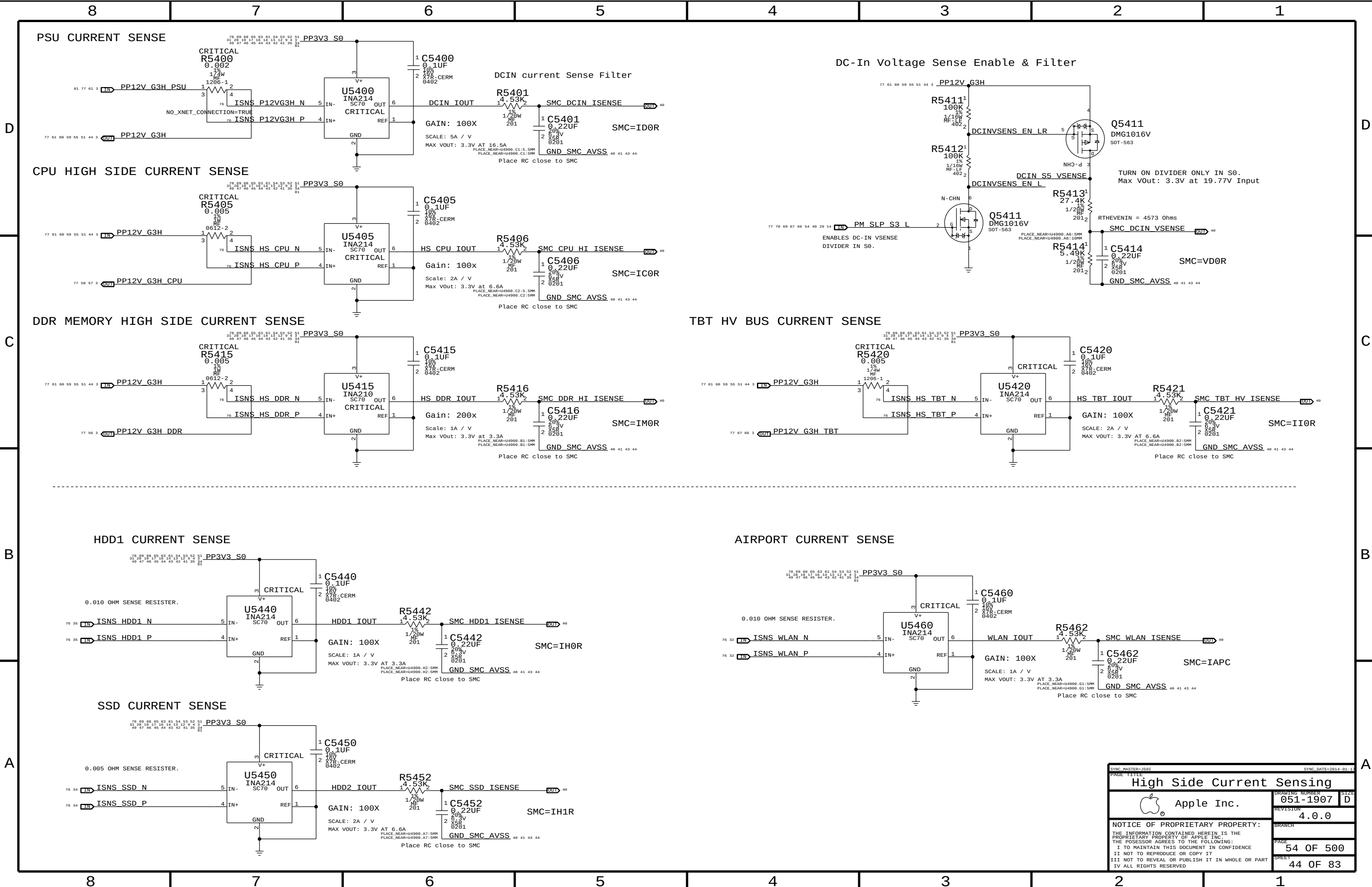


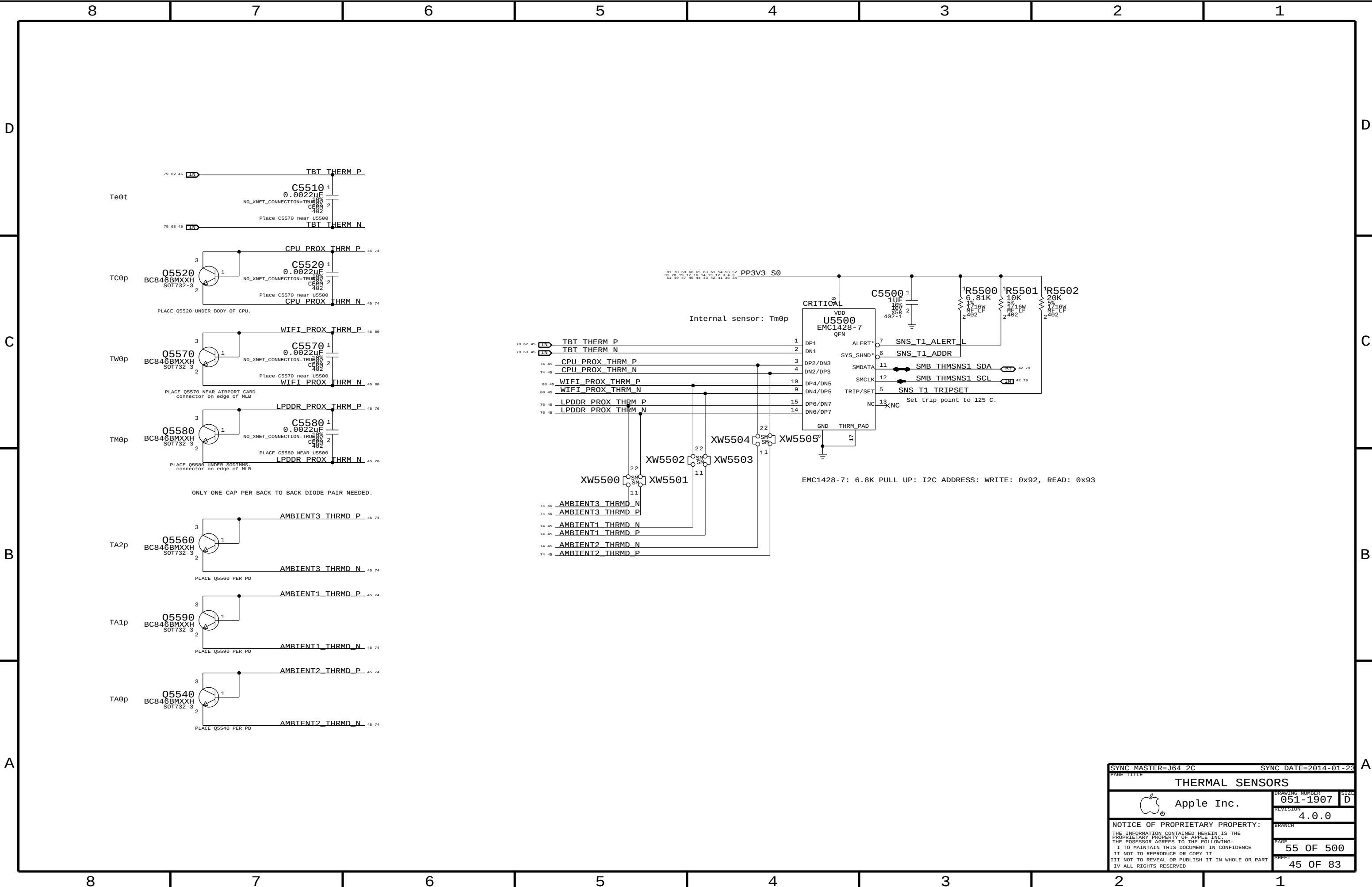
SYNC MASTER=J64 2C		SYNC DATE=2014-01-21	
PAGE TITLE		SMC Support	
Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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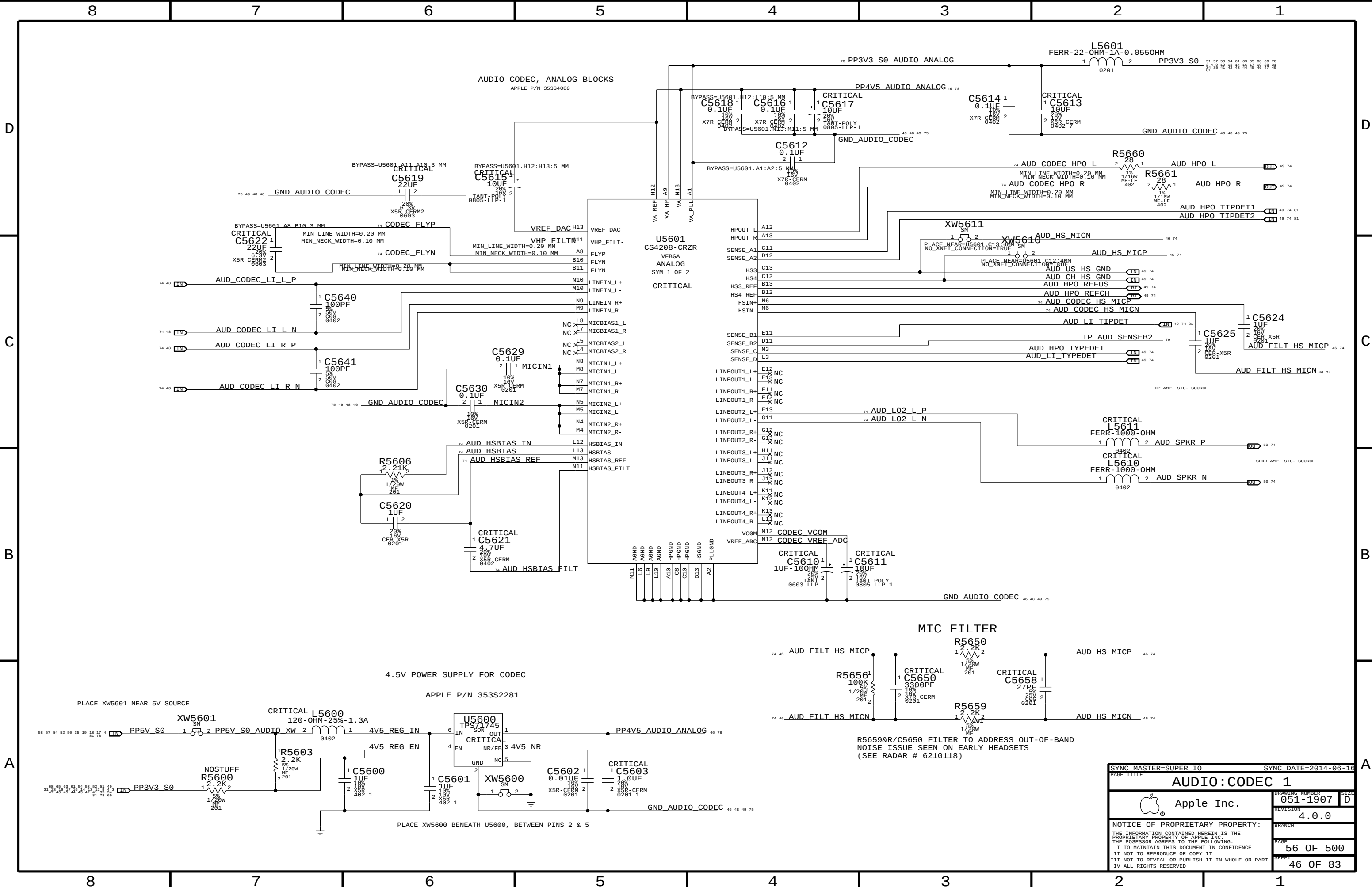




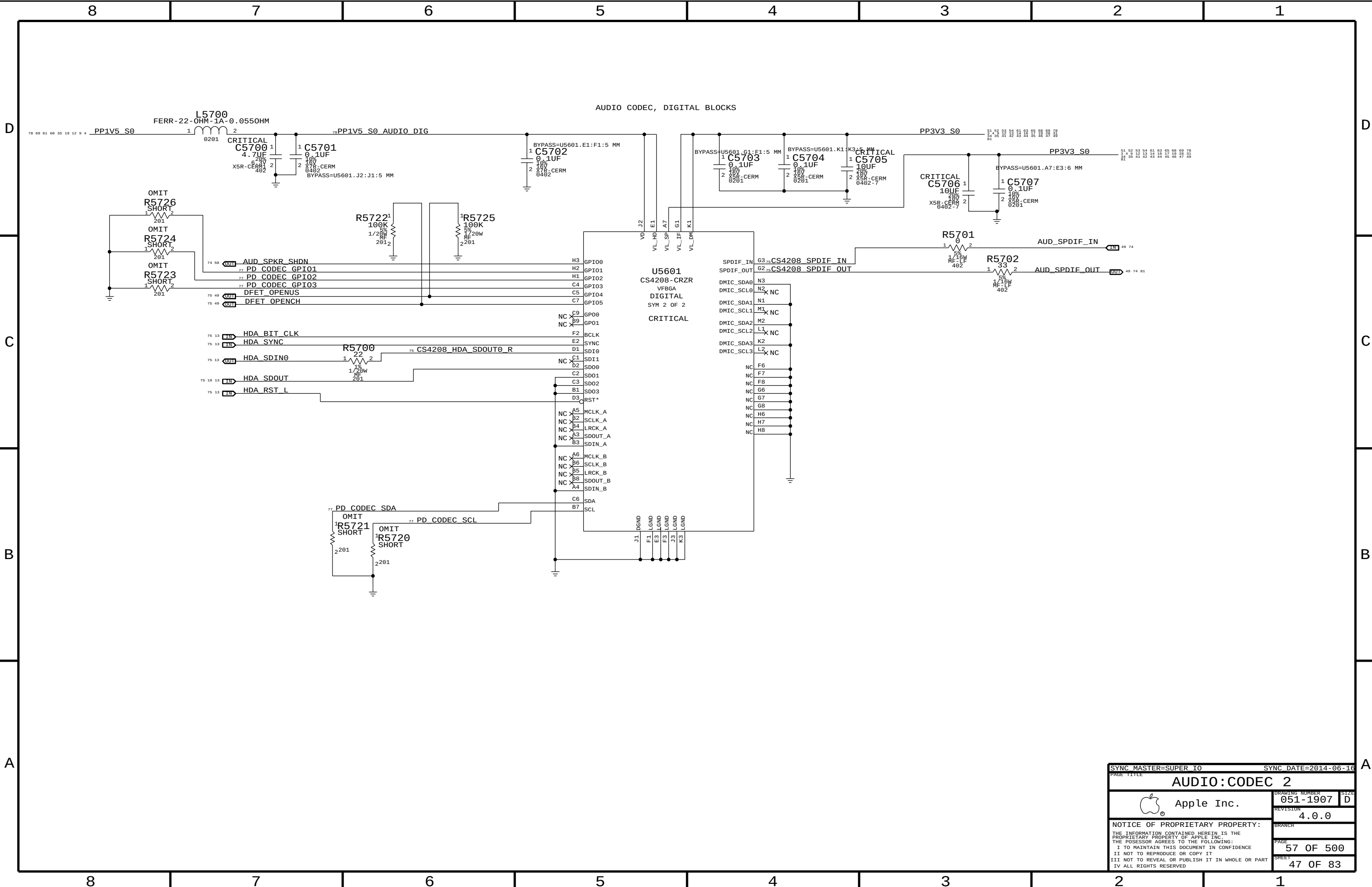
SYNC MASTER=164 2C		SYNC DATE=2014-01-15	
PAGE TITLE		DRAWING NUMBER	
Voltage & Load Side Current Sensing		051-1907	
 Apple Inc.		REVISION	4.0.0
		BRANCH	
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		SHEET	43 OF 83

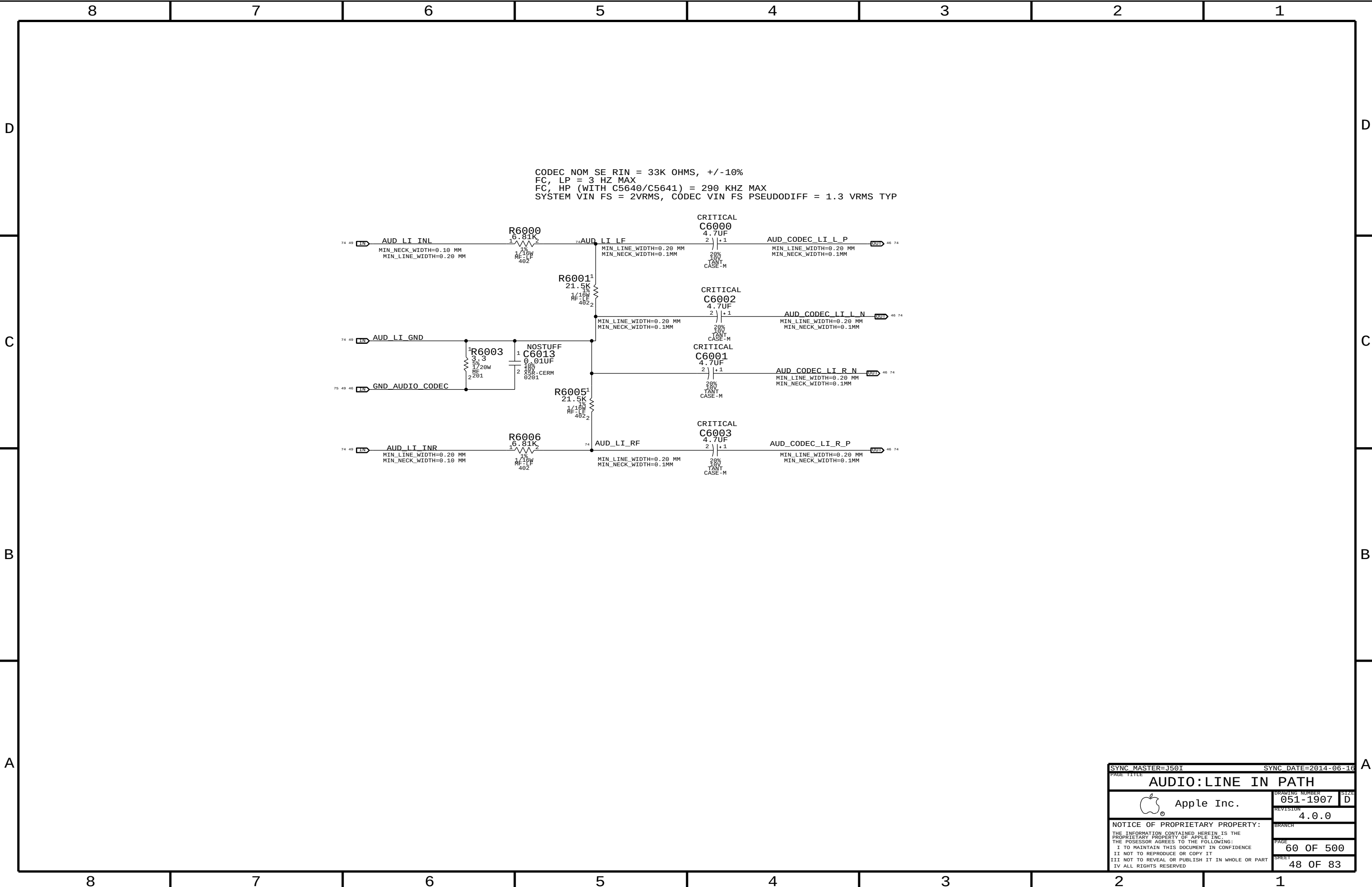


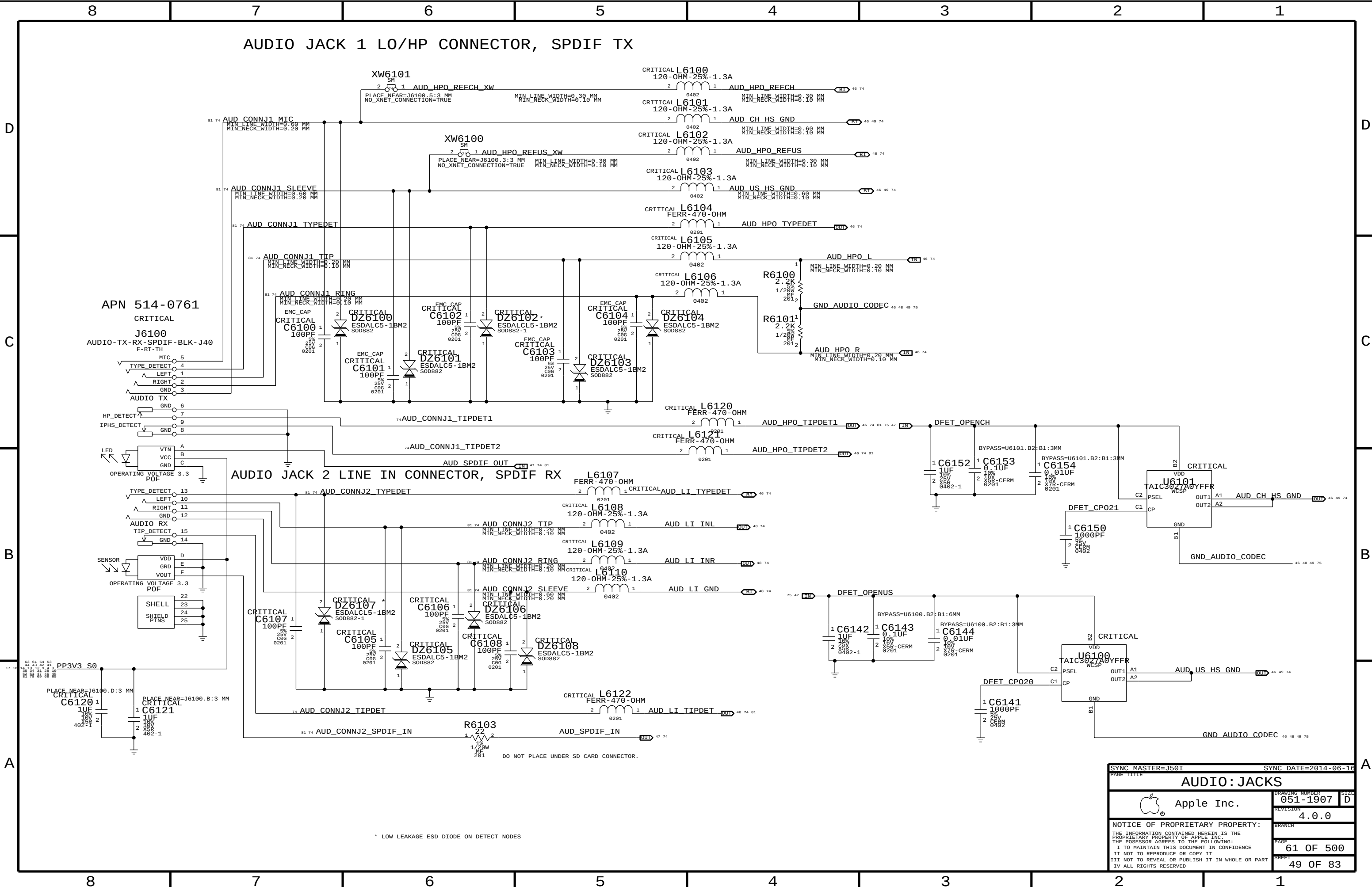




SYNC MASTER=SUPER IO		SYNC DATE=2014-06-16	
PAGE TITLE		AUDIO:CODEC 1	
Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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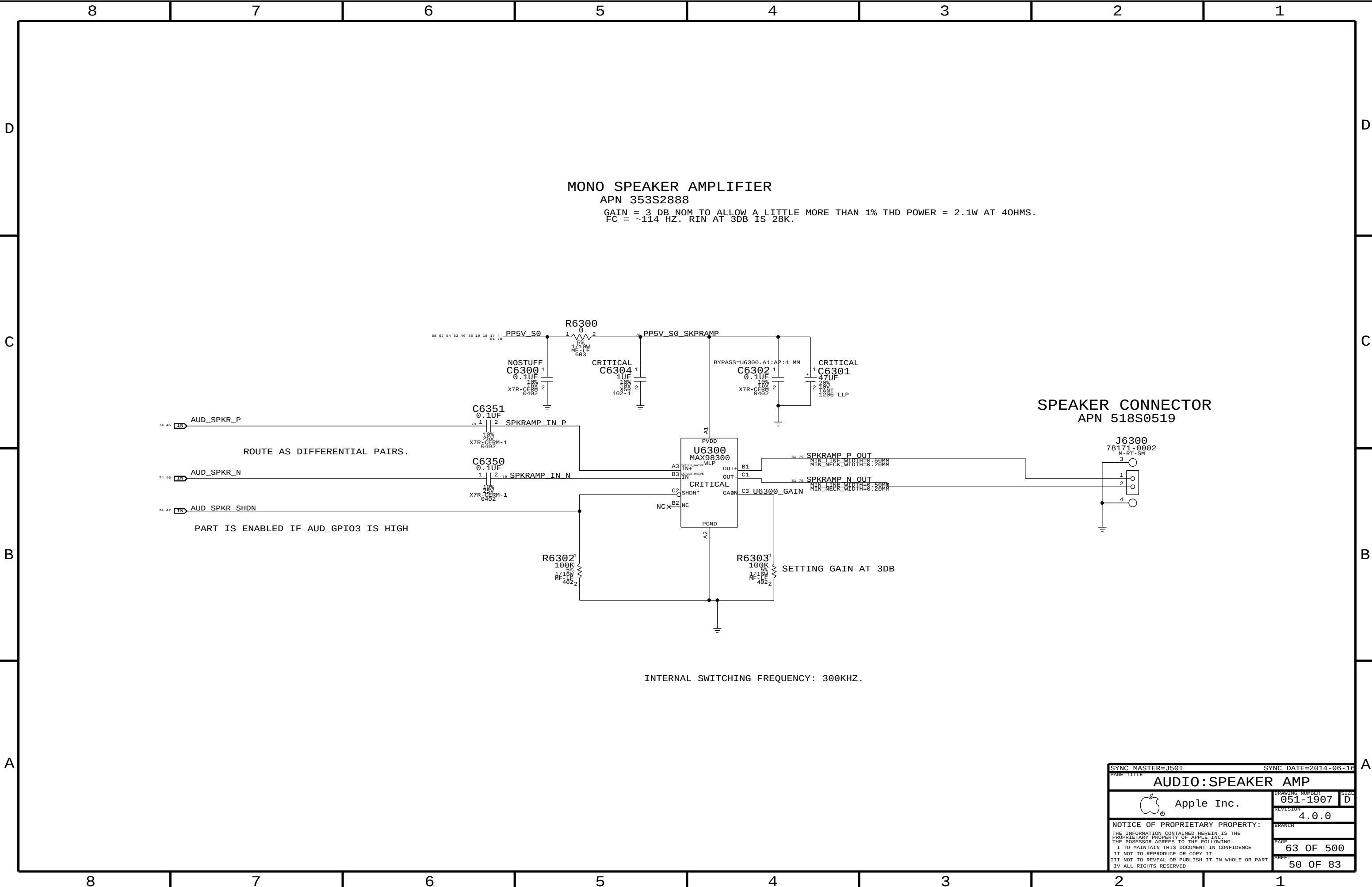


APN 514-0761

CRITICAL
J6100
AUDIO-TX-RX-SPDIF-BLK-J40
F-RT-TH

AUDIO JACK 2 LINE IN CONNECTOR, SPDIF RX

SYNC MASTER=J501		SYNC DATE=2014-06-16	
PAGE TITLE		AUDIO: JACKS	
Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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MONO SPEAKER AMPLIFIER

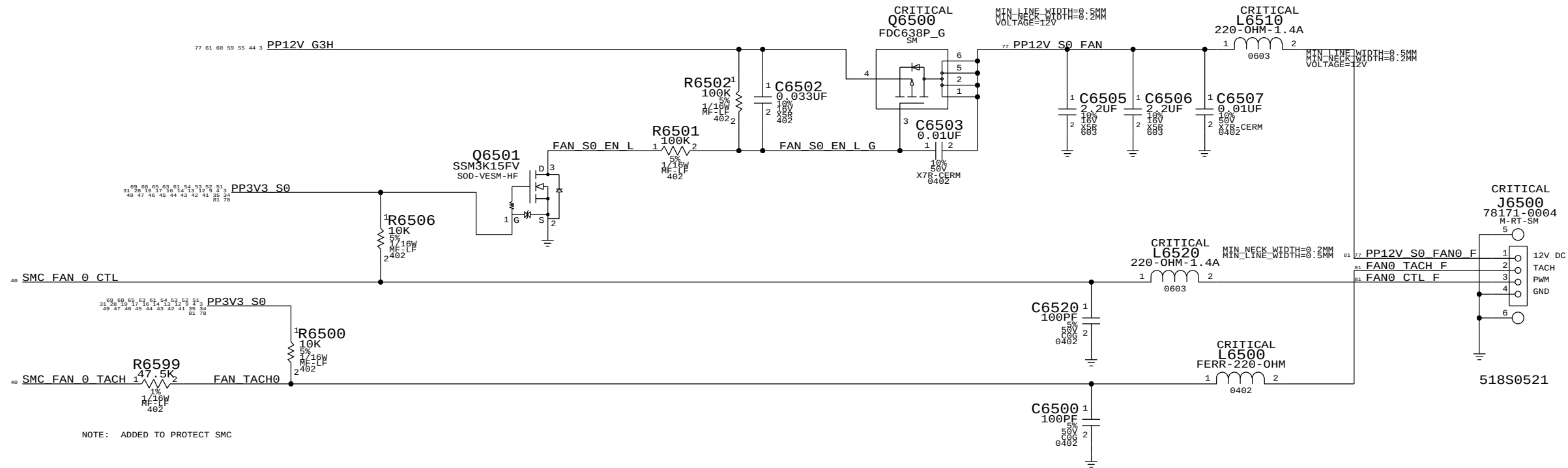
APN 353S2888

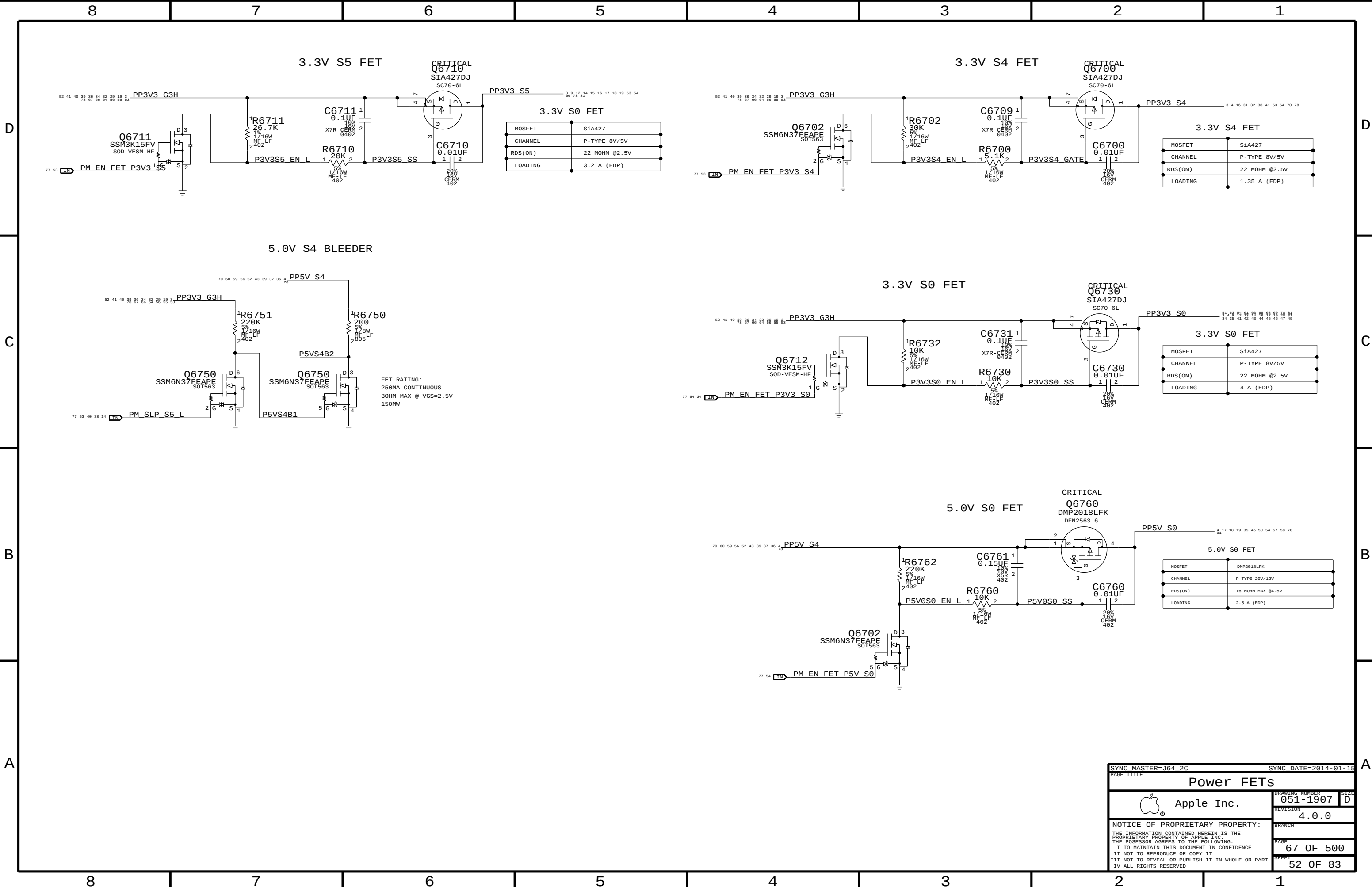
GAIN = 3 DB NOM TO ALLOW A LITTLE MORE THAN 1% THD POWER = 2.1W AT 40HMS.
FC = ~114 HZ. RIN AT 3DB IS 28K.

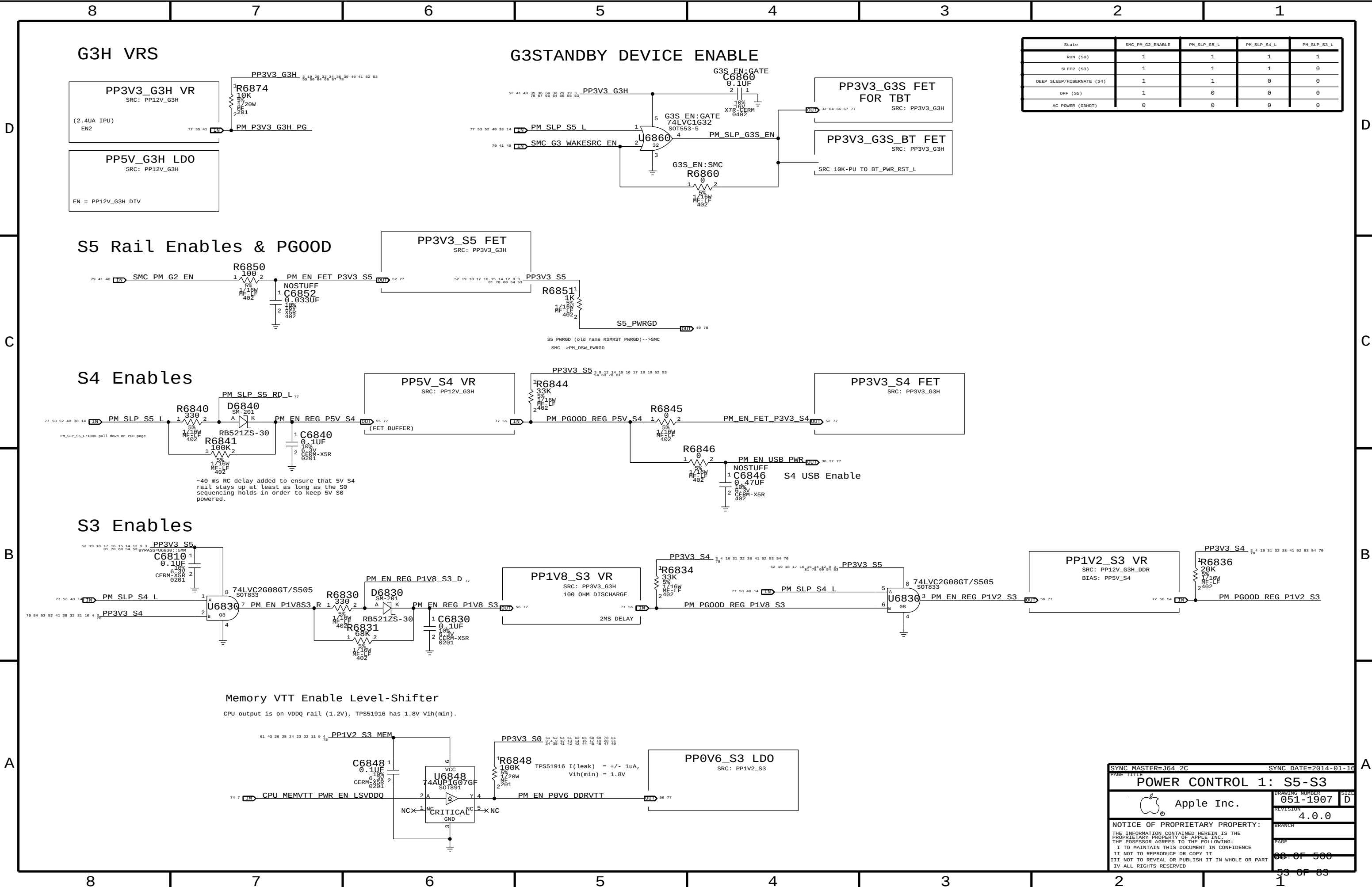
SPEAKER CONNECTOR
APN 518S0519

SYNC MASTER=J501		SYNC DATE=2014-06-16	
PAGE TITLE			
AUDIO:SPEAKER AMP			
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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FAN CONTROL CIRCUIT





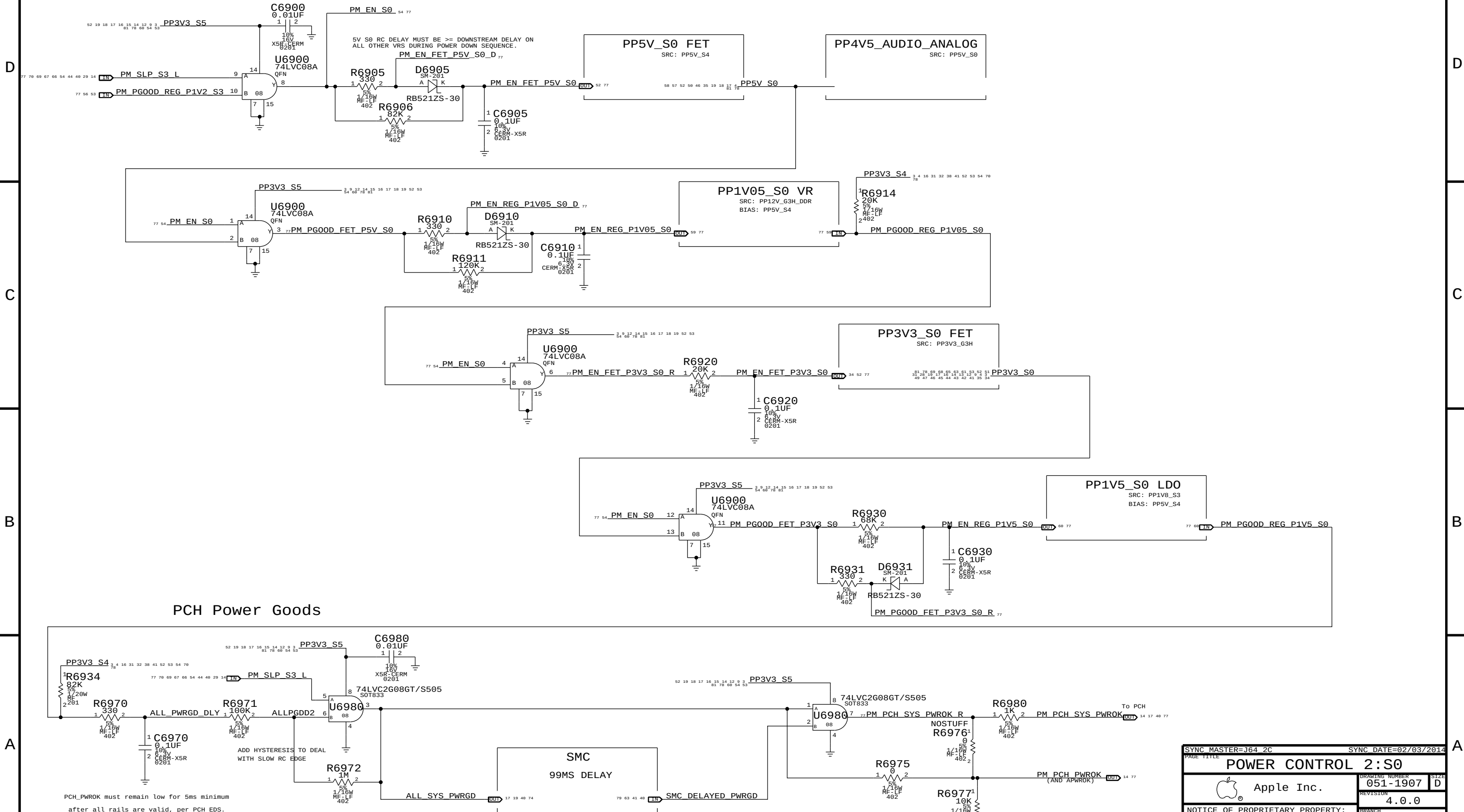


State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
RUN (S0)	1	1	1	1
SLEEP (S3)	1	1	1	0
DEEP SLEEP/HIBERNATE (S4)	1	1	0	0
OFF (S5)	1	0	0	0
AC POWER (G3HOT)	0	0	0	0

SYNC MASTER=J64_2C		SYNC DATE=2014-01-16	
PAGE TITLE			
POWER CONTROL 1: S5-S3			
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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		PAGE	66 OF 500
			53 OF 83

S0 Enables

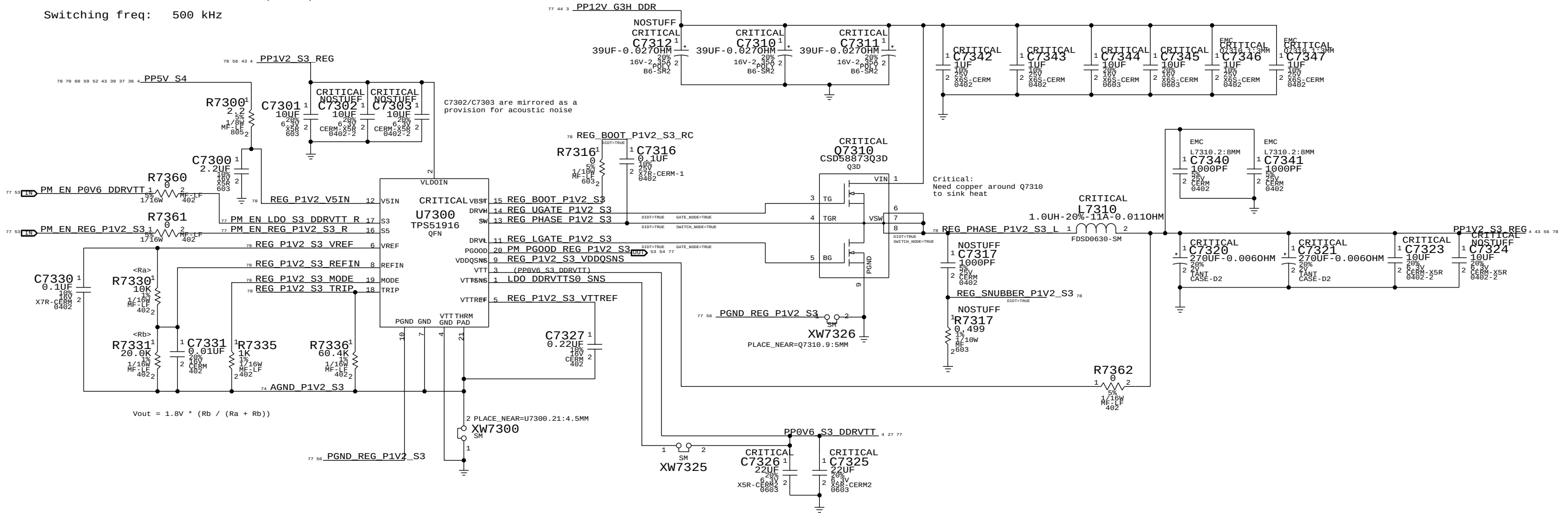
Audio + PCH Sequencing Requirements:
5V (4.5V) -> 1.05V -> 3.3V -> 1.5V -> ALL SYS GOOD
POWER DOWN IN REVERSE ORDER.



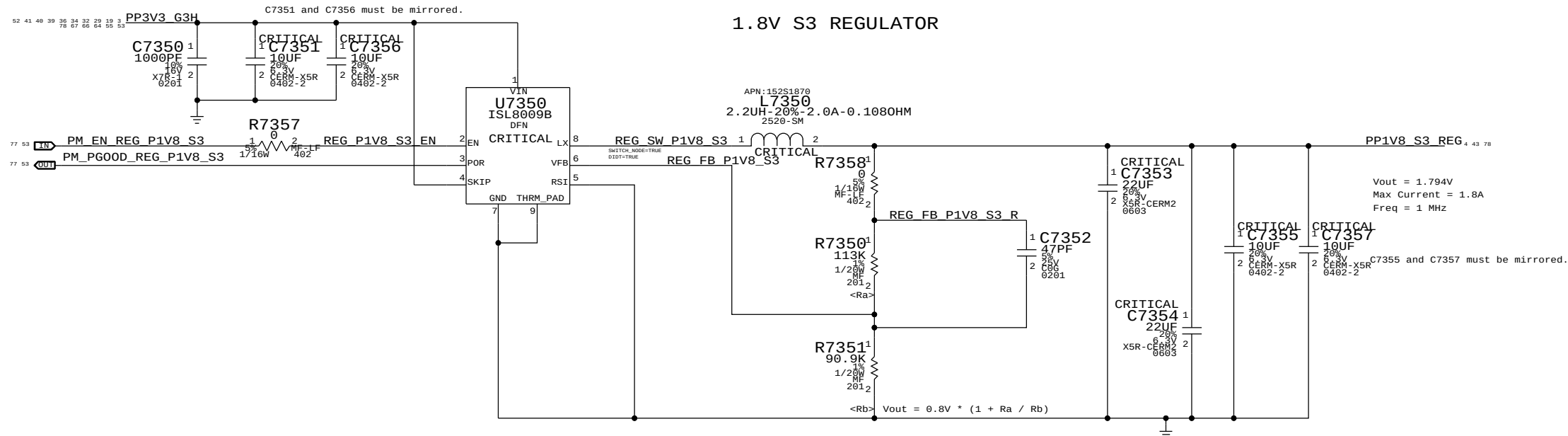
SYNC MASTER=J64 2C		SYNC DATE=02/03/2014	
PAGE TITLE			
POWER CONTROL		2:S0	
 Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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VDDQ 1.2V S3 Regulator

OC trip point: $13.2 \text{ A VDDQ} = \frac{R7336 * 10e-6}{8 * Rds(Q7310)} + \frac{1.08}{L7310 * f(\text{switch})}$
2 A VTT (FIXED)
10 mA VTTREF (FIXED)
Switching freq: 500 kHz



1.8V S3 REGULATOR



SYNC MASTER=J64 2C		SYNC DATE=2014-01-23	
PAGE TITLE		VR VDDQ S3 / 1.8V S3	
Apple Inc.		DRAWING NUMBER	051-1907
		REVISION	4.0.0
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D

C

B

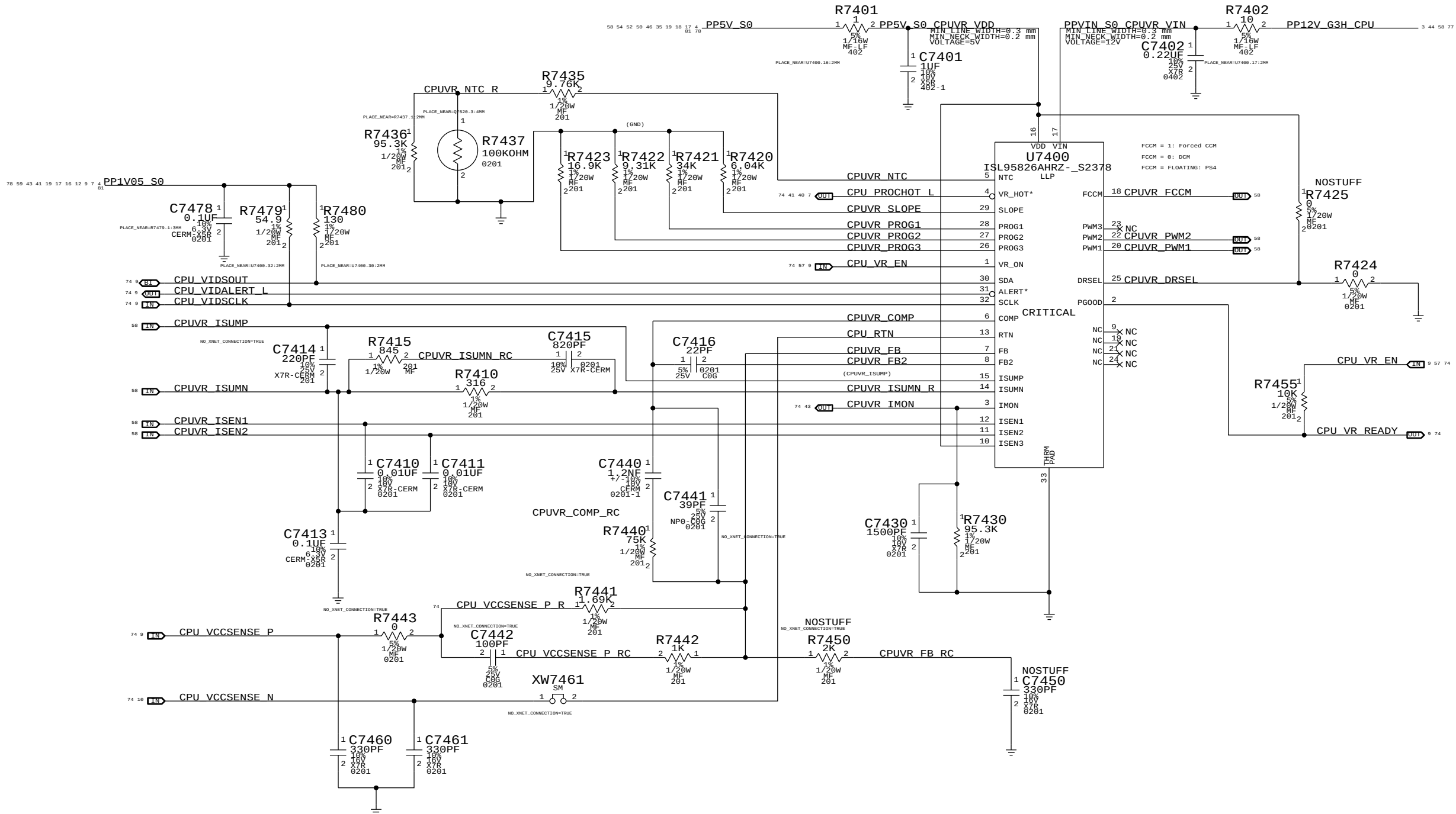
A

D

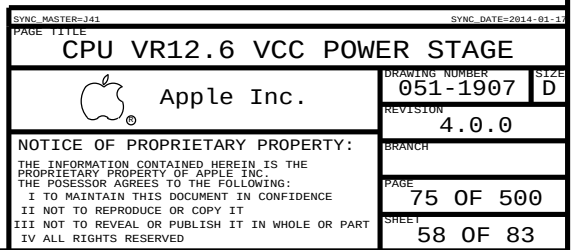
C

B

A



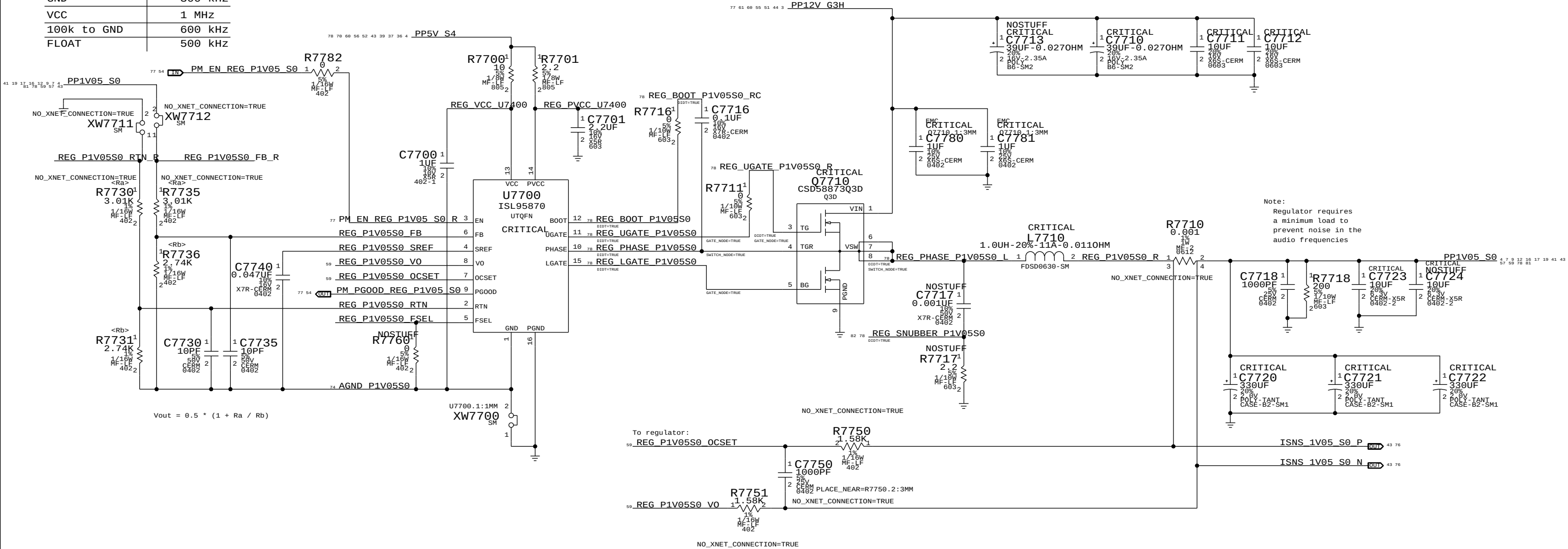
SYNC MASTER=J64 2C		SYNC DATE=2014-01-17	
PAGE TITLE			
CPU VR12.6 VCC Regulator IC			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-1907	D
		REVISION	
		4.0.0	
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		57 OF 83	



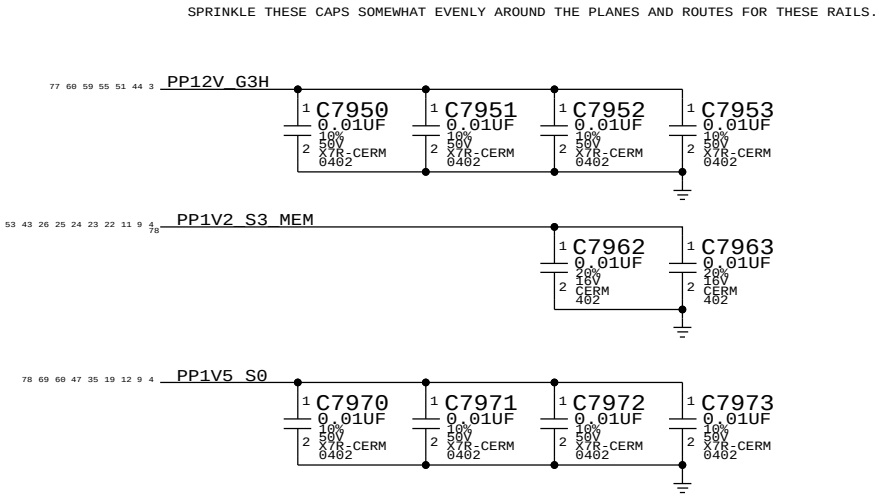
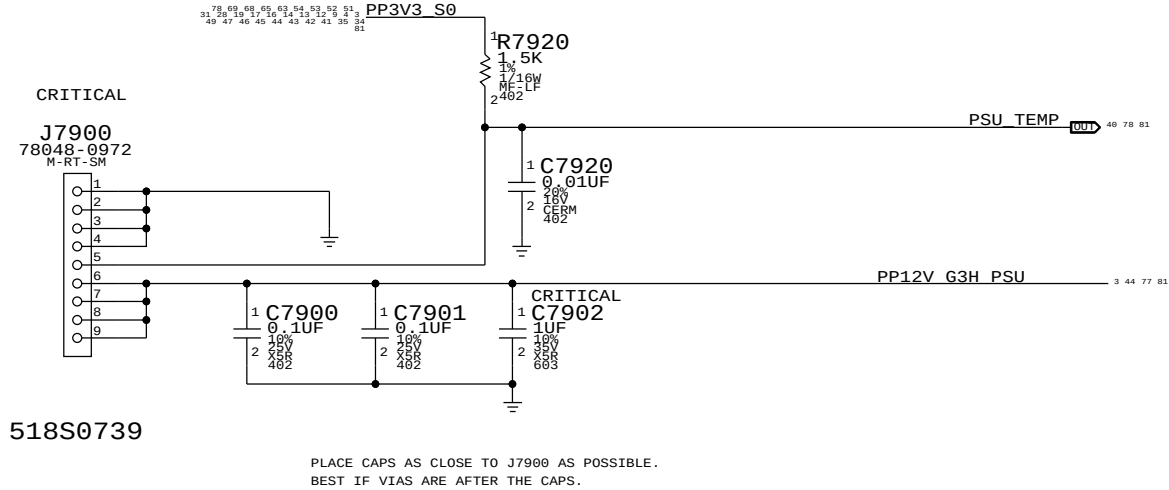
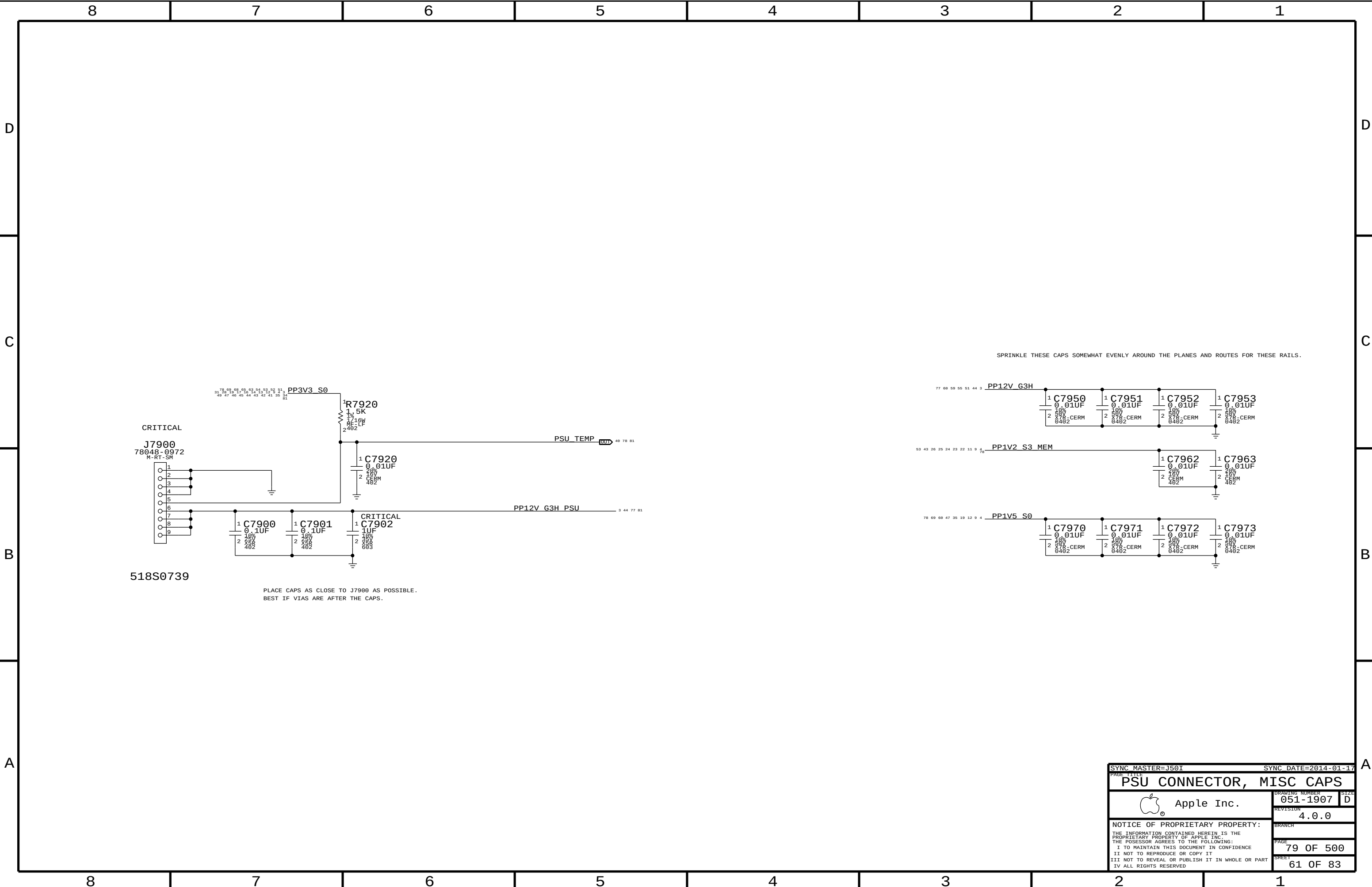
PCH (1.05V) S0 REGULATOR

Switching freq: 500 kHz OC trip point: 12.4 A = $\frac{R7750 \cdot 8.5 \text{ E-6}}{R7710}$

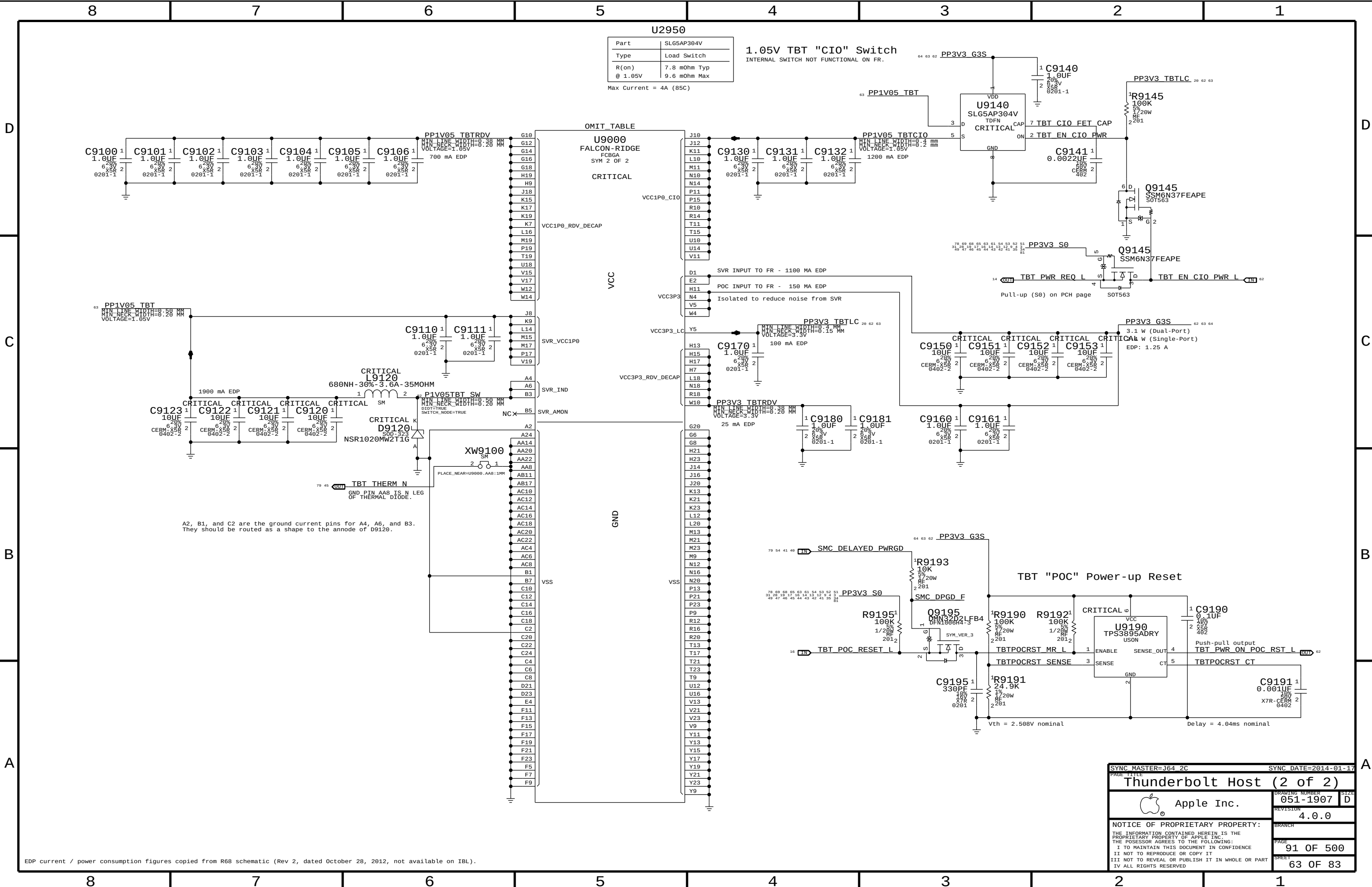
FSEL STRAP	SW FREQ
GND	300 kHz
VCC	1 MHz
100k to GND	600 kHz
FLOAT	500 kHz



Note:
Regulator requires
a minimum load to
prevent noise in the
audio frequencies



SYNC MASTER=J501		SYNC DATE=2014-01-17	
PAGE TITLE		PSU CONNECTOR, MISC CAPS	
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		PAGE	79 OF 500
		SHEET	61 OF 83



Part	SLG5AP304V
Type	Load Switch
R(on)	7.8 mOhm Typ
@ 1.05V	9.6 mOhm Max
Max Current = 4A (85C)	

1.05V TBT "CIO" Switch
INTERNAL SWITCH NOT FUNCTIONAL ON FR.

A2, B1, and C2 are the ground current pins for A4, A6, and B3.
They should be routed as a shape to the anode of D9120.

SYNC MASTER=J64 2C

SYNC DATE=2014-01-17

Thunderbolt Host (2 of 2)

Apple Inc.

051-1907

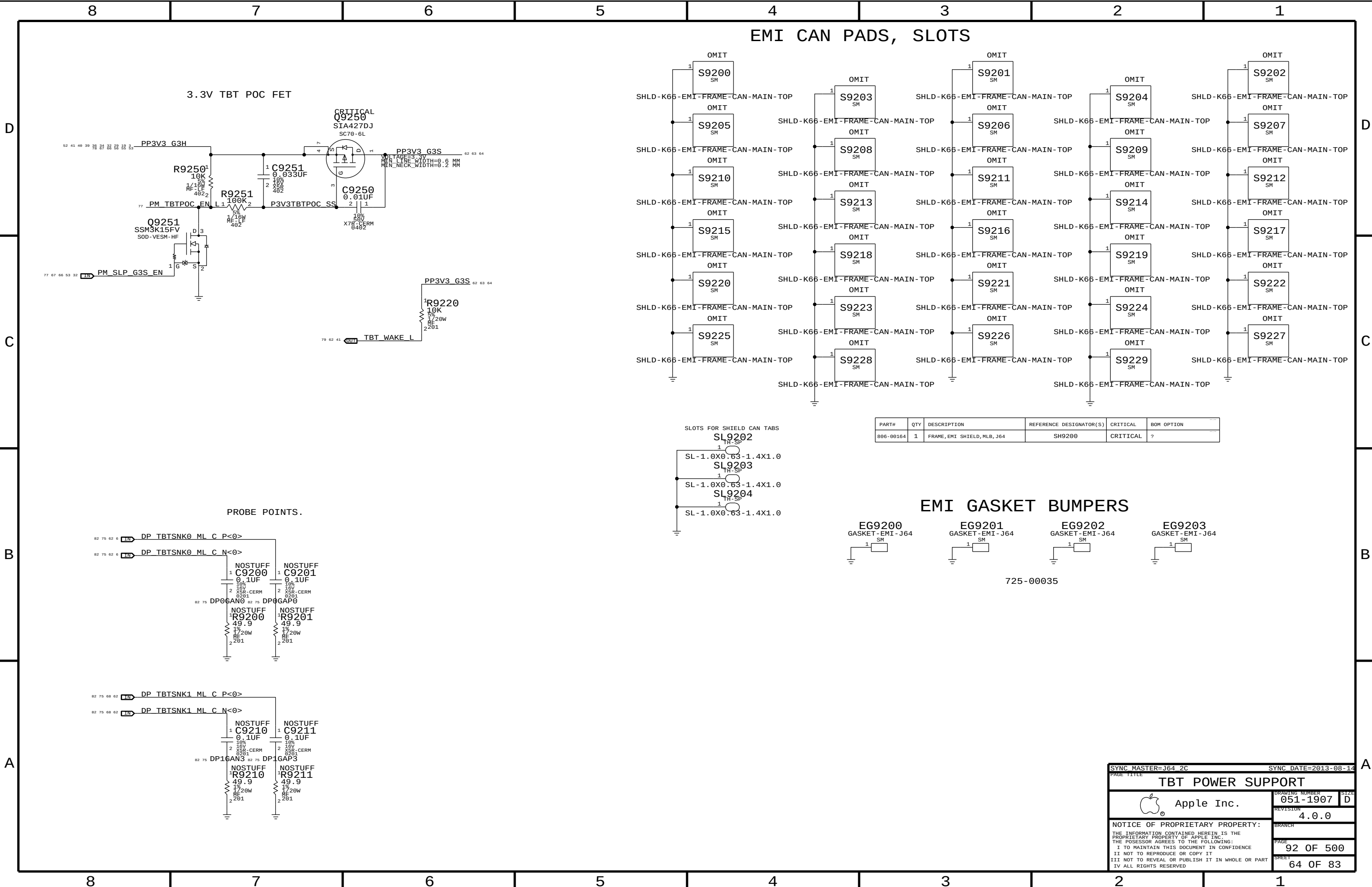
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3.3V TBT POC FET

EMI CAN PADS, SLOTS

EMI GASKET BUMPERS

PROBE POINTS.

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
896-00164	1	FRAME, EMI SHIELD, MLB, J64	SH9200	CRITICAL	?

SYNC MASTER=J64 2C

SYNC DATE=2013-08-14

TBT POWER SUPPORT

Apple Inc.

051-1907

051-1907

4.0.0

4.0.0

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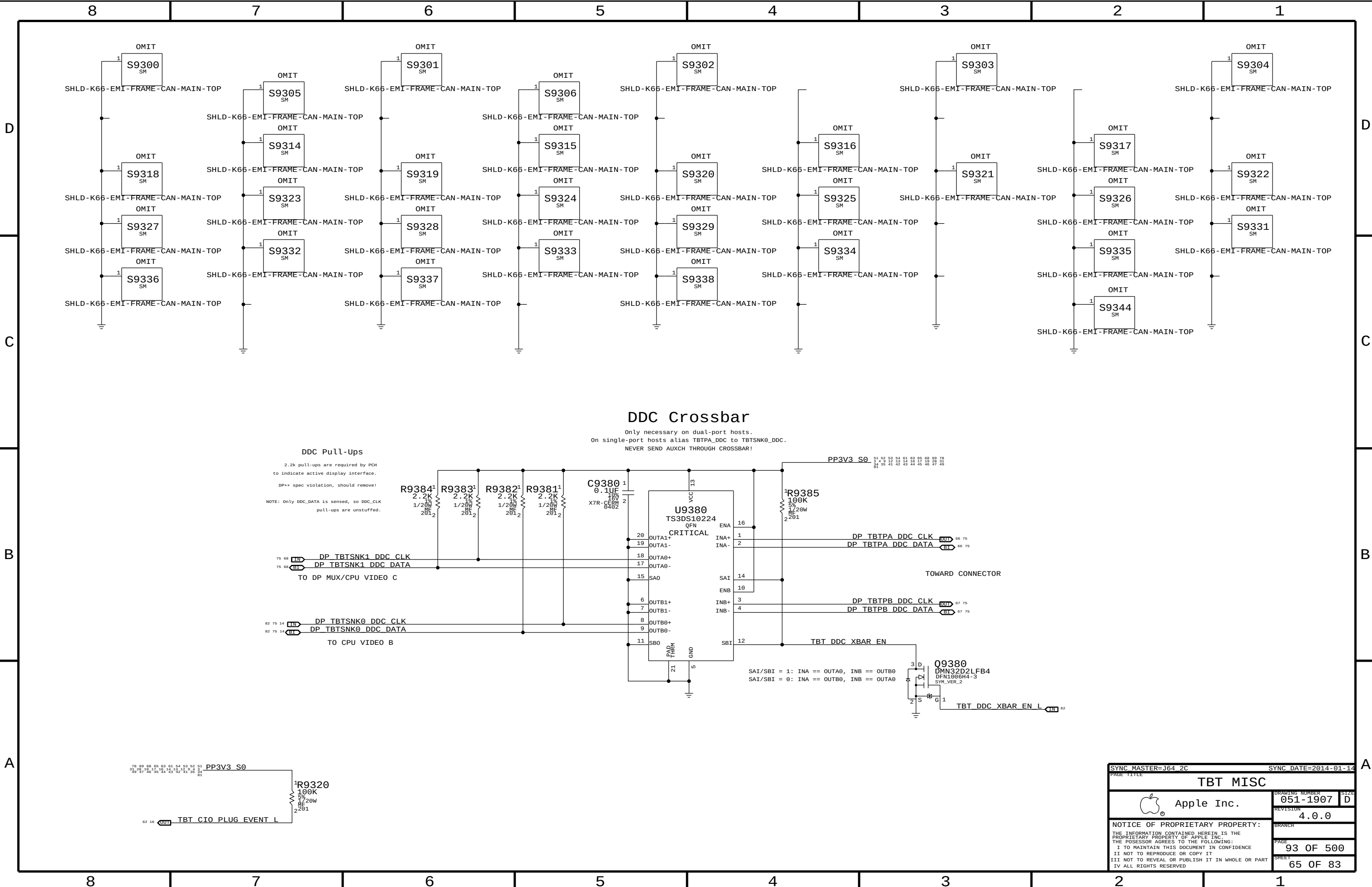
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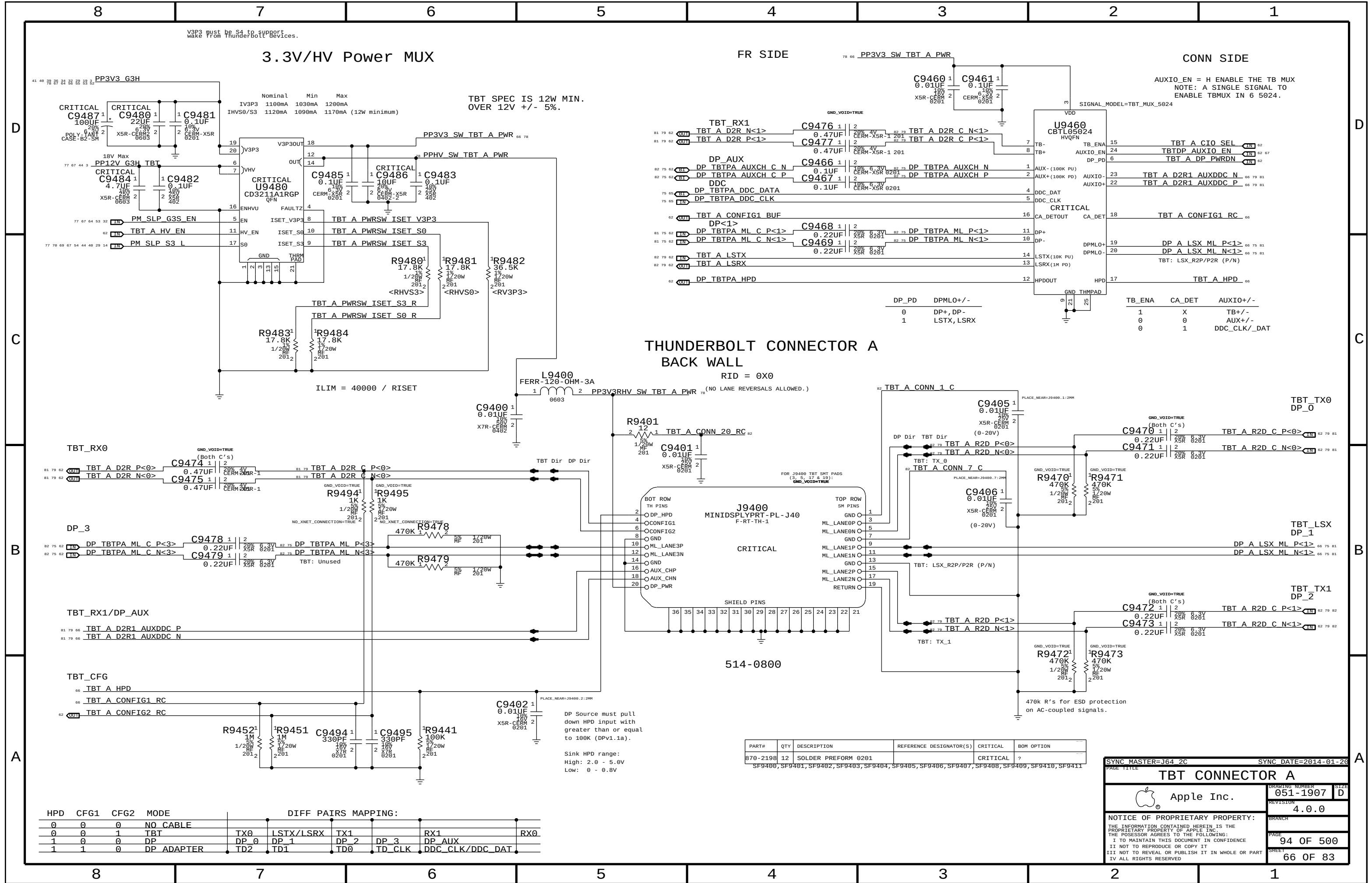
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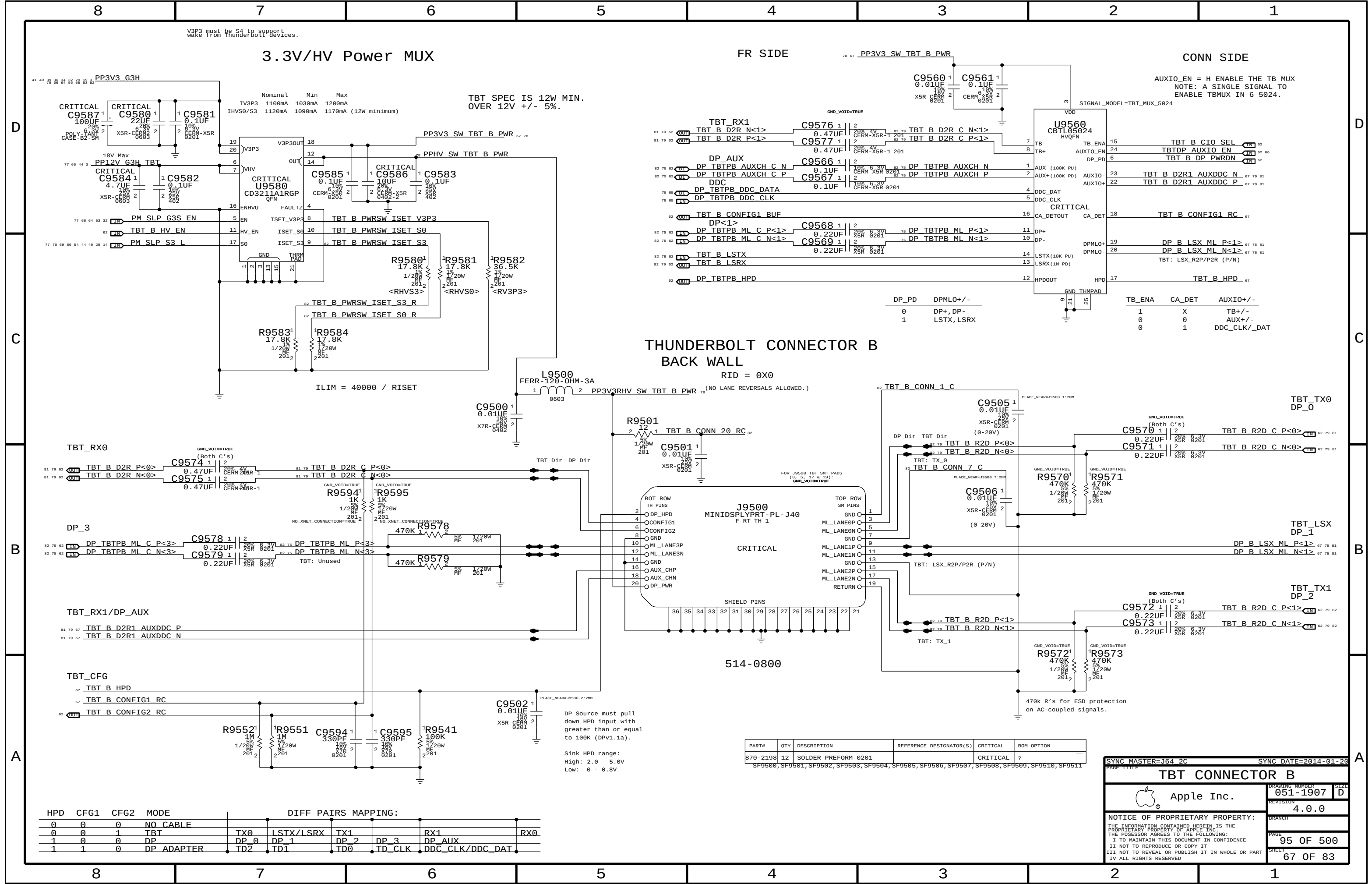
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SYNC MASTER=J64 2C		SYNC DATE=2014-01-14	
PAGE TITLE			
TBT MISC			
 Apple Inc.		DRAWING NUMBER	051-1907
		SIZE	D
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		BRANCH	
		PAGE	93 OF 500
		SHEET	65 OF 83





HPD	CFG1	CFG2	MODE	DIFF PAIRS MAPPING:					
0	0	0	NO CABLE	TX0	LSTX/LSRX	TX1	DP_2	RX1	RX0
0	0	1	TBT	DP_0	DP_1	DP_2	DP_3	DP_AUX	
1	0	0	DP						
1	1	0	DP ADAPTER	TD2	TD1	TD0	TD_CLK	DDC_CLK/DDC_DAT	

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
870-2198	12	SOLDER PREFORM 0201		CRITICAL	?
SF9500, SF9501, SF9502, SF9503, SF9504, SF9505, SF9506, SF9507, SF9508, SF9509, SF9510, SF9511					

SYNC MASTER=J64 2C

SYNC DATE=2014-01-26

TBT CONNECTOR B

Apple Inc.

DRAWING NUMBER

051-1907

SIZE

D

REVISION

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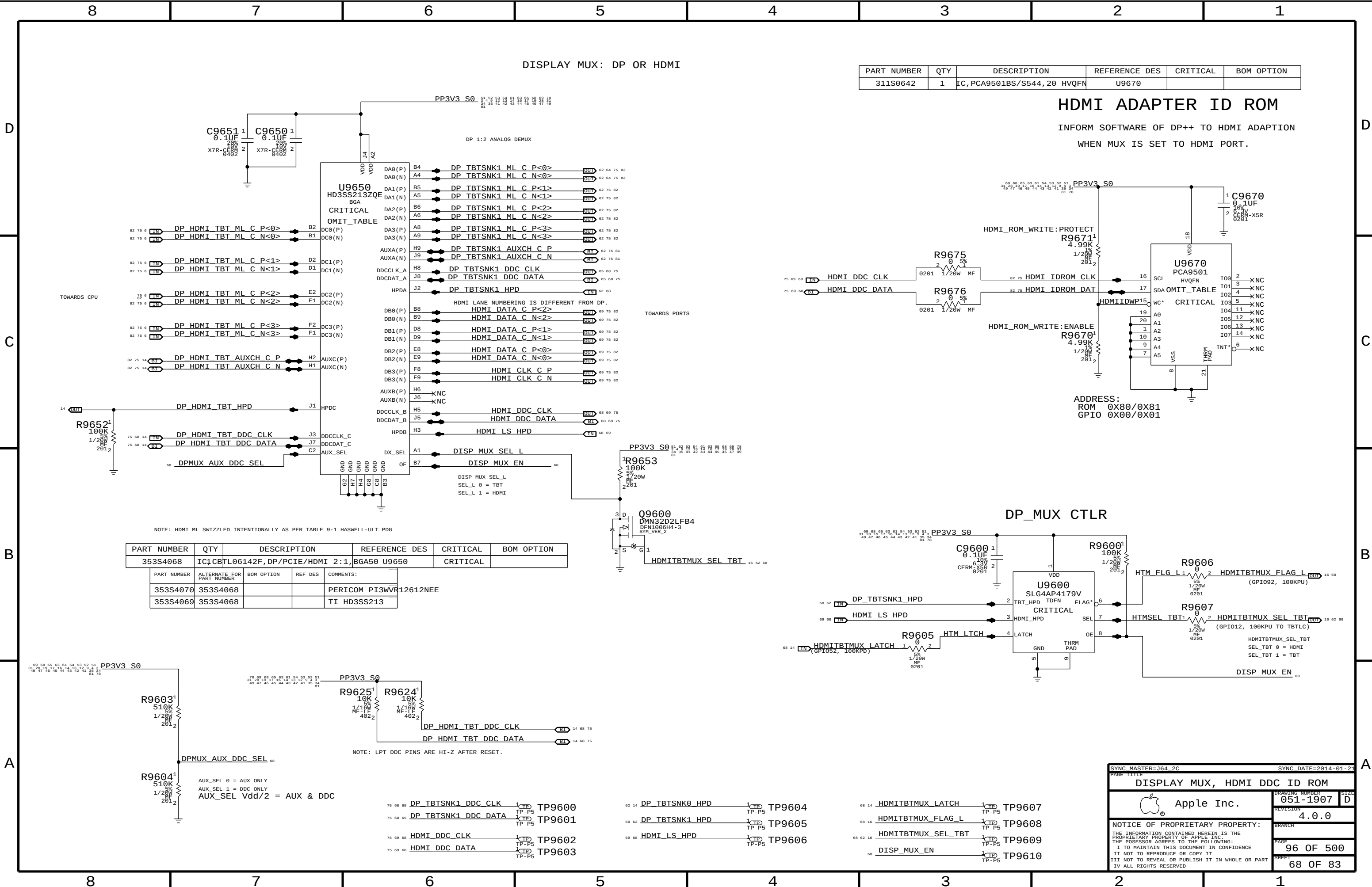
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SHEET

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DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

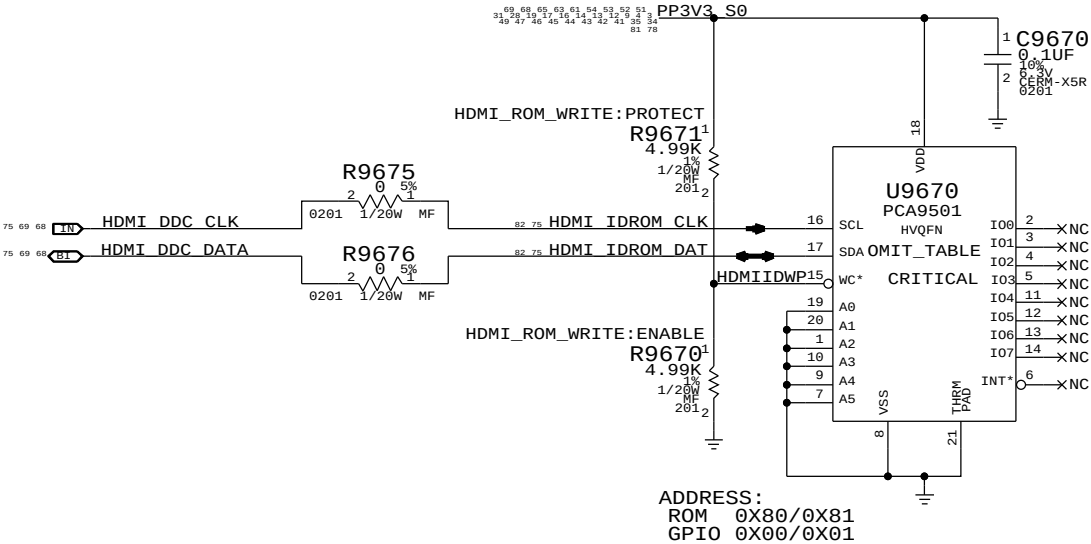


DISPLAY MUX: DP OR HDMI

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
311S0642	1	IC, PCA9501BS/S544, 20 HVQFN	U9670		

HDMI ADAPTER ID ROM

INFORM SOFTWARE OF DP++ TO HDMI ADAPTION
WHEN MUX IS SET TO HDMI PORT.

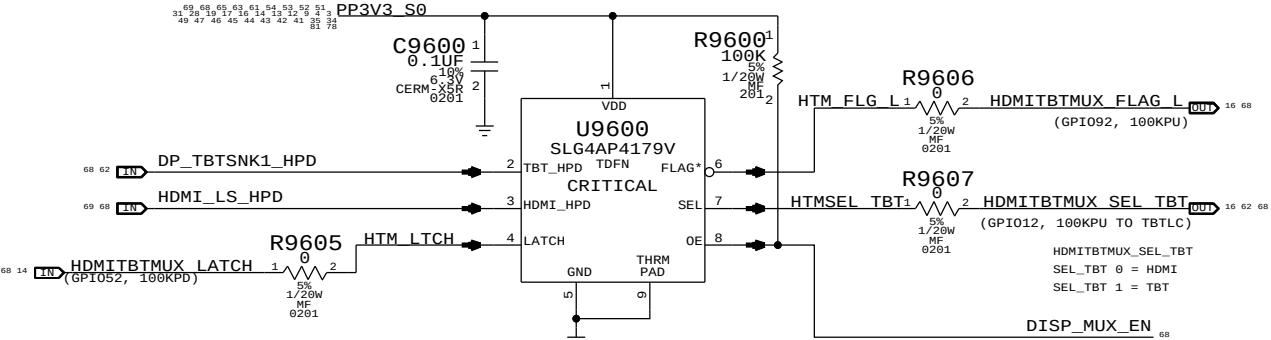


ADDRESS:
ROM 0X80/0X81
GPIO 0X00/0X01

NOTE: HDMI ML SWIZZLED INTENTIONALLY AS PER TABLE 9-1 HASWELL-ULT PDG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S4068	1	IC, CBL06142F, DP/PCIE/HDMI 2:1, BGA50	U9650	CRITICAL	
353S4070	353S4068				PERICOM PI3WVR12612NEE
353S4069	353S4068				TI HD3SS213

DP_MUX CTLR

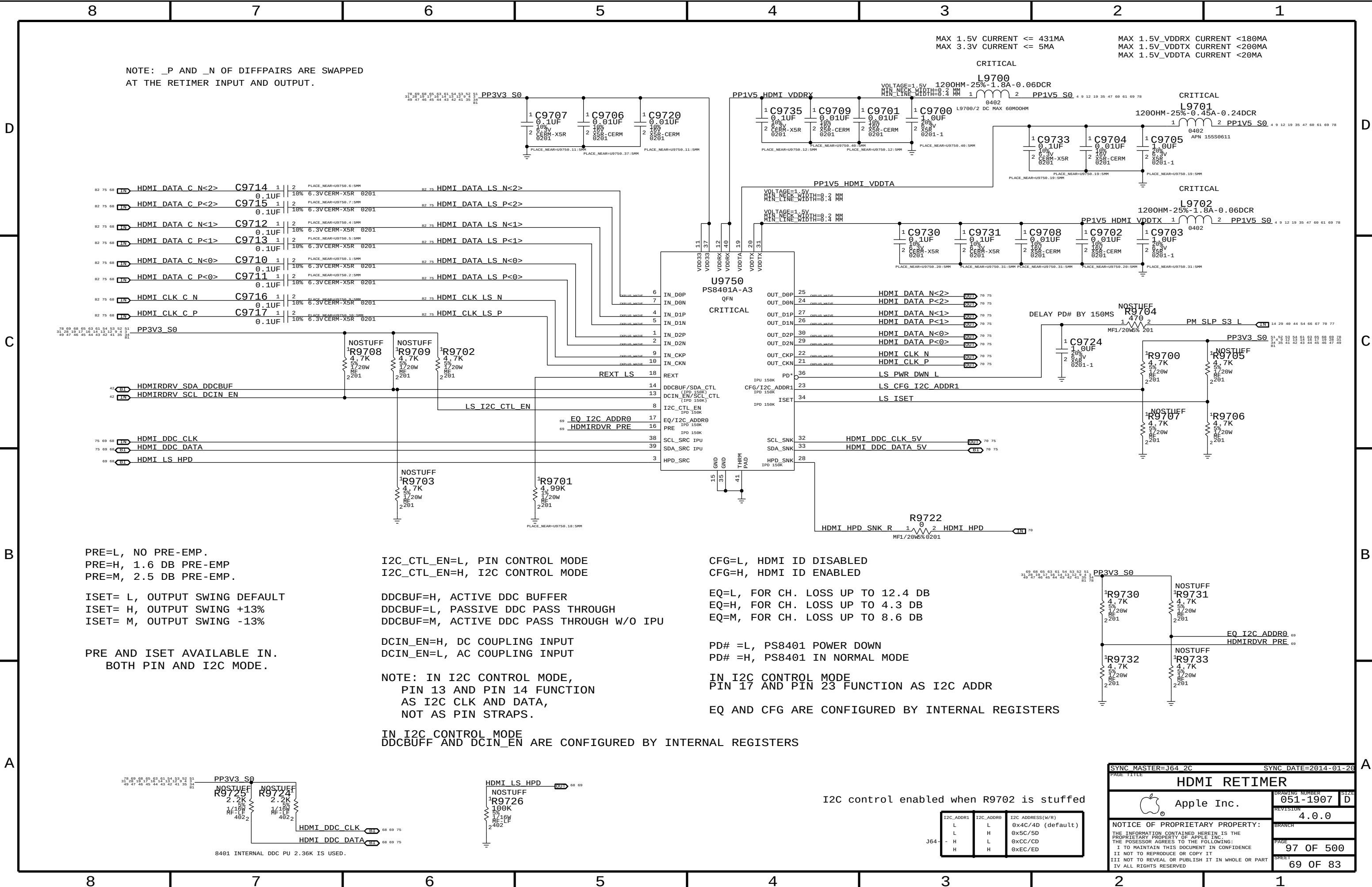


NOTE: LPT DDC PINS ARE HI-Z AFTER RESET.

75 68 65	DP_TBTSNK1_DDC_CLK	1	TP-P5	TP9600
75 68 65	DP_TBTSNK1_DDC_DATA	1	TP-P5	TP9601
75 69 68	HDMI_DDC_CLK	1	TP-P5	TP9602
75 69 68	HDMI_DDC_DATA	1	TP-P5	TP9603
62 14	DP_TBTSNK0_HPD	1	TP-P5	TP9604
68 62	DP_TBTSNK1_HPD	1	TP-P5	TP9605
69 68	HDMI_LS_HPD	1	TP-P5	TP9606

68 14	HDMITBTMUX_LATCH	1	TP-P5	TP9607
	HDMITBTMUX_FLAG_L	1	TP-P5	TP9608
68 16	HDMITBTMUX_SEL_TBT	1	TP-P5	TP9609
68	DISP_MUX_EN	1	TP-P5	TP9610

SYNC MASTER=J64 2C		SYNC DATE=2014-01-21	
PAGE TITLE		DRAWING NUMBER	
DISPLAY MUX, HDMI DDC ID ROM		051-1907	
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		4.0.0	
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NOTE: _P AND _N OF DIFFPAIRS ARE SWAPPED
AT THE RETIMER INPUT AND OUTPUT.

MAX 1.5V CURRENT <= 431MA
MAX 3.3V CURRENT <= 5MA
MAX 1.5V_VDDRX CURRENT <180MA
MAX 1.5V_VDDTX CURRENT <200MA
MAX 1.5V_VDDTA CURRENT <20MA

CRITICAL

CRITICAL

CRITICAL

CRITICAL

CRITICAL

PRE=L, NO PRE-EMP.
PRE=H, 1.6 DB PRE-EMP
PRE=M, 2.5 DB PRE-EMP.

ISET= L, OUTPUT SWING DEFAULT
ISET= H, OUTPUT SWING +13%
ISET= M, OUTPUT SWING -13%

PRE AND ISET AVAILABLE IN.
BOTH PIN AND I2C MODE.

I2C_CTL_EN=L, PIN CONTROL MODE
I2C_CTL_EN=H, I2C CONTROL MODE

DDCBUF=H, ACTIVE DDC BUFFER
DDCBUF=L, PASSIVE DDC PASS THROUGH
DDCBUF=M, ACTIVE DDC PASS THROUGH W/O IPU

DCIN_EN=H, DC COUPLING INPUT
DCIN_EN=L, AC COUPLING INPUT

NOTE: IN I2C CONTROL MODE,
PIN 13 AND PIN 14 FUNCTION
AS I2C CLK AND DATA,
NOT AS PIN STRAPS.

CFG=L, HDMI ID DISABLED
CFG=H, HDMI ID ENABLED

EQ=L, FOR CH. LOSS UP TO 12.4 DB
EQ=H, FOR CH. LOSS UP TO 4.3 DB
EQ=M, FOR CH. LOSS UP TO 8.6 DB

PD# =L, PS8401 POWER DOWN
PD# =H, PS8401 IN NORMAL MODE

IN I2C CONTROL MODE
PIN 17 AND PIN 23 FUNCTION AS I2C ADDR

EQ AND CFG ARE CONFIGURED BY INTERNAL REGISTERS

I2C control enabled when R9702 is stuffed

I2C_ADDR1	I2C_ADDR0	I2C ADDRESS(W/R)
L	L	0x4C/4D (default)
L	H	0x5C/5D
H	L	0xCC/CD
H	H	0xEC/ED

SYNC MASTER=J64 2C

SYNC DATE=2014-01-20

HDMI RETIMER

Apple Inc.

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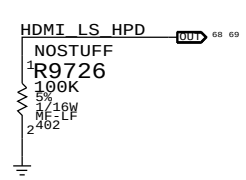
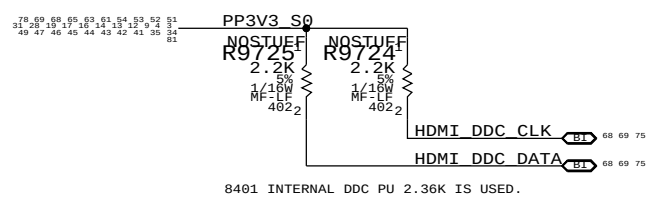
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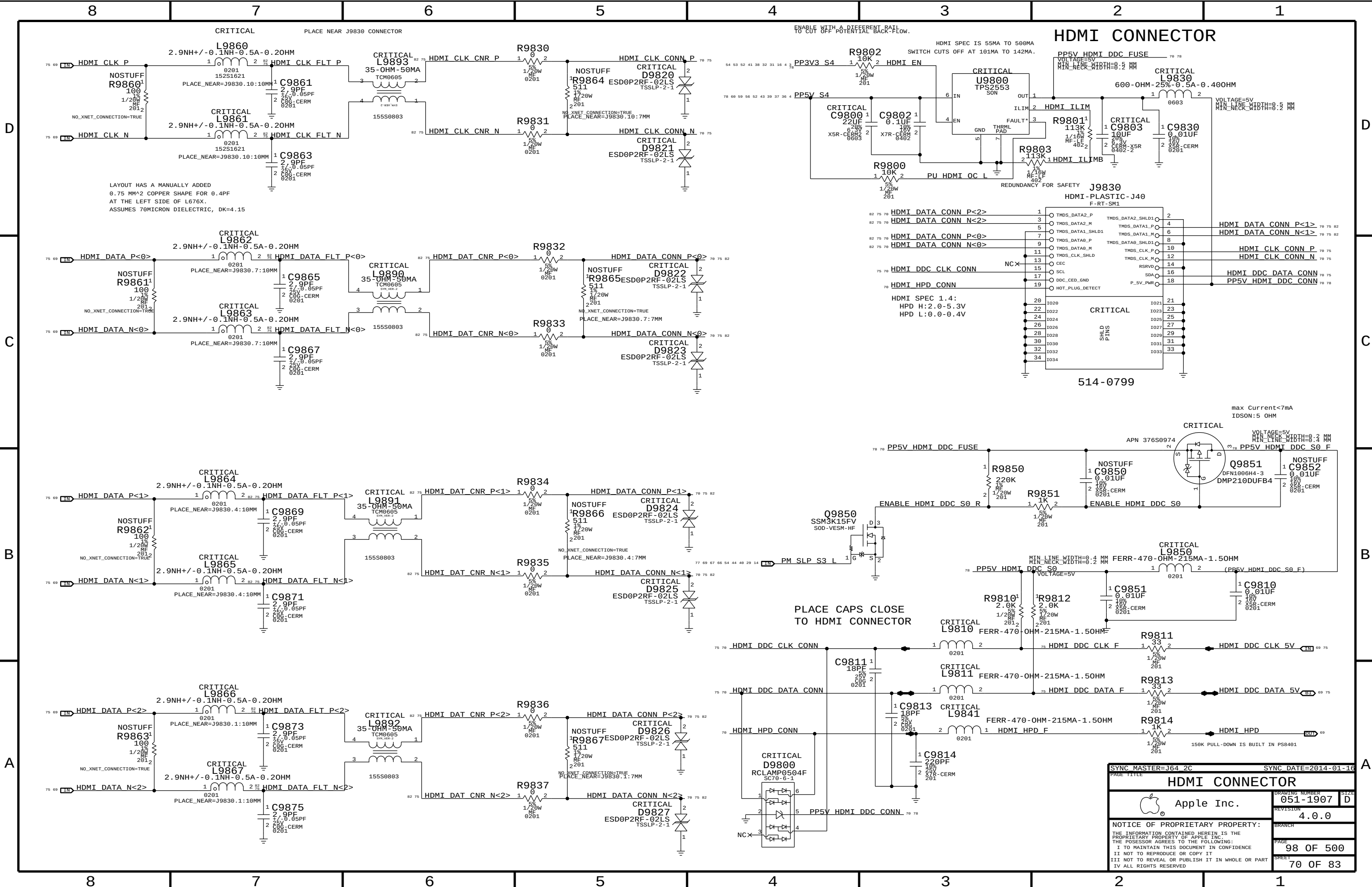
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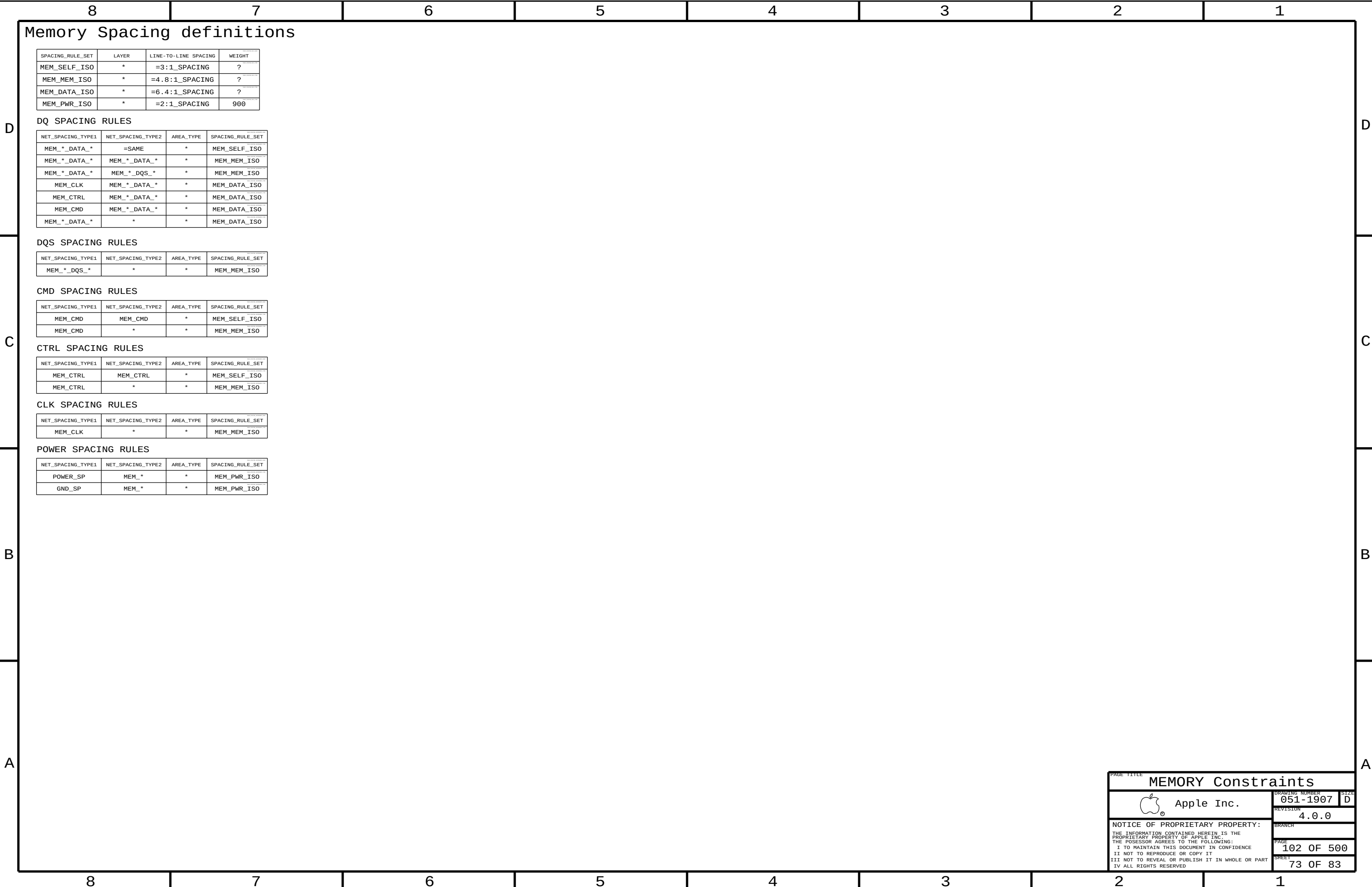
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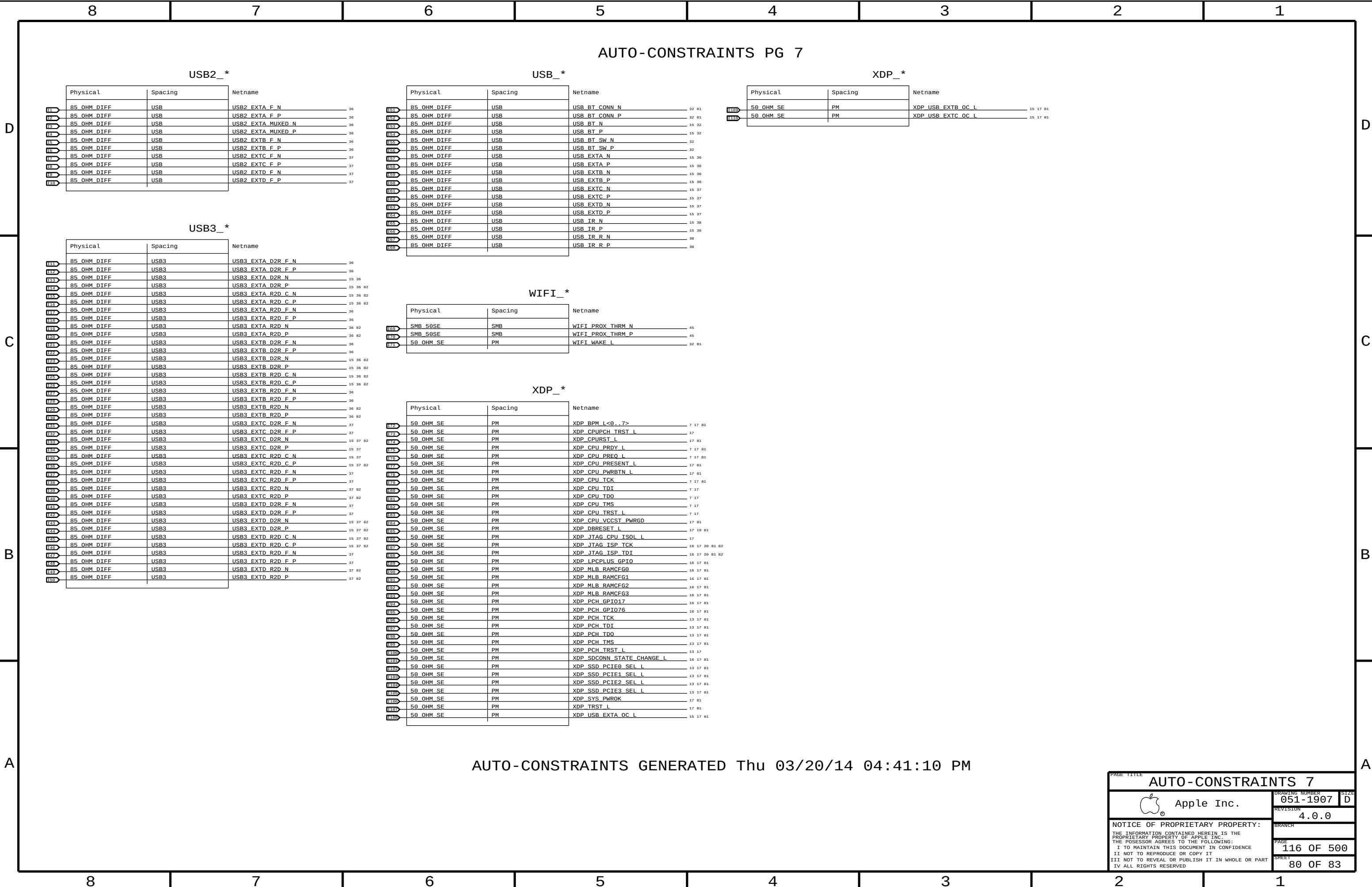
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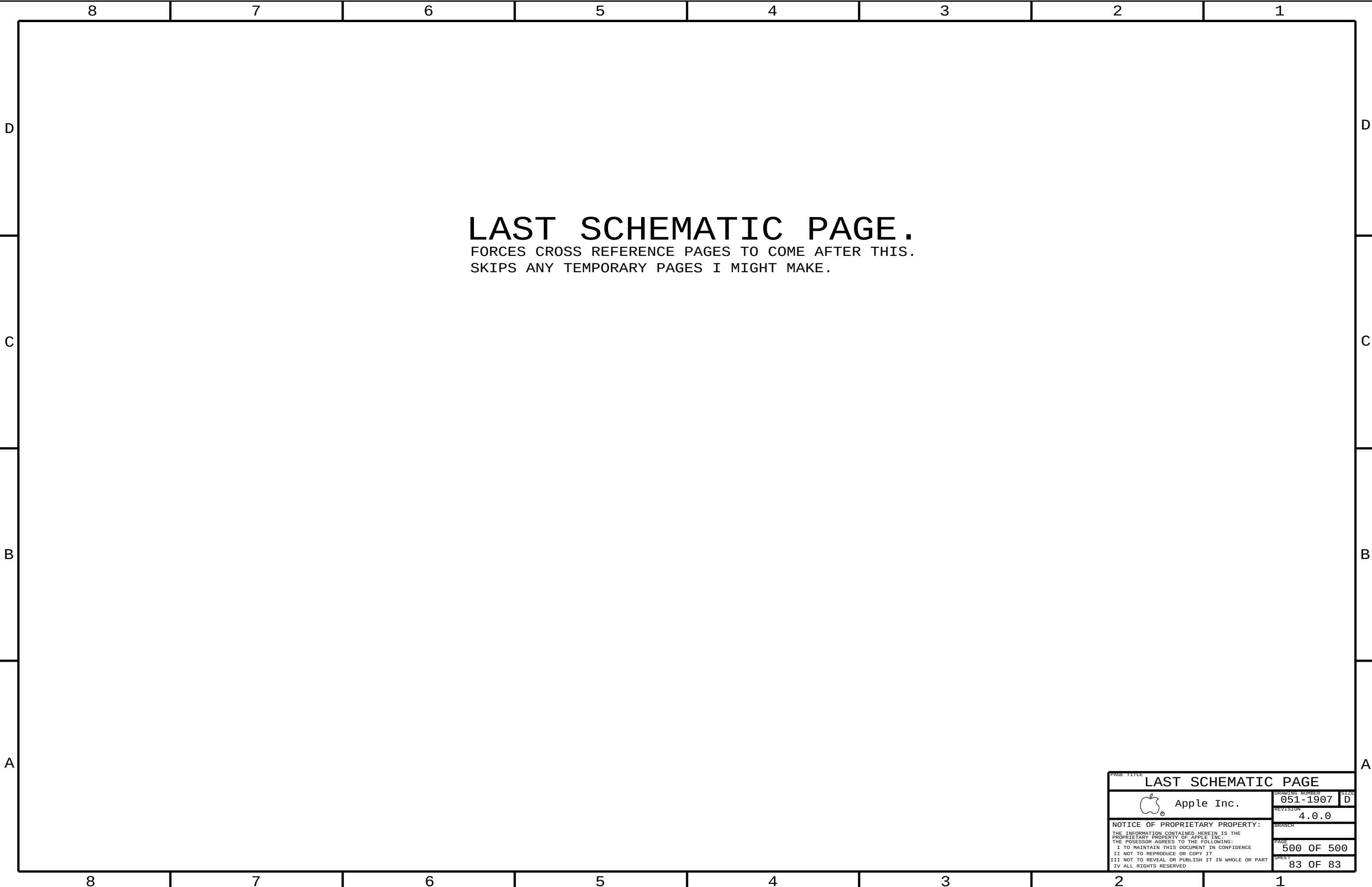




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